Mathematically : $F = \frac{KQ_1Q_2}{r^2}$ where $K = 9\times 10^9$ N. m^2/C^2 is the proportionality constant.

Newton's law of gravitation states : that everybody in the universe attracts every other body with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Mathematically : $F = \frac{Gm_1m_2}{r^2}$ where $G = 6.67\times 10^{-11}$ $N.m^2/kg^2$ is the gravitational constant. Therefore , Coulomb's law for the force between electric charges most closely resembles to Newton's law of gravitation.

∴ ***Answer C***

45.Household light bulbs are normally connected in parallel to a power supply. Suppose a 40 W and a 60W light bulbs are ,instead connected in series. Which bulb is brighter?

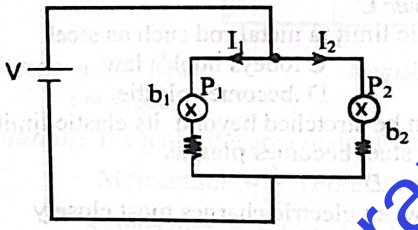
A .The 60 W bulb.
B .The 40 W bulb.
C .The bulbs are equally bright.
D .The bulbs will not give light in series connection.

Given: bulbs = $b_1 = p_1 = 40W$

Bulbs = $b_2 = p_2 = 60W$

Required : Which bulb is brighter when corrected in series = ?

Solution: $P_1 = v\,.I_1$



$$P_1 = \frac{V^2}{R_1}$$

$$R_1 = \frac{V^2}{P_1}$$

$$R_2 = \frac{V^2}{P_2}$$

$$\frac{R_2}{R_1} = \frac{P_1}{P_2}$$

$$\frac{R_2}{R_1} = \frac{40W}{60W}$$

$$R_2 = \frac{2}{3}R_1$$

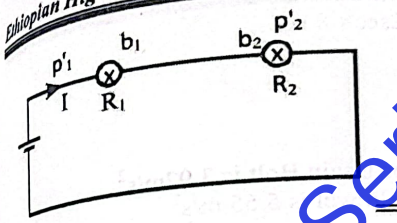
$$\Longrightarrow R_2 < R_1$$

∴ When connected in parallel the 60 w bulb is brighter .

When connected in series: $I_1 = I_2 = I$ ..... series

$P'_1 = I^2{}_1R_1 = I^2.R_1$

$P'_2 = I_2{}^2.R_2 = I^2.R_2$



$$\frac{P'_2}{P'_1} = \frac{R_2}{R_1}$$

$$P'_2 = \frac{R_2}{R_1}P'_1$$

$R_2 < R_1 \Longrightarrow P'_2 < P'_1$

$P'_2 = I^2.R_2$

$P'_1 = I^2.R_1$

$\Longrightarrow R_1 > R_2$.

∴ $\mathbf{P'_1 > P'_2}$

Therefore , the 40 W bulb is brighter than the 60 W bulb.

Answer B

46 .In a 100m race ,Usain Bolt and Assafa Powel cross the finish line in a dead heat, both taking 10 S. Accelerating uniformly, Usain Bolt took 3s and Assafa Powel took 2 s to attain maximum speed, which they maintained for the rest of the race. What was the acceleration of Usain Bolt and Assafa Powel respectively?

A 4.92m/s², 2.25 m/s²
B 3.92m/s² ,5.55m/s²
C.4.92m/s², 6.45m/s²
D.4.92m/s², 5.35 m/s²

Given one : Seed forUsain Bolt ($U_B = 0$)

Acceleration for Usain Bolt a_1 for t_1 =3 sec ,t_2 = (10 - 3)sec =7 sec .

V_1 is his maximum speed .So$V_1 = a_1t_1$.

Required one : Acceleration for Usain Bolt (a_1 = ?)

Solution one: T otal distance for both usain Bolt and Assefa Powel is 100 m and total time for both in 10 sec .

$$S_T = S_1 + S_2$$

$$100m = U_Bt_1 + \frac{1}{2}a_1t^2{}_1 + v_1t_2 = \frac{1}{2}a_1t^2{}_1 + v_1t_2$$

$$100\,m = \frac{1}{2}(a_1)\,(3s)^2 + (a_1t_1)\,t_2$$

$$100m = \frac{9}{2}a_1 + a_1 3Sec \times 7Sec$$

$$\therefore a_1 = 3.92m/s^2$$

Given two : Assafa Powel Speed ($U_A = 0$)

Acceleration for Assafa Powel a_2 for t_1 =2 sec ,t_2 = (10 - 2)s =8 s .

V_2 is his maximum speed .So$V_2 = a_2t_1$

Required two : Acceleration for Assafa Powel (a_2 = ?)

Solution two: His total distance is $S = s_1 + S_2$

$$100\,m = U_At_1 + \frac{1}{2}a_2t^2{}_1 + v_2t_2$$

$$100\,m = \frac{1}{2}a_2t^2{}_1 + v_2t_2$$

$$100\,m = \frac{1}{2}a_2(2sec)^2 + a_2t_1t_2$$

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$100\ m = \frac{1}{2} a_2 \times 4\ sec^2 + a_2 \times 2sec \times 8\ sec$

$2a_2 + 16\ a_2 = 100$

$a_2 = \frac{100}{18}$

$\therefore a_2 = 5.555\ m/s^2$

Therefore, the a cceleration for Usain Bolt is $3.92 m/s^2$ and the acceleration for Assafa Powel is $5.55\ m/s^2$

$\therefore$ ***Answer B***

47. A running man has half the kinetic energy that a boy of half his mass has. The man speeds up by 1m/s and then has the same kinetic energy as the boy. What is the original speed of the man in m/s?

A. $\frac{1}{\sqrt{2}+1}$ C. $\frac{1}{\sqrt{2}+2}$

B. $\frac{1}{\sqrt{2}-1}$ D. $\frac{1}{\sqrt{2}-2}$

Given :The KE_m and KE_b are the original kinetic energies of the man and the boy respectively is given by :

$KE_m = \frac{1}{2} KE_b$

$m_b = \frac{1}{2} m_m$

The man speed V_m up by 1 m/sec his speed is now 1 m/sec + Vom.

$\Longrightarrow KE_m = KE_b$

Required :Original speed of the man in m/sec (Vom) = ?

Solution: $KE_m = \frac{1}{2} KE_b$

$\frac{1}{2} m_m V^2_m = \frac{1}{2}(\frac{1}{2} m_b V^2_b)$

$2m_b V^2_m = \frac{1}{2} m_b V^2_b$

$4V^2 m = V^2_b$

$V_b = 2V_m$

$KE_m = KE$

$\frac{1}{2} m_m V^2_m = \frac{1}{2} m_b V^2_b$

$2m_b V^2 m = m_b V^2_b$

$2V^2_m = V^2_b$

$2(V_m+1)^2 = V_b^2 = (2V_m)^2$

$(V_m+1)^2 = 2V_m^2$

$V_m+1 = \sqrt{2}\, V_m$

$\therefore V_m = \frac{1}{\sqrt{2}-1}$ m/s

$\therefore$ ***Answer B***

48. Consider a long cylindrical charge distribution of radius R with a uniform charge density ρ. What is the magnitude of electric field at distance r from the axis where r < R?

A. $\frac{\rho r}{3\varepsilon_o}$ B. $\frac{\rho r}{2\varepsilon_o}$ C. $\frac{\rho r}{7\varepsilon_o}$ D. $\frac{\rho r}{5\varepsilon_o}$

Given : $\rho = \frac{q}{V} \Longrightarrow q = \rho V$

Required : Magnitude of electric field at r < R (E = ?)

Solution:

$E.A = \frac{\rho V}{\varepsilon_o}$

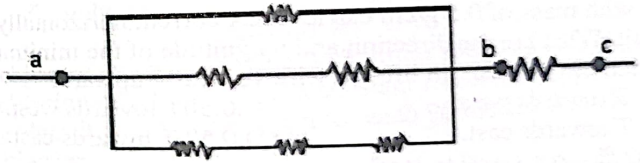
$E \times 2\pi rh = \frac{\rho \pi r^2 h}{\varepsilon_o}$ .....Area of cylinder = $2\pi rh$

$\therefore E = \frac{r\rho}{2\varepsilon_o}$

$\therefore$ ***Answer B***

49. What is the ratio of power dissipated by the combined network of resistors ,each with resistance R ,between terminals a and b to the power dissipated by a resistor of resistance R between terminals b and c, shown in the figure below. The two terminals a and c of the network are connected to a battery such that current I flows from terminals a to terminal c?

A. $\frac{7}{13}$ B. $\frac{6}{13}$ C. $\frac{7}{11}$ D. $\frac{6}{11}$

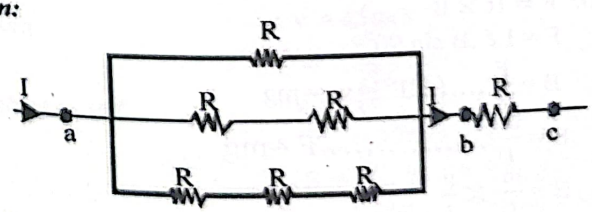


Given : P_{ab} – Power dissipated between a and b

P_{bc} – power dissipated between b and c

Required : Ratio of power dissipated = $\frac{P_{ab}}{P_{bc}} = ?$

Solution:



$P_{ab} = I^2 R_{Tab}$

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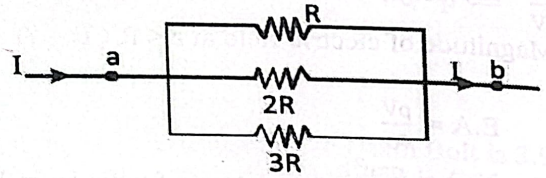

$$\frac{1}{R_{Tab}} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{3R}$$

$$R_{Tab} = \frac{6}{11} R$$

$$P_{ab} = I^2 \times \frac{6R}{11}$$

$$P_{bc} = I^2 \times R$$

$$\frac{P_{ab}}{P_{bc}} = \frac{I^2 \frac{6}{11R}}{I^2 R}$$

$$\therefore \frac{P_{ab}}{P_{bc}} = \frac{6}{11}$$

∴ Answer D

50.A wire with mass of 0.5 g/cm carries a 2A current horizontally towards the south. What are the direction and magnitude of the minimum magnetic field needed to lift this wire vertically upward?

A .0.25 T towards west.
B .0.25 T towards east.
C.0.50T towards west.
D.0.50 T towards east.

Given : $\frac{m}{\ell} = 0.5 \times 10^{-3}$ kg/10^{-2} m

$\frac{m}{\ell} = 5 \times 10^{-2}$ kg/m

$I = 2A$

$g = 10$ m/sec^2

Required: $\vec{B} = ?$

Solution: $\vec{F} = I\vec{\ell} \times \vec{B}$

$F = L\ell .B \sin 90^0$

$B = \frac{F}{I\ell}$ ..... $F = w = mg$

$B = \frac{mg}{\ell I}$ ..... $F = mg$

$B = \frac{m}{\ell} \times \frac{g}{I}$

$\vec{B} = 5 \times 10^{-2}$kg/m $\times \frac{10}{2A}$m/s^2 due east.

$\therefore \vec{B} = 0.25$ T due east.

∴ Answer B

1.Which of the following is a vector quantity?

A .power
B . Momentum
C. Temperature
D. Electric current

Solution: **A vector quantity** is a quantity described by both magnitude and direction.

Examples: Some examples of vector quantities are:

- ✓ displacement
- ✓ weight
- ✓ velocity
- ✓ acceleration
- ✓ momentum
- ✓ torque

✓ **A scalar quantity** has only magnitude and there is no direction associated with it.

Example: Some examples of scalar quantities are:

- ✓ distance
- ✓ mass
- ✓ time
- ✓ electric current
- ✓ volume
- ✓ density
- ✓ speed
- ✓ real numbers
- ✓ temperature
- ✓ work
- ✓ energy
- ✓ power
- ✓ potential difference

∴ Answer B

2.An object of mass m moving with initial speed of 15m/s got an acceleration of 10 m/s^2 after a constant force F is applied to it . How long does it take for the object to reach 90 meters from its initial position?

A .3.22S
B .0.75 S
C .3S
D .2.5S

Given:

u = 15m/s

a = 10m/s^2

S = 90m

Required : t = ?

Solution:

$v^2 = u^2 + 2sa$

$v = \sqrt{(15\ m/s)^2 + 2 \times 90m \times 10m/s^2}$

$v = \sqrt{225 + 1800}$

$\therefore v = 45$m/s

$V_{av} = \frac{u+v}{2}$

$V_{av} = \left(\frac{15+45}{2}\right)$

$\therefore V_{av} = 30$m/s

$S = V_{av} \times t \Rightarrow t = \frac{S}{V_{av}}$

$\therefore t = \frac{90m}{30m/s}$

$\therefore t = 3$ sec

∴ Answer C

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3. The zeroth law of thermodynamics states that:
   A .two bodies that are separately in thermal equilibrium with a third body must be in thermal equilibrium with each other.
   B .the increase in internal energy of a system is equal to the sum of the energy entering the system through heating, and the work done on the system.
   C . no process is possible in which there is an overall decrease in the entropy of the universe.
   D .no process or cycle is perfectly reversible and no refrigerator or heat engine can be perfectly efficient.

Solution: If a body A is in thermal equilibrium with body B, and body C is in thermal equilibrium with body B, then body A is in thermal equilibrium with body C. This is called **the zeroth law of thermodynamics.**

∴ ***Answer A***

4. Which of the following statements is **NOT** true?
   A. The magnitude of average angular velocity is directly proportional to the angular displacement during a given time.
   B. Average angular velocity and instantaneous angular velocity are the same for uniform angular motion.
   C. The direction of the average angular velocity is determined by the right hand rule.
   D. The direction of average angular velocity is parallel to a tangent line along the trajectory of the object.

Solution: Using the right hand rule if one curls, his right Hand fingers in the direction of tangential motion of a particles, his thumb indicates the direction of angular velocity.

Therefore, $\vec{\omega} \perp \vec{V_T}$.

∴ ***Answer D***

5. Which one of the following statements is true about action and reaction forces referred to in Newton's third law of motion?
   A . They act upon the same body.
   B .They are equal in magnitude but need not have the same line of action.
   C .They act upon two different bodies.
   D. They are not equal in magnitude but have the same line of action.

Solution: **Newton's third law** of motion state that "every action, there is an equal and opposite reaction; action and reaction forces act on different bodies"

∴ ***Answer C***

6. A body has moved from its initial position at origin along x –axis to a final position x =5 meters under the influence of a force given by $\vec{F} = (4x)\hat{i}$ Newton's. What is the work done by the force in moving the body from its initial position to its final position?
   A. 100J B .50J C .25J D .20J

Given: $\vec{F} = (4x)\hat{i}$ N

$x_0 = 0$ m

$x = 5$m

Required : Work done =W =?

Solution: $F(0) = 4\times 0 = 0N$

$F(5) = 4\times 5m = 20N$

$Fav = \frac{F_{min}+F_{max}}{2}$

$Fav = (\frac{0+20}{2})N$

$\therefore F_{av} = 10N$

$W = F_{av} \times \Delta x$ ..... $\theta = 0$

$W = 10N\ (5m - 0m)$

$\therefore W = 50J$

∴ ***Answer B***

7. A wave front is
   A . a line that represents the direction of propagation of light.
   B .the locus of all points of travelling wave that are in phase.
   C .the minimum distance between two points that are in phase.
   D .the number of waves crossing a given plane in a unit time.

Solution: A **wave Front is the locus** of all points of a travelling wave that are in phase.

∴ ***Answer B***

8. Which of the following statements is **NOT** true?
   A. An electric current can be induced in a circuit with a change in the cross- sectional area of the wire.
   B. The induced emf in a circuit is directly proportional to the time rate of change of magnetic flux.
   C. Emf can be induced in a ciruit by change in the area enclosed by the loop
   D. An electric current can be induced in a circuit with a changing of magnetic field

Solution : If the current in a coil changes, the flux through the coil due to the current also changes .As a result , the changing current in a coil induces an emf in that same coil.

Because an induced emf ε is proportional to $\frac{\Delta\phi}{\Delta t}$ and because $\Delta\phi$ is proportional to Δi , where i is the current that cases the flux,

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$\varepsilon = -(\text{constant})\frac{\Delta i}{\Delta t} = -N\frac{\Delta\phi}{\Delta t} = \frac{-N\Delta BA}{\Delta t}$

Therefore , emf can be induced in a circuit be charge

i) Magnetic field (B).
ii) Area (A) of the loop.
iii) Magnetic flux (ϕ)

∴ ***Answer A***

9. Uniform circular motion is a:
A .motion with constant velocity.
B .motion in one dimension.
C .motion with acceleration of constant magnitude.
D .motion in which net force is constant.

Solution: **Uniform circular motion** the velocity changes in direction and is constant in magnitude. Thus the velocity is not constant in two dimensional motion of a particle about a fixed axis (radius) at a constant tangential speed.

∴ ***Answer C***

10. One of the following statements is **NOT** characteristics of electric field.
A .An electric field is the region around a charged object where another charged object will experience a force.
B .Lines of electric field have a direction which move away from negative and towards positive.
C .The direction of electric field is the direction of force acting on a small positive test charge.
D .Lines of electric flux do not cross over each other.

Solution: **Electric lines of** originate at positive charge end at positive directed .Wave from positive end to a negative charge.
They do not cross each other. The field is strong where the field lines are closed.

∴ ***Answer B***

11. Which pair of forces **CANNOT** give a resultant of 5N?
A .1N and 4N
B .1N and 6N
C .3N and 4N
D .1N and 7N

Solution: Give two forces such that F_1 =1N and F_2 = 7N . Their resultant is :. $F_{min} = |F_2 - F_1|$ and $F_{max} = |F_2 + F_1|$

$|7 - 1| \le F_R \le |7 + 1|$

$6\ N \le F_R \le 8\ N$

Therefore ,the resultant force cannot be 5 N .

∴ ***Answer A***

12. A sample of gas is expanded to twice its original volume of $1m^3$ in a quasi-static process for which $p = \propto V$, with $\propto = 5 \times 10^{+5}\frac{N}{m^5}$. How much work was done by the expanding gas?
A .0.75MJ
B .1.0 MJ
C .7.5 MJ
D .1.5 MJ

Solution:

Given :

$P = \propto V$

$\propto = 5\times 10^{+5}\frac{N}{m^5}$

$V_1 = 1m^3$

$V_2 = 2v_1 = 2m^3$

Required:

work done= W =?

$p_1 = \propto V_1 = 5 \times 10^{+5}\frac{N}{m^5} \times 1m^3$

$P_1 = 5\times 10^5 Pa$

$P_2 = \propto V_2 = 5\times 10^5 \times 2m^3$

$P_2 = 10\times 10^5 Pa$

$p_{av} = \frac{P_1+P_2}{2} = (\frac{5+10}{2}) \times 10^5 Pa$

$p_{av} = 7.5 \times 10^5$ pa

$W = P_{av} \times \Delta V$

$W = Pav\ (V_2 - V_1)$

$W = 7.5 \times 10^5 Pa\ (2-1)\ m^3$

$W = 7.5 \times 10^5 J = 0.75 \times 10^6 J$

∴ **W =0.75 MJ**

∴ ***Answer A***

13. A ball is thrown down ward (not dropped) from the top of a tower. After being released , its down ward:
A .velocity will be uniform
B .acceleration will be equal to g.
C .acceleration will be greater than g.
D .acceleration will be smaller than g.

Solution: When a body thrown down ward its velocity increases and its acceleration remains constant and equal to g.

∴ ***Answer B***

14. A block Slides on a rough 45^0 incline. The coefficient of friction between block and incline is μ_K .What is the ratio of acceleration when the block accelerates down the incline to the acceleration when the block is projected up the incline?
A . $\frac{\mu k-1}{\mu k+1}$
B . $\frac{\frac{\mu_k}{\sqrt{2}}-1}{\mu k+1}$
C. $\frac{1-\mu K}{1+\mu K}$
D. $\frac{1+\mu k}{1-\mu k}$

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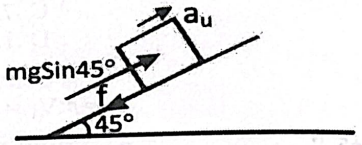
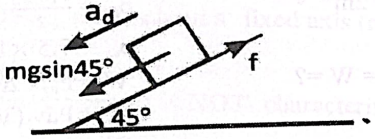
Given: $\theta = 45°$ ***Required:*** $\frac{a_d}{a_u} = ?$

a_d = acceleration of the block down the plane
a_u = acceleration of the block up the inclined plane.

Solution:



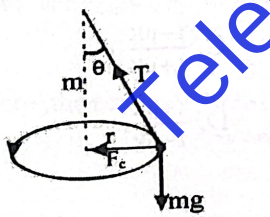
$$ma_u = mg\sin 45^0 + \mu_k mg\cos 45^0$$
$$a_u = 0.707g\,(1+\mu_k)$$



$$ma_d = \mu_k mg\cos 45^0 - mg\sin 45^0$$
$$a_d = 0.707g(\mu_k - 1)$$
$$\frac{a_d}{a_u} = \frac{0.707\,g(\mu_k - 1)}{0.707g(1+\mu_k)}$$
$$\frac{a_d}{a_u} = \frac{\mu_k - 1}{\mu_k + 1}$$

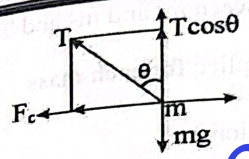
∴ Answer A

15. When a small mass m is suspended from one end of a vertical string and then whirled in a horizontal circle at a constant speed V,
A .the string stays vertical
B .the string becomes inclined to the vertical.
C .there is no force on m except its weight.
D .the angle of inclination to the vertical of the string does not depend on V.



Solution:

$$mg = T\cos\theta = W$$
$$\frac{mV^2}{r} = T\sin\theta = F_c$$
$$V = \sqrt{rg\tan\theta}$$
$$\sin\theta = \frac{mV^2}{rT}$$



Increasing the speed and increase the Tension as well as the angle of inclination.

∴ Answer B

16. Suppose $\vec{A} = 3\hat{i} - 2\hat{j}$, $\vec{B} = -\hat{i} - 4\hat{j}$ and $\vec{C}$ are three vectors in xy – plane with the property that $\vec{A} + \vec{B} - \vec{C} = 0$ .Which of the following is a unit vector along $\vec{C}$?

A .$2\hat{i} - 6\hat{j}$
B .$-2\hat{i} + 6\hat{j}$
C.$-\sqrt{\frac{1}{10}}\,\hat{i} + \frac{6}{\sqrt{10}}\hat{j}$
D. $\sqrt{\frac{1}{10}}\,\hat{i} - \frac{3}{\sqrt{10}}\hat{j}$

Given :

$\vec{A} = 3\hat{i} - 2\hat{j}$

$\vec{B} = -\hat{i} - 4\hat{j}$

And $\vec{C}$ all in xy – plane

$\vec{A} + \vec{B} - \vec{C} = 0 \Rightarrow \vec{C} = \vec{A} + \vec{B}$

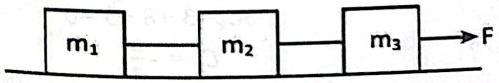
Required : $\overrightarrow{U_C} = ?$

Solution:

$$\overrightarrow{U_C} = \frac{\vec{C}}{|\vec{C}|}$$
$$\vec{C} = \vec{A} + \vec{B}$$
$$\vec{C} = (3-1)\hat{i} + (-2-4)\hat{j}$$
$$\vec{C} = 2\hat{i} - 6\hat{j}$$
$$|\vec{C}| = \sqrt{2^2 + (-6)^2}$$
$$|\vec{C}| = 2\sqrt{10}$$
$$\overrightarrow{U_C} = \frac{\vec{C}}{|\vec{C}|} = \frac{2\hat{i}-6\hat{j}}{2\sqrt{10}}$$
$$\therefore \overrightarrow{U_C} = \sqrt{\frac{1}{10}}\,\hat{i} - \frac{3}{\sqrt{10}}\hat{j}$$

∴ Answer D

17.Three blocks of mass m_1,m_2, and m_3 are connected by thread and placed on a smooth horizontal surface as shown in the figure below. A force F is applied on the block of mass m_3 in forward direction as shown .Neglecting friction, find the tension in the thread connecting m_1 and m_2.



A. $\frac{m_1 F}{m_1+m_2+m_3}$
B. $\frac{m_2 F}{m_1+m_2+m_3}$
C. $F - \frac{m_3 F}{m_1+m_2+m_3}$
D. $\frac{(m_2+m_3)F}{m_1+m_2+m_3}$

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Solution: Let T_1 is the tension between m_1 and m_2 and T_2 is the tension between m_2 and m_3.

By Newton's second law can be applied for each mass

For mass one (m_1):

$T_1 = m_1a$ .....Equation 1

For mass two (m_2):

$T_2 - T_1 = m_2a$

$T_2 = m_2a + T_1$ Equation2

For mass three (m_3):

$F - T_2 = m_3a$

$F = m_3a + T_2$ .....Equation 3

Now $T_2 = m_2a + T_1$

$T_2 = m_2a + m_1a$

$F = m_2a + m_2a + m_1a$

$F = (m_3 + m_2 + m_1)\,a$

$\therefore a = \frac{F}{m_1+m_2+m_3}$

Now $T_1 = m_1a = m_1\left(\frac{F}{m_1+m_2+m_3}\right)$

$\therefore T_1 = \frac{m_1F}{m_1+m_2+m_3}$

∴ Answer A

18. Suppose $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{B} = -\hat{i} - 4\hat{j} + 3\hat{k}$ and $\vec{C}$ is a vector lying along x –axis with the property that $\vec{A}\cdot(\vec{B} + \vec{C}) = 0$. Then vector $\vec{C}$ is given by :

A. $\frac{3}{\sqrt{10}}\hat{j}$ B. $-\frac{11}{3}\hat{i}$ C. $\frac{6}{\sqrt{10}}\hat{j}$ D. $-\frac{8}{3}\hat{i}$

Given:

$\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$

$\vec{B} = -\hat{i} - 4\hat{j} + 3\hat{k}$

$\vec{C} = C_x\hat{i}$

$\vec{A}\cdot(\vec{B} + \vec{C}) = 0$

Required:

$\vec{C} = C_x\hat{i}$ = ?

Solution:

$\vec{B} + \vec{C} = -\hat{i} - 4\hat{j} + 3\hat{k} + C_x\hat{i}$

$\vec{B} + \vec{C} = (C_x - 1)\hat{i} - 4\hat{j} + 3\hat{k}$

$\vec{A}.(\vec{B} + \vec{C})$

$= (3\hat{i} - 2\hat{j} + k).(C_x - 1)\hat{i} - 4\hat{j} + 3\hat{k})$

$3(C_x - 1) + -2 \times -4 + 1 \times 3 = 0$

$3C_x - 3 + 8 + 3 = 0$

$C_x = -\frac{8}{3}$

$\vec{C} = C_x\hat{i}$

$\therefore \vec{C} = -\frac{8}{3}\hat{i}$

∴ Answer D

19. A cart moving at a speed V collides with an identical stationary cart on an air track and the two stick together after collision. What is their velocity after collision?

A .-0.5V B .Zero C. 0.5 V D .V

Given : $m_1 = m_2 = m$

Required: Common velocity after collision ($V_1 = V_2 = V'$) ?

$u_1 = v$

$u_2 = 0$

Solution: Using conservation of linear momentum

$\overrightarrow{P_{final}} = \overrightarrow{P_{initial}}$ or $\overrightarrow{P_{before}} = \overrightarrow{P_{after}}$

$m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$

$m_1v' + m_2v' = mv$

$v'(m_1 + m_2) = m_1v$ .....$u_1 = v$

$v' = \frac{mv}{m+m}$ .....$m_1 = m_2 = m$

$v' = \frac{mv}{2m}$

$\therefore v' = \frac{1}{2}V$

∴ Answer C

20. A 4kg block is given an initial Speed of 8m/s at the bottom of 30^0 inclined plane. The frictional force that retards its motion is 15N. If the blo directed up the inclined plane, how far will it move before it stops?

A .3.35m B .3.66m C .4.40m D .4.67m

Given: m = 4kg

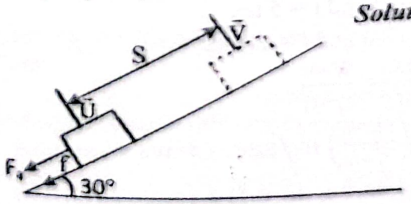
u =8m/s

$\theta = 30^0$

Frictional force =f =15N

V = 0

Required: S =?



Solution: $F_a = mg\sin 30^0 = 20N$

$-f.s - SF_a = \Delta KE$

$S = \frac{\Delta KE}{-(f+fa)}$

$S = \frac{\frac{1}{2}mv^2 - \frac{1}{2}mu^2}{-(f+f_a)}$

$S = \frac{mu^2}{2(f+f_a)}$

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$$S = \frac{4kg\,(8\ m/s)^2}{2(15+20)N}$$

$\therefore S = 3.657N$

∴ Answer B

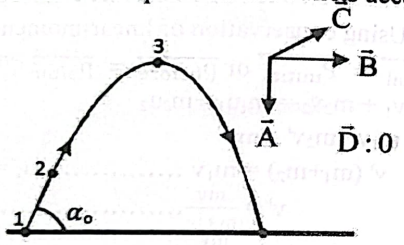
21. A bullet is fired from cannon at point 1 and follows the trajectory shown in the figure below. Air resistance may be neglected. Four possible vectors are also shown in the figure; the letter $\vec{D}$ represents the zero vector. Which vector best represents the bullets acceleration at point 2?

A. $\vec{A}$

B. $\vec{B}$

C. $\vec{C}$

D. $\vec{D}$



Solution: A projectile is a free fall acted by gravity .Its acceleration $\vec{a} = \vec{g}$ that is directed to the center of the earth.

∴ Answer A

22. A car initially at rest and at the origin has moved a distance of 5,10 ,15 ,20 ,25,30,40, 50,60,70 meters during the 1,2,3,...10 seconds periods respectively. What is the average speed during the first 5 seconds and the second half period from t = 6 seconds to t =10 seconds in m/s respectively ?

A.4,10 B. 4,9 C.5,9 D 5,10

Given: Displacement of a car in time.

t(s)	0	1	2	3	4	5	6	7	8	9	10
S(m)	0	5	10	15	20	25	30	40	50	60	70

Required: V_{av} = ? Between t = 0 and t = 5 sec

V_{val} = ? Between t =6 and t = 10 sec

Solution: $V_{av} = \frac{\Delta S}{\Delta t} = \frac{V-u}{t_f - t_i}$

$\therefore V_{av} = \left(\frac{25-0}{5-0}\right) m/sec = 5\ m/sec$

Therefore, the average speed during the first 5 seconds is 5 m/s.

$$V_{val} = \frac{\Delta S}{\Delta t} = \frac{V-u}{t_f - t_i}$$

$\therefore V_{val} = \left(\frac{70-30}{10-6}\right) m/sec$

$\therefore V_{val} = 10\ m/sec$

Therefore, the average speed during the second half period from t = 6second to t =10 second is 10 m/sec.

∴ Answer D

23. A disk rolls without slipping along a horizontal surface. The center of the disk has a translational speed V. The uppermost point on the disk has a translational speed

A. Zero B. V C.2V D.2π

Solution: A disk rolls without slipping the tangential speed ($\vec{V_T}$) is equal to the center of the disk a translational speed(V). or $\Rightarrow \vec{V_T} = V$.

∴ Answer B

24. Light with frequency of 6×10^5 GHz passes through two narrow slits of width 0.2mm apart and forms an interference pattern on a screen 2.0 m away from the slits. What is the distance between the fringes in the interference pattern ?(The speed of light $C = 3\times10^8$ m/s).

A.3mm B.5 mm C .4 mm D3.5 mm

Given: $f = 6\times10^5$ GHz $= 6\times10^{14}$ Hz ***Required:*** $\Delta y = ?$

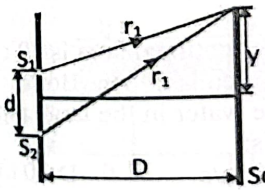
$d = 2\times10^{-4}$ m

$D = 2m$

$C = 3\times10^8$ m/s

Solution: From Young's double slit Experiment, we have :

$$y_m = \frac{mD\lambda}{d}$$



$$\Delta y = y_{m+1} - y_m = \frac{D\lambda}{d}$$

But $\lambda = \frac{C}{f} = \frac{3\times10^8}{6\times10^{14}} = 5\times10^{-7}m$

$$\therefore \Delta y = \frac{2\times5\times10^{-7}}{2\times10^{-4}}$$

$\Delta y = 5\times10^{-3}m = 5mm$

Therefore, the distance between the fringes in the interference pattern is 5mm.

∴ Answer B

25. The angular momentum of a particle is :

A. independent of the specific origin of coordinates.

B. zero when its position and momentum vectors are parallel.

C. zero when its position and momentum vectors are perpendicular.

D. not zero for non zero position and momentum vectors.

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Solution: The magnitude of angular momentum of a body about a given axis is equal to the product of moment of inertia (I) of the body and its angular speed (ω) about that axis.

∴ Angular momentum = moment of inertia × angular speed or

$$L = I\omega = (mr^2) \times \left(\frac{V}{r}\right) = mVr = pr.$$

Therefore ,p× r is zero when $\vec{r} \parallel \vec{p}$.

∴ Answer B

26 .A small car and a bus move with velocities given by $\vec{V}_c = (3\hat{\imath} - 2\hat{\jmath})$m/s and $\vec{V}_b = (-5\hat{\imath} - 4\hat{\jmath})$m/s respectively according to an observer on the ground . What is the velocity of the bus according to a car driver ?

A. $(-2\hat{\imath} + 6\hat{\jmath})$ m/s
B. $(-2\hat{\imath} - 6\hat{\jmath})$ m/s
C. $(-8\hat{\imath} + 2\hat{\jmath})$m/s
D. $(-8\hat{\imath} - 2\hat{\jmath})$ m/s

Given: $\vec{V}_b = (-5\hat{\imath} - 4\hat{\jmath})$m/s

$\vec{V}_c = (3\hat{\imath} - 2\hat{\jmath})$m/s

Required: The velocity of the bus according to a car driver = $\overrightarrow{V_{b.c}}$ = ?

Solution: $\overrightarrow{V_{b.c}} = \vec{V}_b - \vec{V}_c$,

$\overrightarrow{V_{b.c}} = (-5\hat{\imath} - 4\hat{\jmath})$m/s $-$ $(3\hat{\imath} - 2\hat{\jmath})$m/s

$\therefore \overrightarrow{V_{b.c}} = (-8\hat{\imath} - 2\hat{\jmath})$ **m/s**

Therefore, the velocity of the bus according to a car driver is $(-8\hat{\imath} - 2\hat{\jmath})$ **m/s**

∴ Answer D

27.Water flows through a pipe whose cross- sectional area is 50 cm^2 into another pipe whose cross- sectional area is 10 cm^2.Both pipes are horizontal. If the velocity of the water in the large pipe is 12 m/s, the velocity in the small pipe is:

A.0.24 m/s B.1.2 m/s C. 60 m/s D.30 m/s

Given: $A_1 = 50\ cm^2 = 5\ A_2$

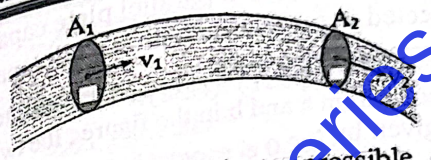
$V_1 = 12$ m/s

$A_2 = 10cm^2$

Required: Velocity in Small pipe = V_2 = ?

Solution: Consider the steady flow of a liquid through a pipe of varying cross – sectional area as shown in the figure below.

- Let A_1 and A_2 be cross – sectional areas of the pipe at A and B respectively.
- Let $V_1$and V_2 be velocities of the liquid at A and B respectively.
- ρ_1 and ρ_2 be densities of the liquid at A and B respectively.



Since liquids are practically incompressible ,$\rho_1 = \rho_2$.

Therefore ,$A_1V_1 = A_2V_2$ = constant.

$$V_2 = \frac{A_1V_1}{A_2} = \frac{5\,A_2V_1}{A_2} = 5V_1 = 5\times 12\ m/s$$

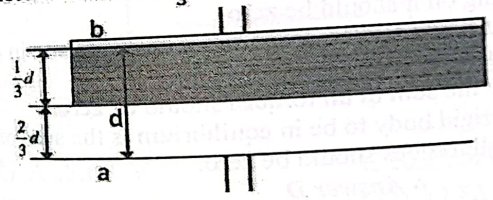
$\therefore V_2 = 60$ **m/s**

∴ Answer C

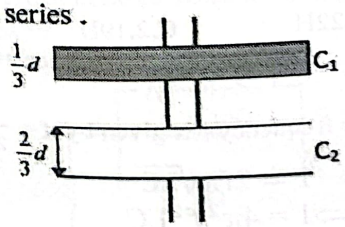
28. A parallel –plate capacitor with a plate separation d has a capacitance C in the absence of a dielectric. What is the capacitance when a slab of dielectric material of dielectric constant K and thickness $\frac{1}{3}$ d is inserted between the plates ?

A. $\frac{3K}{2K+1}$ C
B. $\frac{2K+1}{3K}$ C
C. $\frac{3K}{2K-1}$ C
D. $\frac{2K-1}{3K}$ C

Solution: A parallel –plate capacitor of plate separation d has partially filled dielectric thickness $\frac{1}{3}$ d .



The equivalent circuit of the capacitor consists of two capacitor connected in series .



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This capacitor is equivalent to two parallel plate capacitors of the same area A connected in series one with a plate separation $\frac{1}{3}$d and the other plate separation $\frac{2}{3}$d. The upper plate capacitor C_1 and the lower plate capacitor C_2 from a and b in the figure , the two capacitances are given by :

$C_1 = \frac{K\epsilon_o A}{\frac{d}{3}}$ and $C_2 = \frac{\epsilon_o A}{\frac{2d}{3}}$

For two capacitor combined in series we get : $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$

$$\frac{1}{C_o} = \frac{1}{\frac{K\epsilon_o A}{\frac{d}{3}}} + \frac{1}{\frac{\epsilon_o A}{\frac{2d}{3}}}$$

$$\frac{1}{C_o} = \frac{d}{3\epsilon_o A}\left(\frac{1}{K} + 2\right)$$

$$\therefore C_o = \left(\frac{3K}{2K+1}\right)\frac{\epsilon_o A}{d}$$

Since the capacitance without the dielectric material is given by $c = \frac{\epsilon_o A}{d}$. $\Rightarrow C_o = \frac{3K}{2K+1}C$

∴ Answer A

29.The necessary and sufficient condition for a rigid body to be in equilibrium is the sum of :

A. all torques acting on it is zero.
B. all forces acting on it should be zero.
C all forces should be equal in magnitude , but opposite in direction to the sum of all torques acting on it.
D. all forces and the sum of all torques should be zero.

Solution: For a rigid body to be in equilibrium is the sum of all forces and the sum of all torques should be zero.

∴ Answer D

30.What is the inductance of an LC circuit that oscillates at 120Hz when the capacitance is 8.0 μF ?

A.0.22 μH B.0.22H C.2.19H D.2.19 μH

Given: C =8. 0 μF f =120Hz ***Required:*** Inductance = L = ?

Solution: The resonance frequency f is given by : $f = \frac{1}{2\pi\sqrt{LC}}$

$$\Rightarrow 1 = 2f\pi\sqrt{LC}$$
$$\Rightarrow 1 = 4\pi^2 f^2 LC$$

$$\Rightarrow L = \frac{1}{4\pi^2 f^2 C}$$

$$\Rightarrow L = \frac{1}{4(3.14)^2(120)^2 \times 8\times10^{-6}}$$

$$\Rightarrow L = 0.22H$$

Therefore, the Inductance is 0.22H.

∴ Answer B

31.The resistance of an Ohmic conductor is independent of:

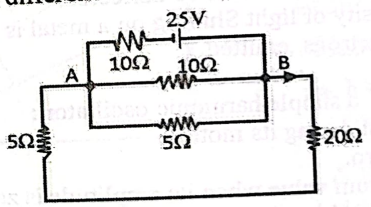
A. the size of the material. C .the temperature.
B. the nature of the material. D. the voltage applied

Solution: The resistance R of a wire of length L and cross – sectional area A is: $R = \frac{\rho L}{A}$ where ρ = Resistivity.

If a wire has a resistance R_o at a temperature T_o , then its resistance R at a temperature T is : $R = R_o + \alpha R_o (T - T_o) = R_o(1 + \alpha \Delta T)$ where α is the temperature coefficient of resistance of the material of the wire. Therefore , the resistance of an ohmic conductor is independent of the voltage applied.

∴ Answer D

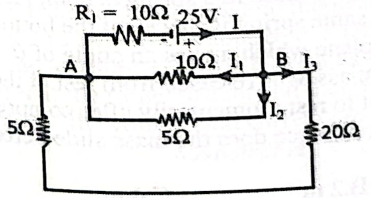
32.Consider the circuit shown below .What are the current in 20 Ω resistor and the potential difference between points A and B ?



A.0.227 A ,5.68 V C .0.215A ,4.36V
B.0.251A , 4.36V D.0.227A, 4.36V

Required: Current in 20 Ω resistor = $I_{20\Omega} = I_3 = ?$
$V_{BA} = ?$

Solution:



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$$\frac{1}{R_{T1}} = \frac{1}{25\Omega} + \frac{1}{5\Omega} + \frac{1}{10\Omega} \Rightarrow R_{T1} = \frac{50}{17}\Omega$$

$$R_T = R_1 + R_{T1}$$

$$R_T = 10\,\Omega + \frac{50}{17}\Omega = \frac{220}{17}\Omega$$

$$I = \frac{V}{R_T} = \frac{25\,V}{\frac{220}{17}\Omega} = 1.93\ A$$

$$-10I_1 - 10I + 25 = 0$$

$$-10I_1 - 10 \times 1.93 = 0.57\ A$$

$$-5I_2 + 10I_1 = 0$$

$$I_2 = 2I_1 = 2 \times 0.57\ A = 1.14\ A$$

$$-25I_3 + 5I_2 = 0$$

$$I_3 = 0.228A$$

$$\therefore V_{BA} = -10I + 25 = 10I_1 = 5I_2 = 25I_3 = 5.7\ V$$

∴ Answer A

33. When the intensity of light Shining on a metal increases ,
   A. the energy of photoelectrons , emitted , will increase.
   B. the number of photoelectrons ,emitted , will increase.
   C. the threshold frequency for photoelectron emission will increase.
   D. the photoelectrons emitted will be deflected.

Solution: The intensity of light Shining on a metal is proportional to the number of photoelectrons ,emitted.

∴ Answer B

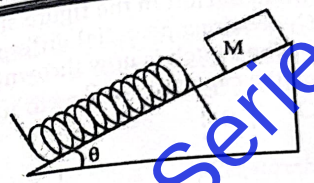
34.The acceleration of a simple harmonic oscillator :
   A. remains constant during its motion.
   B. remains non- zero.
   C. attains its maximum value when its amplitude is zero.
   D. attains minimum value when its velocity is maximum.

Solution: A simple harmonic motion whose acceleration directly proportional to the displacement(restoring force) ,opposite to the displacement and takes place in a straight line. When the velocity is maximum ,the displacement ,hence the acceleration is maximum.

∴ Answer D

35.As shown below ,an ideal mass less spring is compressed 1 m by a force of 100 N .This same spring is placed at the bottom of a frictionless inclined plane which makes an angle of $\theta = 30°$ with the horizontal . A 10 kg mass M is released from rest at the top of the incline and is brought to rest momentarily after compressing the spring 2.0 m. Through what distance does the mass slide before coming to rest ?

A.1 m B.2 m C.4 m D.6m



Required: $S = S_1 + X = ?$

Given: $F = 100N$

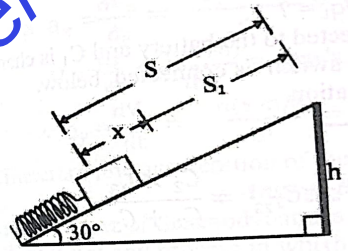
$x = 1\ m$

$K = \frac{F}{x} = \frac{100\ N}{1\ m} = 100N/m$

$m = 10\ kg$

$\theta = 30°$

$x = 2\ m$



Solution:

Since the plane is frictionless ,

$$mgh = \frac{1}{2}Kx^2$$

$$\Rightarrow h = \frac{Kx^2}{2mg}$$

$$\Rightarrow h = \frac{100 \times (2\ m)^2}{2 \times 10 \times 10} = 2m$$

From the figure:

$$\sin 30° = \frac{h}{s}$$

$$\Rightarrow h = S \sin 30°$$

$$\Rightarrow 2 = S \times \frac{1}{2}$$

$$\therefore S = 4m$$

∴ Answer D

36.The phase angle in an alternating current circuit is zero degree when the circuit contains :
   A. resistance only .
   B. capacitance only .
   C . inductance only.
   D. series combination of resistance and inductance.

Solution: When an ac voltage is applied to a pure **resistance** , the voltage across the resistance and the current through it attains their maximum values at the same instant and their phase angle zero values at the same instant , the voltage and current are said to be **in phase.**

∴ Answer A

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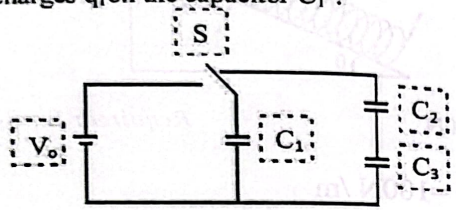
37. When switch S is thrown to the left in the figure shown below, the plates of the capacitor C_1 acquire a potential difference V_o, C_2 and C_3 are initially uncharged. The switch is now thrown to the right. What are the final charges q_1 on the capacitor C_1 ?



A. $\left(\frac{C_1C_2+C_2C_3}{C_1C_2+C_1C_3+C_2C_3}\right)C_1V_0$

C. $\left(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3}\right)C_2V_0$

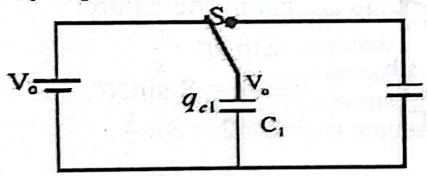
B. $\left(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3}\right)C_3V_0$

D. $\left(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3}\right)C_1V_0$

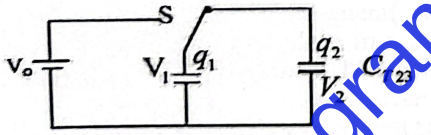
Required: Final charge on $C_1 = q_1 = ?$

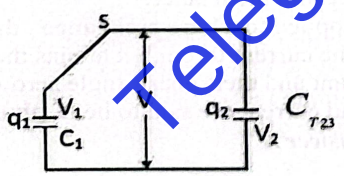
Solution: First switch S is connected to the battery and C_1 is charged to potential of V_o. $q_{C1} = C_1V_o$, the switch is connected below.

$V_1 = V_2 = V$ ..... Parallel combination



$$C_{T_{23}} = \frac{C_2 \times C_3}{C_2 + C_3}$$





$$\frac{q_1}{C_1} = \frac{q_2}{C_{T_{23}}}$$

$$q_2 = \frac{C_{T_{23}} \times q_1}{C_1}$$

$$q_1 + q_2 = q_{C1} = C_1V_o$$

$$q_1 + \frac{C_{T_{23}} \times q_1}{C_1} = C_1V_0$$

$$q_1\left(1 + \frac{C_2C_3}{\frac{C_2+C_3}{C_1}}\right) = C_1V_o$$

$$q_1\left(\frac{C_1C_2+C_1C_3+C_2C_3}{C_1C_2+C_1C_3}\right) = C_1V_o$$

$$q_1 = \left(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3}\right)C_1V_o$$

∴ ***Answer D***

38. The x and y coordinates of a particle at any time t is given by $x = 9t + 5t^2$ and $y = 5t + 8$ where x and y are in meters and t in seconds. The acceleration of the particle at t =5 seconds is :

A. zero B. $10m/s^2$ C. $14m/s^2$ D. $20m/s^2$

Required: $a = ?$

Given: $x = 9t + 5t^2$

$y = 5t + 8$

$t = 5$ seconds

Solution: $a = \frac{dv}{dt}$

$v = \frac{ds}{dt}$

$v_x = \frac{dx}{dt} = \frac{d(9t+5t^2)}{dt}$

$v_x = 9 + 10t$

$\therefore a_x = \frac{dv_x}{dt} = \frac{d(9+10\,t)}{dt} = 10m/s^2$

$v_y = \frac{dy}{dt} = \frac{d(5t+8)}{dt} = 5m/s$

$\therefore a_y = \frac{dv_y}{dt} = \frac{d(5\,m/s)}{dt} = 0\ m/s^2$

Therefore ;the acceleration of the particle at t =5 seconds is 10 m/s^2.

∴ ***Answer B***

39. The second law of thermodynamics does NOT imply that :

A. there is not any process in which there is an overall decrease in the entropy of the universe.

B. the complete conversion of energy from a hot source into work is not possible.

C. heat flows from colder body to hotter body as in the refrigerator without work done on the system.

D. perfectly reversible processes are impossible.

Solution: The second law of thermodynamics has different version:

a. heat flows spontaneously from a body of higher temperature to a body of lower temperature but can be forced to flow in the reverse direction by doing work.

b. heat cannot be converted directly to work.

c. there is no perfect engine.

d. there is no perfect refrigerator.

e. there is no any process in which there is an overall decrease in entropy of the universe.

∴ ***Answer C***

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40.A key phenomenon that demonstrates the particle nature of light is:
A. photoelectric effect. C . dispersion of light
B. Doppler effect D. interference
Solution: **Photoelectric effect** shows that the particle nature of light.
∴ ***Answer A***

41.Two discs of moments of inertia I_1 and I_2 about their respective axes (which are normal to the disc and passing through the centre), and rotating with angular speeds ω_1 and ω_2 are brought into contact face-to –face with their axes of rotation coincident. Take $\omega_1 \neq \omega_2$. What is the angular speed of the two –disc system after contact ?

A. $\frac{I_1\omega_1+2I_2\omega_2}{I_1+2I_2}$ C. $\frac{I_1\omega_1+2I_2\omega_2}{I_1-2I_2}$

B. $\frac{I_1\omega_1+I_2\omega_2}{I_1+I_2}$ D. $\frac{I_1\omega_1+I_2\omega_2}{I_1-I_2}$

Required: Angular speed of the two disc system after contact $\omega_{f1} = \omega_{f2} = \omega_f = ?$

Solution: Using conservation of Angular momentum: $\vec{L_f} = \vec{L_i}$

$$I_1\omega_{f1} + I_2\omega_{f2} = I_1\omega_1 + I_2\omega_2$$
$$I_1\omega_f + I_2\omega_f = I_1\omega_1 + I_2\omega_2$$
$$\omega_f(I_1 + I_2) = I_1\omega_1 + I_2\omega_2$$
$$\therefore \omega_f = \frac{I_1\omega_1+I_2\omega_2}{I_1+I_2}$$

∴ ***Answer B***

42.A particle that undergoes simple harmonic motion has a period of 0.4 S and an amplitude of 12mm .The maximum velocity of the particle is :
A.3 cm/s B.9.42 cm/s C.12 cm /s D .18.84 cm/ s

Given: T = 0.4 S ***Required:*** Maximum velocity $=V_{max} = V_o = ?$

A = 12mm $=12 \times 10^{-3}$ m

Solution: The maximum velocity is given by : $V_o = A\sqrt{\frac{K}{m}}$.....Equation1

The period (T) is given by : $T = 2\pi\sqrt{\frac{m}{K}}$..... Equation2

$$\Rightarrow \sqrt{\frac{K}{m}} = \frac{2\pi}{T} \quad \text{.....From equation 2}$$
$$\Rightarrow V_o = A\sqrt{\frac{K}{m}} = \frac{2\pi}{T} \times A$$
$$\Rightarrow V_o = \frac{2\times 3.14}{0.4s} \times 12\times 10^{-3}\ m$$

$\Rightarrow V_o = 0.1884$ m/s $= 18.84$ cm /s
∴ ***Answer D***

43.When does a light bulb carry more current ?
A. Just when it is turned on during which the filament has low temperature and high resistance.
B. Long after it is turned on as the filament warms and has high resistance.
C. Just when it is turned on during which the filament has low resistance.
D. Long after it is turned on as the filament warms and has low resistance.
Solution: A light bulb carry more current the filament has low resistance.

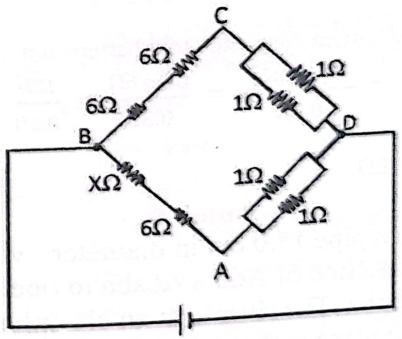
$$\Rightarrow R = \frac{V}{I}$$

A light bulb carry more current when the filament has low temperature and low resistance.

$$\Rightarrow P = I^2R = \frac{Q}{t}$$
$$\Rightarrow Q = I^2\ Rt$$

∴ ***Answer C***

44.For what value of X will the potential difference between B and D be zero in the arrangement shown below ?
A.2Ω
B. 3Ω
C. 4Ω
D. 6Ω



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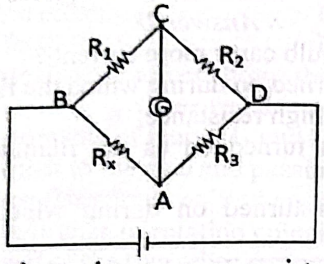
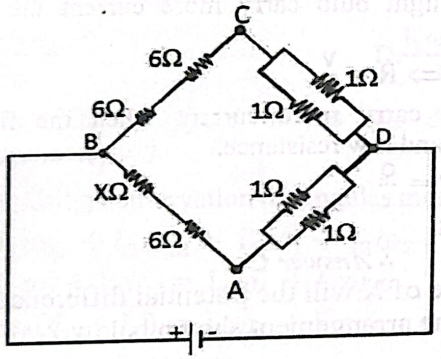
Required: $X\Omega = ?$ When $V_{DB} = ?$

Solution:



A **Wheatstone bride** is used to measure resistance X. At balanced the current through the galvanometer G is zero It implies $V_{CA} = 0$.



$V_{BD} \neq 0$,when connected to a battery but V_{CA} can be zero.

$$\frac{R_x}{R_1} = \frac{R_3}{R_2} = \frac{R_X+6\Omega}{0.5\Omega} = \frac{6\Omega+6\Omega}{0.5\Omega} = \frac{12\Omega}{0.5\Omega}$$

$$\Rightarrow R_X = 6\Omega$$

∴ Answer D

45.Through a pipe 15.0 cm in diameter , water is pumped from the river outside of Addis Ababa to one of the sub- cities of Addis Ababa .The river is at an elevation of 564 m and the sub – city is at an elevation of 2096 m. What is the minimum pressure at which the water must be pumped if it is to arrive at the sub- city ?

A.15.32 MPa
B.15.01 MPa
C. 14.99 MPa
D.16.03 MPa

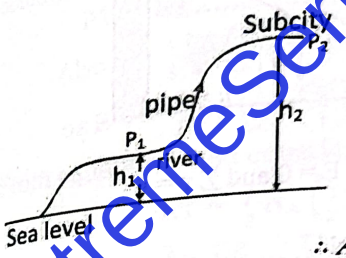
Given: $d = 15.0\times10^{-2}$ m ***Required:*** minimum pressure, $P_1 = ?$

$h_1 = 564$ m

$h_2 = h_{sub-city} = 2096$ m

Solution:

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g h_2$$



P_1 is minimum if $V_1 = V_2$ if $P_2 = 0$

$P_1 = \rho g(h_2 - h_1)$

$P_1 = 10^3\times10(2096-564)$

$P_1 = 15.32\times10^6$ Pa

$\therefore P_1 = 15.32$MPa

∴ Answer A

46.As a satellite moves in an elliptical orbit around the earth , the gravitational attraction of the earth for the satellite :

A. is greater than the satellites gravitational attraction for the earth.
B. is less than the satellites gravitational attraction for the earth.
C. Varies inversely as the square to the distance between the earth and the satellite.
D. remains constant.

Solution: **Newton's Law of universal gravitation states that** "everybody in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of distance between their centers ".

$\Rightarrow F \alpha m_1 m_2$ **Equation1**

And $F \alpha \frac{1}{r^2}$ **Equation2**

Combining equation 1 and 2 ,we have

$\Rightarrow F \alpha \frac{m_1 m_2}{r^2}$ or $F = G\frac{m_1 m_2}{r^2}$ where G is a constant of proportionality and is called **universal gravitational constant.**

∴ Answer C

47.A uniform ladder of length l and weight W = 50 N rests against a smooth , vertical wall. If the coefficient of static friction between ladder and ground is $\mu = 0.5$,find the minimum angle θ_{min} between the ladder and ground such that the ladder will not slip.

A.45° B.53° C .60° D.37°

Given: $W_1 = 50$ N

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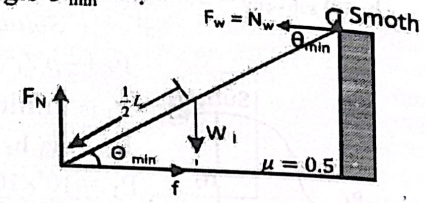
Coefficient of static friction = μ_s =0.5

Required: Minimum angle θ_{min} =?

Solution:



Since the ladder is in equilibrium :$\sum F = 0$ and $\sum \tau = 0$

Vertical force are : $F_N - W_l = 0$

$$\Rightarrow F_N = W = 50\ N$$

Horizontal forces: $F_W - f = 0$

$$F_W = f \text{ but } f = \mu F_N = 0.5 \times 50\ N = 25N$$

Using the second condition of equilibrium: $\sum \tau = 0$

$$\Sigma_o \tau = 0 \Rightarrow F_W L \sin\theta_{min} - W_l \frac{L}{2}\cos\theta_{min} = 0$$

$$\tan\theta_{min} = \frac{W_l}{2F_w} = \frac{50N}{2\times 25N} = 1$$

$$\theta_{min} = \tan^{-1}(1)$$

$$\therefore \theta = 45^0$$

∴ Answer A

48.The impedance of an 8 microfarad capacitor at 1000Hz is :

A.600Ω B.22.7Ω C.19.9Ω D.0.44Ω

Given: C= 8μ F ***Required:*** Impedance = X_C = ?

f= 1000 Hz

Solution: $X_C = \frac{1}{2\pi fc}$

$$X_C = \frac{1}{2\times 3.14\times 10^{3}\times 8\times 10^{-6}}$$

$$X_C = \frac{1}{50.2\times 10^{-3}}$$

$$\therefore X_C = 19.9\Omega$$

∴ Answer C

49.Which of the following statements follows from kinetic theory of gases ?

A. The pressure of the gas is proportional to the average of square of molecular speed of particles.

B The molecules undergo inelastic collisions with each other and with the walls of the container.

C. The forces between molecules of gas are strong at all times.

D . Newton laws of motion is a consequence of kinetic theory of gases.

Solution: Consider one mole of a gas . Let P ,V and T be the pressure ; volume and temperature of the gas respectively. Clearly the gas will contain N_A molecules. according to perfect gas equation:

$$PV = RT \ldots\ldots\ldots\ldots \textbf{Equation1}$$

$$\text{Also } P = \frac{2}{3}\frac{N_A}{V}\left(\frac{1}{2}m\overline{V^2}\right) \ldots\ldots\ldots\ldots \textbf{Equation2}$$

$$\text{or } PV = \frac{2}{3}N_A\left(\frac{1}{2}m\overline{V^2}\right)$$

From equation 1 and equation 2 , we have :

$$RT = \frac{2}{3}N_A\left(\frac{1}{2}m\overline{V^2}\right) \text{or}$$

$$\frac{1}{2}m\overline{V^2} = \frac{3}{2}\frac{RT}{N_A}$$

Both R and NA are universal constants .Therefore $\frac{R}{N_A} = K$ is a constant called **Boltzmann's constant** K =1.38 × 10^{-23} JK^{-1} and is the gas constant per molecule.

∴ Average kinetic energy of a gas molecule is $\overline{KE} = \frac{3}{2}KT$

Thus the average kinetic energy of a molecule in an ideal gas is directly proportional to the absolute temperature of the gas.

According to perfect gas equation : PV = nRT... But $\overline{KE} \sim T = \frac{1}{2}m\overline{V^2}$

Here n is the number of moles of the gas ,suppose N is total number of molecules in gas .So that N = nN_A where N_A is Avogadro's number PV = nRT

$$PV = \frac{N}{N_A}\left(\frac{2}{3}N_A\left(\frac{1}{2}m\overline{V^2}\right)\right)$$

$$PV = \frac{N}{3}m\overline{V^2}$$

$$P = \frac{N}{3}\frac{m}{V}\overline{V^2}$$

$$P = \frac{N}{3}\rho\overline{V^2} = \frac{1}{3}\rho\overline{V^2} \ldots\ldots\ldots\ldots \rho \text{ constant}$$

$$\therefore P \propto \overline{V^2}$$

∴ Answer A

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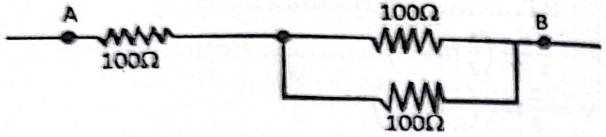
50.Three 100 Ω resistors are connected as shown in the figure below .The maximum power that can be delivered to any one resistor is 25.0 W. What is the maximum voltage that can be applied between the terminals A and B ?



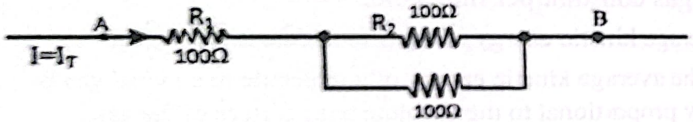
A.50V B.75V C.100V D.125V

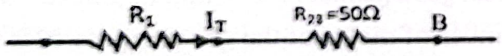
Given:$R_1 = R_2 = R_3 = 100\ \Omega$

Maximum power = $P_{max} = 25.0$ W

Required: Maximum voltage between the terminals A and B = $V_{BA} = ?$

Solution:





The current in the circuit is $P = I^2R$

$$I^2 = \frac{P}{R}$$

$$I = \sqrt{\frac{25}{100}} = 0.5A$$

The total resistance of the circuit is $R_T = R_1 + R_{23}$

$R_T = 100\Omega + 50\Omega = 150\Omega$

$V_{BA} = IR_T$ ...I $= I_T$

$V_{BA} = 0.5 \times 150\Omega$

$\therefore V_{BA} = 75V$

∴ Answer B

1.The sum and difference of two non-zero vectors $\vec{A}$ and $\vec{B}$ are equal in magnitude. What can you conclude about these two vectors?

A . $\vec{A}$ and $\vec{B}$ have the same direction.

B . $\vec{A}$ and $\vec{B}$ have opposite directions.

C $\vec{A}$ and $\vec{B}$ have the same magnitude.

D . $\vec{A}$ and $\vec{B}$ are perpendicular to each other.

Solution: $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$ or $|\vec{A}+\vec{B}|^2 = |\vec{A}-\vec{B}|^2$ ... Equation 1

We know that dot product of a vector with itself is equal to the square of the magnitude of the vector .Therefore ,expression given in equation I can be written as :

$(\vec{A}+\vec{B})\,.\,(\vec{A}+\vec{B}) = (\vec{A}-\vec{B})\,.\,(\vec{A}-\vec{B})$

$\Rightarrow A^2+B^2+2AB\cos\theta = A^2+B^2-2AB\cos\theta$

$\Rightarrow 2AB\cos\theta = -2AB\cos\theta$

$\Rightarrow 4AB\cos\theta = 0$

$\Rightarrow \theta = \cos^{-1}(0)$

$\therefore \theta = 90^0$

Therefore , the vectors $\vec{A}$ and $\vec{B}$ are perpendicular to each other.

∴ Answer D

2 .A projectile is fired at 60^0 with the horizontal. At the highest point of its path, the vertical component of its velocity is momentarily zero. What is the magnitude of its acceleration at this point?

A .20m/s^2 B.10m/s^2 C.5m/s^2 D .0m/s^2

Solution: The acceleration of a projectile motion is always constant and at any point its magnitude is equal to g =10m/s^2 .

∴ Answer B

3.The speed of a mass is increased by two fold. By what factor does the kinetic energy change?

A .2 B. 4 C .8 D .16

Given : V =2Vo

Required: By what factor does the KE change = ?

Solution: $KEo = \frac{1}{2}mv_o^2$

$KE = \frac{1}{2}mv^2$

$KE = \frac{1}{2}m(2vo)^2$

$KE = 4(\frac{1}{2}mvo^2)$

$\therefore$ **KE = 4KEo**

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Therefore , kinetic energy is increased by a factor of four.

∴ *Answer B*

4 . What is the SI unit of Electric field?

A . Am^{-1} B. Vm^{-1} C. Cm D. NC^{-1}

Solution: **The Electric field E** at a point is equal to the electric force experienced by a unit positive test charge placed at that point ,i.e force per unit charge.

- The Electric field is sometimes called **electric intensity or electric field intensity .**
- The SI unit of force is **Newtons (1N).**
- The SI unit of charge is **coulombs (1C).**
- Electric field $= \frac{\text{Force}}{\text{charge}} \Rightarrow E = \frac{F}{C}$
- Therefore , the SI unit of Electric field is **1N/C or $1NC^{-}$**
- **Electric field** equals to the potential difference between the plates divided by the distance between them.
- In SI units , where the unit of potential difference is 1V and the unit of distance is 1 m, the unit of electric field is **1 volt per meter (1 Vm^{-}).**
- In practice ,the volt per meter is most commonly used as the unit of E.
- Notice that the SI units for electric field ,V/m and N/C are identical.

∴ *Answer B*

5.Which statement is correct about a purely inductive circuit?

A .The current leads the voltage by $\pi/2$.

B .The current is in phase with the voltage.

C .The current lags behind the voltage by $\pi/2$.

D .The current lags by $\pi/2$ at low frequencies and leads by $\pi/2$ at high frequencies.

Solution :In a purely inductive circuit the voltage is leading the current by $\frac{\pi}{2}$ radians or 90^0.

- The current lags behind the voltage by $\frac{\pi}{2}$ radians or 90^0.

∴ *Answer C*

6. Suppose $\vec{A} = 3\hat{i} - 2\hat{j}$, and $\vec{B} = -\hat{i} - a\hat{j}$ are two vectors in xy- plane. What is the value of a such that $\vec{A} + \vec{B} = 2\hat{i}$?

A .3 B. 2 C .-2 D .-3

Given :

$\vec{A} = 3\hat{i} - 2\hat{j}$

$\vec{B} = -\hat{i} - a\hat{j}$

$\vec{A} + \vec{B} = 2\hat{i}$

Required: a= ?

Solution:

$\vec{A} + \vec{B} = 2\hat{i}$

$(3\hat{i} - 2\hat{j}) + (-\hat{i} - a\hat{j}) = 2\hat{i}$

$(3 - 1)\hat{i} + (-2 - a)\hat{j} = 2\hat{i}$

$\Rightarrow (-2 - a)\hat{j} = 0$

∴ a = -2

∴ *Answer C*

7 .The temperature of a gas is due to

A .the kinetic energy of the molecules.

B .the potential energy of the molecules.

C .the attractive force between the molecules.

D . the repulsive forces between the molecules.

Solution: The temperature of a gas is due to the kinetic energy of the molecules.

∴ *Answer A*

8 .If the amplitude of a sine wave is doubled , the intensity

A .remains the same.

B .increases by a factor of 2.

C .increases by a factor of 4.

D .increases by a factor of 16.

Given : $I_1 = I_o$ and $S_1 = S_o$ ***Required*** : $I_2 = ?$

$S_2 = 2S_1$

Solution: The intensity of a sinusoidal wave is given by:

$I = 2\pi^2 \times f^2 \rho u S_o^2$

where :f – is frequency

ρ - density of the medium.

u – wave speed

S_o – amplitude of oscillation of the medium as the wave propagates.

$I_1 = 2\pi^2 f^2 \rho u (S_1)^2$

$I_2 = 2\pi^2 f^2 \rho u (2S_1)^2$

$I_2 = 4(2\pi f^2 \rho u S_1^2)$

∴ $I_2 = 4I_1$

∴ *Answer C*

9 .What is the magnitude of the electric force between two protons separated by 2×10^{-15} m ? (charge of proton is 1.6×10^{-19}C).

A .67.5 N C .57.5 N

B .115.0 N D .83.4 N

Given : $q_1 = q_2 = e^+ = 1.6 \times 10^{-19}$ C

$r = 2 \times 10^{-15}$m

Required : Magnitude of the electric force = F =?

Solution: Two proton repel each other by :

$F = \frac{Kq_1 \cdot q_2}{r^2}$ .....Coulombs law

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$$F = k\frac{e^+ . e^+}{r^2}$$

$$F = 9\times 10^9 \frac{Nm^2}{c^2} \times \frac{1.6\times10^{-19}\times 1.6\times10^{-19}}{(2\times10^{-15})^2}$$

$$\therefore F = 57.6N$$

Therefore, the magnitude of the electric force between two protons are 57.6N

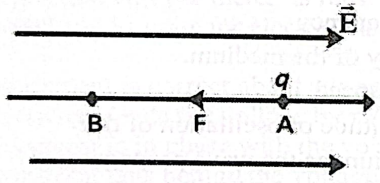
∴ Answer C

10. An ion accelerated through a potential difference of 115 V experiences an increase in kinetic energy of 7.37×10^{-17} J. What is the charge on the ion?

A .6.41×10^{-19}C
B .2.32×10^{-15}C
C .4.51×10^{-15}C
D. 2×10^{-21}C

Given : $V_{B.A} = 115V$
$\Delta KE = 7.37\times10^{-17}J$

Required : q = ?



Solution :

$$V_{B.A} = \frac{W_{A\rightarrow B}}{q}$$

$$q = \frac{W_{A\rightarrow B}}{V_{B.A}}$$

$$W_{A\rightarrow B} = \Delta KE$$

$$q = \frac{\Delta KE}{V_{B.A}}$$

$$q = \frac{7.37\times10^{-17}J}{115V}$$

$$\therefore q = 6.409\times10^{-19}C$$

∴ Answer A

11. A car accelerates steadily from 0 m/s to 30 m/s in a distance d and a time t .Another car takes a time 3t to accelerate steadily from stationary to the same final velocity. What distance does the second car cover during the new acceleration?

A .3d
B .d
C. d/3
D. d/9

Given :
$u_A = 0$
$v_A = 30m/s$
$S_A = d$
$t_A = t$

Solution:

$$S_A = V_{av} . t_A$$

$$S_A = \left(\frac{u_A+v_A}{2}\right)t$$

$$S_A = \frac{0+30}{2}\times t$$

$$d = 15m/s . t$$

$$S_B = V_{av}.t_B$$

$$S_B = \left(\frac{u_B+v_B}{2}\right)t$$

$$S_B = \left(\frac{0+30}{2}\right)m/s \times 3t$$

$$S_B = 3(15m/s . t)$$

$$\therefore S_B = d_2 = 3d$$

$u_B = 0$
$v_B = 30m/s$
$t_B = 3t$
Required : $S_B = d_2$?

Therefore, the distance of the second car is 3d.

∴ Answer A

12. A cart on an air track is moving at 0.5 m/s when the air is suddenly turned off. The cart comes to rest after travelling 1m.The experiment is repeated, but now the cart is moving at 1m/s when the air is turned off. How far does the cart travel before coming to rest?

A .1m
B. 2m
C .3m
D. 4m

Given for first case:
$u = 0.5m/s$
$v = 0$ m/s
$S = 1$ m

Solution for the first case:

$$S = \frac{V^2-U^2}{2a}$$

$$2as = V^2 - U^2$$

$$a = \frac{V^2-U^2}{2S}$$

$$\therefore a = \frac{-0.25}{2} m/s^2$$

Given for second case:
$u_1 = 1$ m/sec
$v_1 = 0$ m/s

Required : distance = s = ?

Solution for the second case:

Assuming acceleration is the same for both cases :

$$S = \frac{V_1^2-U_1^2}{2a}$$

$$S = \frac{(-1)^2}{2(\frac{-0.25}{2})}$$

$$\therefore S = 4m$$

∴ Answer D

13. A car (mass m_1)is driving at velocity v ,when it smashes into an unmoving car (mass m_2), locking bumpers. Both cars move together at the same velocity. The common velocity will be given by:

A. $\frac{m_1 v}{m_2}$
B. $\frac{m_2 v}{m_1}$
C. $\frac{m_1 v}{(m_1+m_2)}$
D. $\frac{(m_1+m_2)v}{m_1}$

Given : $m_1, u_1 = V$
$m_2, u_2 = 0$

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Required : $v_1 = v_2 = v = ?$

Solution: Using the conservation of linear momentum

$$P_f = P_i$$

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

$$v(m_1+m_2) = m_1 v \ldots\ldots u_2 = 0$$

$$\therefore V = \left(\frac{m_1}{m_1+m_2}\right) v$$

∴ Answer C

14. What does the value of Reynolds number signify?

A .It describes whether the flow is vertical or horizontal.
B .It describes the magnitude of the viscosity of the liquid.
C .It describes whether the flow is stream lined or turbulent.
D .It describes the magnitude of the surface tension of the liquid.

Solution : **Reynolds number**

When the velocity of a fluid flowing in a pipe exceeds a certain critical value , the nature of the flow becomes **turbulent**. Experiments indicate that a combination of four factors determines whether the flow of a fluid through a tube or pipe is laminar or turbulent .This combination is known as **Reynolds number** N_R and is given by $N_R = \frac{\rho V D}{\eta}$ where

ρ = density of the fluid.
v= Average velocity of the liquid
D = diameter of the tube or pipe.
η = coefficient of viscosity of the liquid.

Experiments show that Reynolds number is of primary importance in determining whether the flow is turbulent or laminar.
Observations show that if N_R is less than 2000 the laminar flow occurs and if N_R is greater than 3000, the turbulent flow occurs.
At values between 2000 and 3000 ,the flow is unstable and may change back and forth from one type of flow to another.

∴ Answer C

15. What does a curve drawn between two points on P- V diagram represent?

A .The state of the system.
B . A thermodynamic process.
C .Work done in a cyclic process.
D .Work done on or by the system.

Solution: A graph of pressure P against volume V shows the **thermodynamic process**

∴ Answer B

16. A 1 kg mass is attached to a spring of stiffness constant 100N/m. What is the maximum displacement of the mass from its equilibrium position if its speed is 3m/s at the equilibrium point?

A . 0.1 m B. 0.3m C.3 m D.10 m

Required : Maximum displacement =A = ?

Given :
m =1kg
K =100N/m
u_o = 3 m/s

Solution : Using conservation of energy the total energy at the equilibrium position is equal to the energy at the amplitude position.

$$PE = KE_O$$

$$\frac{1}{2}KA^2 = \frac{1}{2}mu_o^2$$

$$A = \sqrt{\frac{m}{k}}.u_o$$

$$A = \sqrt{\frac{1kg}{100N/m}} \times 3m/s$$

$$\therefore \mathbf{A = 0.3\ m = 30\ cm}$$

Therefore, the maximum displacement is 0.3m.

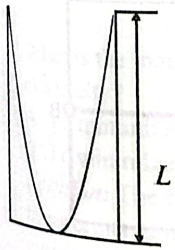
∴ Answer B

17. A pipe, 68cm long , is open at one end and closed at the other. When air is blown across the open end sound is produced at 110Hz. What is the velocity of sound along the pipe?

A .598.4 m/s B .334.6m/s C. 299.2m/s D. 149.6 m/s

Given : L = 68 cm
f=110HZ

Required: velocity = V =?



$$L = \frac{\lambda}{4}$$

Solution: The first mode of vibration occurs for n= 1 and the corresponding wavelength λ is given by: $\lambda = \frac{4L}{2n-1}$ … for n=1

$$\lambda = 4L$$

$$\lambda = 4 \times 0.68\ m$$

$$\lambda = 2.72\ m$$

$$V = \lambda . f$$

$$V = 2.72\ m \times 110\ Hz$$

$$\therefore \mathbf{V = 299.2\ m/s}$$

Therefore , the velocity of sound along the pipe is 299.2 m/sec.

∴ Answer C

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18. Which phenomenon is responsible for the colors seen on a thin oil film on the water surface?

A .Diffraction.
B .Interference
C. Super position
D . Refraction.

Solution: The colors seen on a thin oil film on the water surface due to the interference of the wave.

∴ *Answer B*

19.A cell supplies a current of 0.9A through a 2Ω resistor and a current of 0.3A through a 7Ω resistor.The internal resistance of the cell is

A .0.5Ω B . 1.0Ω C .1.2Ω D.2.0Ω

Given: $I_1 = 0.9A$, $R_1 = 2\Omega$, $I_2 = 0.3\ A$, $R_2 = 7\Omega$ ***Required:*** Internal resistance= r = ?

Solution : $\varepsilon = I_1R_1 + I_1r$ ..... Equation 1

$\varepsilon = I_2R_2 + I_2r$ ..... Equation 2

Equating Equation 1 and 2 we get

$\Rightarrow I_2R_2 + I_2r = I_1R_1 + I_1r$

$(I_2 - I_1)\, r = I_1R_1 - I_2R_2$

$$r = \frac{I_1R_1 - I_2R_2}{I_2 - I_1}$$

$$r = \frac{0.9A \times 2\Omega - 0.3\ A \times 7\Omega}{0.3\ A - 0.9\ A}$$

∴ $r = 0.5\Omega$

Therefore ,the internal resistance of the cell is 0.5Ω.

∴ *Answer A*

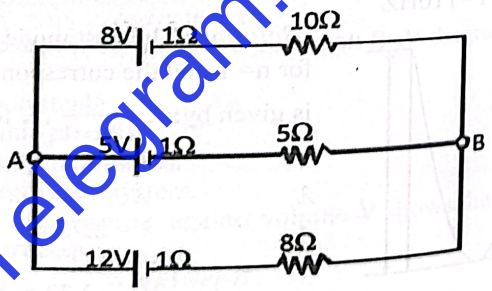
20.Determine the current in the 5Ω resistor in the figure shown below.

A .7.85 A

B .1.09 A

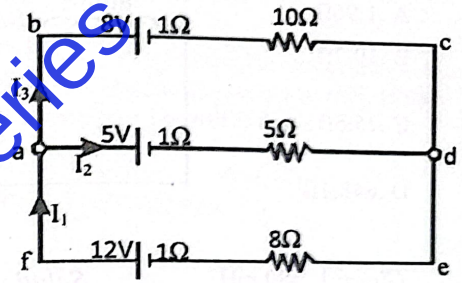
C .0.48 A

D .0.18 A



Required: Current through 5Ω resistor $I_{5\Omega} = I_2 = ?$

Solution:



At junction a : $I_1 = I_2 + I_3$

loop abcda

$-8 - 11\ I_3 + 6I_2 + 5 = 0$

$6I_2 - 11\ I_3 - 3 = 0$. Equation 1

Loop adef a

$-5 - 6\ I_2 - 9I_1 + 12 = 0$

$-6I_2 - 9\ (I_2 + I_3) + 7 = 0$

$-15I_2 - 9I_3 + 7 = 0$..... Equation 2

Solve Equation 1 and Equation 2 simultaneously we get I_2.

∴$I_2 = 0.48\ A$

Therefore,$I_2 = 0.48$ A is the current through the 5Ω resistor.

∴ *Answer C*

21.Find the magnitude of the force felt by an electron travelling at an angle of 60^0 to the earth's magnetic field at 1.4×10^7m/s.(The charge on an electron is 1.6×10^{-19}C and the magnitude of the earth's magnetic field is 5×10^{-5}T).

A .0N
B .9.7×10^{-17} N
C. 1.1×10^{-16}N
D .8.6×10^{-17}N

Given: $\theta = 60^0$, $V = 1.4\times 10^7$m/s, $\vec{B} = 5\times 10^{-5}$ T, $e = 1.6\times 10^{-19}$C

Required :$F_e = ?$

Solution :

$\vec{F} = q\,(\vec{V} \times \vec{B})$

$F_e = qVB \sin 60^0$

$F_e = 1.6\times 10^{-19} \times 1.4\times 10^7 \times 5\times 10^{-5}\,T \times 0.866$

∴ $F_e = 9.699\times 10^{-17}N \approx 9.7\times 10^{-17}N$

∴ *Answer B*

22.What is the main reason why alternating current is used to transmit electricity?

A .To minimize power gain.
B .To minimize energy loss.
C .To maximize voltage gain.
D .To maximize current gain.

Solution: The flux produced by AC changes, due to this property a transformer is used to transfer AC at a higher voltage and Lower current to minimize energy lose.

∴ *Answer B ,C*

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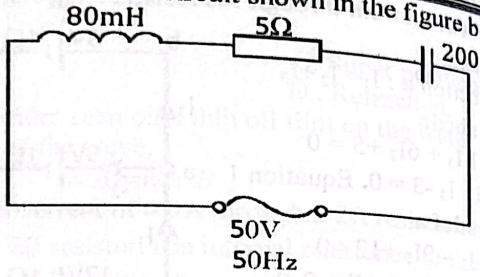
23. Find the impedance for the RLC circuit shown in the figure below.

A .1.96Ω

B .19.2Ω

C .25.5Ω

D .648.1Ω



Given: L =80 mH
R =5Ω
C =200μF
$\mathcal{E}$ = 50 V
f =50Hz

Required: Z =?

Solution:

$X_L = \omega L$

$X_L = 2\pi fL$

$X_L = 6.28 \times 5 \times 10\ Hz \times 80 \times 10^{-3} H$

$\therefore X_L = 25.1\ \Omega$

$X_C = \frac{1}{\omega c}$

$X_c = \frac{1}{2\pi fc}$

$X_c = \frac{1}{6.28 \times 50Hz \times 200 \times 10^{-6} F}$

$\therefore X_C = 15.9\ \Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$Z = \sqrt{(5\Omega)^2 + (25.1\Omega - 15.9\Omega)^2}$

$\therefore Z = 10.4\ \Omega$

∴ No Answer

24 .An object moving with uniform acceleration has a velocity of 12m/s in the positive x direction when its x coordinate is 3 cm. If its X coordinate 2s later is – 5 cm, what is its acceleration?

A .-12m/s^2 B .-13m/s^2 C.-16m/s^2 D. 12m/s^2

Given :
u = 12m/s
S_o = 3 cm
S_2 = -5 cm
t =2 s

Required: a = ?

Solution:

$\Delta S = S = S_2 - S_o$

S = -5 cm – 3 cm = -8 cm

$S = ut + \frac{1}{2}at^2$

$a = \frac{2(s - ut)}{t^2}$

$a = \frac{2(-0.08 - 12 \times 2)}{(2s)^2}$

$\therefore a = -12 m/s^2$

Therefore, the acceleration of an object is -12m/s^2.

∴ Answer A

25. A 4 kg block is given an initial speed of 8m/s at the bottom of 30^0 inclined plane. The frictional force that retards its motion is 15N. If the block is directed up the inclined plane, how far will it move before it stops.

A .3.66 m B. 4.40m C. 3.35m D. 4.67m

Given : m =4kg
u =8m/s
$\theta = 30^0$
frictional force = f =15 N

Required : distance = S= ?

Solution:

$W_g + W_f + \Delta KE = 0$ ....Conservator of energy

$(-mg \sin 30^0 - 15N)s = -\Delta KE$

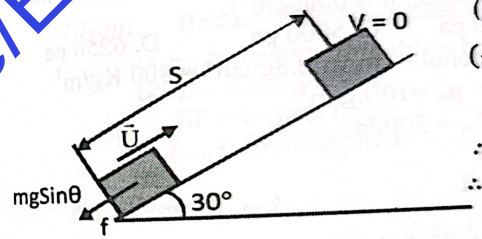
$(-4 \times 10 \times \frac{1}{2} - 15)S = -\frac{1}{2}mu^2$

$(-35)S = -\frac{1}{2}4kg\,(8\ m/s)^2$

$S = \frac{128}{35} m$

$\therefore S = 3.66\ m$

∴ Answer A



26.A wheel starts from rest and rotates with constant angular acceleration to reach an angular speed of 12 rad/s in 3 s. What is the magnitude of the angular acceleration of the wheel in rad/s^2 and the angle in radians through which it rotates in this time interval respectively?

A .4, 18 B .3, 18 C .18, 5 D .18 , 4

Given:
$\omega_o = 0$
ω =12rad/s
t =3 sec

Required:
$\propto$ = ?
θ = ?

Solution:

$\propto = \frac{\omega - \omega_o}{t}$

$\propto = \frac{(12-0)}{3S}$ rad/s

$\therefore \propto = 4\ rad/s^2$

$\theta = \omega_o t + \frac{1}{2}\propto t^2$

$\theta = 0.3\ s + \frac{1}{2}4 rad/S^2 \times (3S)^2$

$\therefore \theta = 18\ rad$

∴ Answer A

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27 .A ladder stands on the ground , leaning against a wall. It is safer climbing up the ladder if

A the ground is frictionless and the wall is rough.

B .the ground is rough and the wall is frictionless.

C .the ground is frictionless and the wall is frictionless.

D .the ladder leans at 45 degree and the ground is frictionless.

Solution: As the ladder press the wall perpendicularly the bottom of the ladder is acted to the left and accelerates unless acted by frictional forces of the ground . It is safer to climb the ladder when Both the wall and the ground are frictional .But the frictional Surface of the ground is necessary condition.

∴ Answer B

28 .The pressure at the bottom of a cylindrical tube filled with water was measured to be 5000 pa . If the water in the tube were replaced with ethyl alcohol, what would be the new pressure at the bottom of the tube? (The density of ethyl alcohol is $0.8g/cm^3$).

A .4000pa B. 4800 pa C .5000 pa D. 6250 pa

Given: density of ethyl alcohol = $\rho_A = 0.8g/cm^3 = 800\ Kg/m^3$

density of water = $\rho_w = 10^3 kg/m^3$

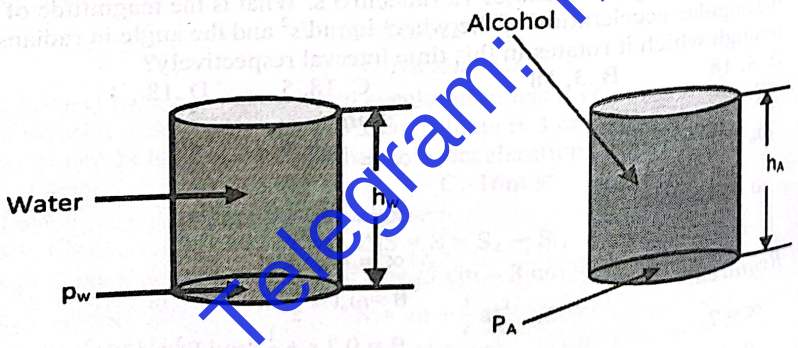
pressure of water $P_w = 5000p$

h_w = depth of water

h_A = depth of alcohol

$h_w = h_A$

Required: pressure of ethyl alcohol = P_A = ?



Solution:

$P_w = \rho_w g\, h_w$ .....Equation 1

$P_A = \rho_A .g .h_A$.....Equation 2

Dividing Equation 2by Equation 1

$$\frac{P_A}{P_w} = \frac{\rho_A g h_A}{\rho_w g h_w}$$

$$P_A = \left(\frac{\rho_A}{\rho_w}\right) \times P_w$$

$$P_A = \frac{0.8\ g/w^3}{1g/cm^3} \times P_w$$

$P_A = 0.8\ P_w$

$P_A = 0.8 \times 5000Pa$

$\therefore P_A = 4000Pa$

∴ Answer A

29.A table – tennis ball has a diameter of 3.80 cm and average density of $0.084g/cm^3$.What force is required to hold it completely submerged under water?

A .0.521 N B .0.258 N C .0.300 N D .0.510N

Given: Radius of table – tennis ball = $r_b = \frac{d}{2} = 1.9 \times 10^{-2} m$

Average density of table – tennis ball = $\rho_b = 0.084g/cm^3$

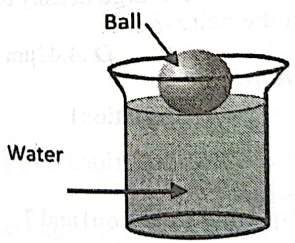
or $\rho_b = 84kg/m^3$

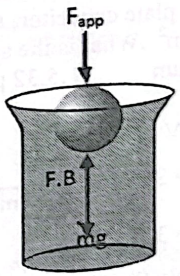
BF = boyant force

F_{app} = force applied

Required: F_{app} = ?

Solution:





$\rho_b < \rho_w \Longrightarrow$ Float in water.

$BF - F_{app} - W_b = ma$

$F_{app} = BF - W_b$

$F_{app} = W_{dw} - W_b$

$F_{app} = m_{dw} .g - m_b .g$

$F_{app} = V_{dw} .\rho_w .g - V_b .\rho_b .g$

$F_{app} = V_{bg} (\rho_w - \rho_b)$ ..... $V_{dw} = V_b$

$F_{app} = \frac{4}{3}\pi r_b^{\,3} \times g(\rho_w - \rho_b)$

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$F_{app} = \frac{4}{3} \times 3.14 \times (1.9 \times 10^{-2})^3 (1000 - 84) kg/m^3$

$\therefore F_{app} = 0.263\ N \approx 0.258\ N$

∴ Answer B

30. The total energy of a particle executing Simple harmonic motion is
A .Zero
B .directly proportional to the amplitude.
C .directly proportional to the square of the amplitude.
D .inversely proportional to the square of the amplitude.

Given: Particle of mass m under SHM .

$U_T = ME = KE + PE$

Required: Quantity on which the total energy.

Solution: Using the conservation of energy: $ME_1 = ME_2$

$KE_1 + PE_1 = KE_2 + PE_2$

$KE_1 + PE_1 = U_T$ (total energy)

$U_T = \frac{1}{2} mv^2_2 + \frac{1}{2} kx^2$..... x =A and $u_2 = 0$

$U_T = \frac{1}{2} KA^2 = K'A^2$

$\therefore U_T \sim A^2$ .....Since $\frac{1}{2} K = K'$

∴ Answer C

31. When a potential difference of 150 V is applied to the plates of a parallel- plate capacitor, the plates carry a surface charge density of 30 nc /cm^2 .What is the spacing between the plates?

A .2.24 μm B .5.32 μm C .12.24μm D .4.42μm

Given :

V =150 V

$\frac{Q}{A} = \lambda = 30 \times 10^{-9} \frac{C}{(10^{-2} m)^2}$

$\frac{Q}{A} = \lambda = 3 \times 10^{-4}\ c/m^2$

Required : Spacing between the plates = d = ?

Solution :

Q = CV.....Equation1

Q = λ A.....Equation2

Equating Equation1and 2

$CV = \lambda A$

$\frac{\varepsilon_o A}{d} V = \lambda . A$

$\Rightarrow d = \frac{\varepsilon_o V}{\lambda}$

$d = \frac{8.85 \times 10^{-12}\ F/m \times 150V}{3 \times 10^{-4} C/m^2}$

$d = 4.425 \times 10^{-6} m$

$\therefore$ **d =4.425 μm**

∴ Answer D

32. A rod is made of two materials connected to each other. Each conductor has a square cross- section of 3mm on a side. The first material has resistivity of $\rho_1 = 4 \times 10^{-3}$ Ωm and is 25 cm long while the second material has resistivity of $\rho_2 = 6 \times 10^{-3}$ Ωm and is 40 cm long. What is the resistance between the ends of the rod?

A .378Ω B .267Ω C .436Ω D .363Ω

Given : $\ell = \ell_1 + \ell_2$

$\ell = 25\ cm + 40\ cm = 65\ cm$

$a = 3mm = 3 \times 10^{-3} m$

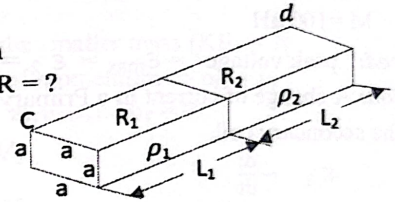
$a = a^2 = (3 \times 10^{-3} m)^2 = 9 \times 10^{-6} m^2$

$\rho_1 = 4 \times 10^{-3}$ Ω.m

$\rho_2 = 6 \times 10^{-3}$ Ωm

Required: Resistance = R = ?

Solution:



The resistance between c and d is $R = R_1 + R_2$

$R_1 = \rho_1 \frac{\ell_1}{A_1} = \frac{4 \times 10^{-3}\ \Omega m \times 0.25\ m}{9 \times 10^{-6} m^2}$

$\therefore$ **R_1 = 111.11Ω**

$R_2 = \frac{\rho_2 \ell_2}{A_2} = \frac{6 \times 10^{-3}\ \Omega m \times 0.4\ m}{9 \times 10^{-6} m^2}$

$\therefore$ **R_2 = 266.66Ω**

$R = R_1 + R_2$

$R = (111.11 + 266.66)\ \Omega$

$\therefore$ **$R = 377.77\Omega \approx 378\ \Omega$**

∴ Answer A

33. A conducting rod of length ℓ moves with a constant velocity v perpendicular to a long , straight wire carrying current I. What is the magnitude of the emf generated between the ends of the rod?

A . $\frac{\mu_o VI}{2\pi r} \ell$ B . $\frac{\mu_o VI}{2\pi \ell} r$ C. $\frac{\mu_o VI}{2\ell} r$ D. $\frac{\mu_o VI}{2r} \ell$

Solution: The magnitude of the field to the wire

$B = \frac{\mu o I}{2\pi r}$ Equation1

$\varepsilon = BV\ell$.....Equation2

Substituting Equation1 and in Equation2 we get :

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$$\varepsilon = \left(\frac{\mu o I}{2\pi r}\right) V\ell$$

$$\therefore \varepsilon = \frac{\mu o V I}{2\pi r}\ell$$

∴ *Answer A*

34. Two coils, held in fixed positions, have a mutual inductance of 100μH. What is the peak voltage in one when the sinusoidal current given by I (t) = (10.0 A) sin (1000 t) flows in the other?

A .1.0 μV B .100.0V C .10.0 V D .1.0 V

Given: I (t) = 10.0A sin (1000 t)

M =100 μH

Required: peak voltage $= \varepsilon_{max} = \varepsilon_2 = ?$

Solution: A change of current in a Primary coil (i_p) induces an emf ε_2 in the secondary coil.

$$\varepsilon_2 \sim \frac{\Delta i}{dt}$$

$$\varepsilon_2 = M\frac{di}{dt}$$

$$\varepsilon_2 = Md\,\frac{10.0A\sin 1000t}{dt}$$

$$\varepsilon_2 = M\times 10000.0A\cos 1000\,t)$$

$$\varepsilon_2 = 100\times10^{-6}\,H\times\ 10000.0A\cos 1000t$$

$$\varepsilon_2 = 1.0\ V\cos 1000t$$

$$\varepsilon_2 = \varepsilon_{max}\cos\omega t.$$

$$\therefore\ \varepsilon_{max} = 1.0\ V$$

∴ *Answer D*

35. In RLC circuit, given R= 7.6Ω , L =2.2 mH and C = 1.8 μF, what is the frequency of the damped oscillation of the circuit?

A .2.5×10^3 Hz C. 12.5×10^3Hz

B .15.72×10^3 Hz D. 35.0×10^3Hz

Given : R = 7.6Ω

L = 2.2mH = 2.2×10^{-3} H

C =1.8 μF =1.8×10^{-6} F

Required: Frequency of the damped oscillation = f = ?

Solution : $f = \frac{1}{2\pi\sqrt{LC}}$

$$f = \frac{1}{2\times3.14\sqrt{2.2\times10^{-3}\times1.8\times10^{-6}}} = 2.5\times10^3 Hz$$

∴ *Answer A*

36. A stationary mass explodes into two parts of masses 0.4 kg and 4 kg. If the larger mass has a kinetic energy of 100 J, what is the kinetic energy of the smaller mass?

A . $4\sqrt{50}$ J

B . 100J

C . 1000 J

D . The change in kinetic energy is the same for both particles.

Given : $m_1 = 0.4kg$

$m_2 = 4$ kg

Total mass $= m_T = m_1 + m_2 = 4.4$ kg

$KE_2 = 100$ J

Required: Kinetic energy of the smaller mass ($KE_1 = ?$)

Solution: Using conservation of momentum $\Rightarrow p_f = p_i$

$$m_1v_1 + m_2v_2 = m_Tu_T \ldots\ldots u_T = 0$$

$$V_1 = \frac{-m_2}{m_1}V_2$$

$$V_1 = \frac{-4kg}{0.4\ kg}V_2$$

$$\therefore\ V_1 = 10\ V_2$$

$$KE_2 = \frac{1}{2}m_2v_2^2 \ldots\ldots \text{Equation 1}$$

$$KE_1 = \frac{1}{2}m_1v_1^2 \ldots\ldots \text{Equation 2}$$

Dividing Equation 2 by Equation 1 we get:

$$\frac{m_1}{m_2} = \frac{0.4kg}{4kg}$$

$$\therefore m_1 = 0.1m_2$$

$$KE_1 = \frac{1}{2}m_1v_1^2$$

$$KE_1 = \frac{1}{2}\times 0.1\times m_2\times (10v_2)^2$$

$$KE_1 = \frac{1}{2}m_2v_2^2\,(0.1\times 10^2)$$

$$KE_1 = 10\ KE_2$$

$$KE_1 = 10\times 100J$$

$$\therefore KE_1 = 1000J$$

∴ *Answer C*

37. Which of the following objects has the largest moment of inertia assuming they all have the same mass and radius?

A .A solid disk C .A circular loop.

B . A solid sphere. D .A solid cylinder.

Solution:

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Materials	I (kg.m^2)
❖ Solid disk	$I = \frac{1}{2}MR^2$
❖ Solid sphere	$I = \frac{2}{5}MR^2$
❖ Circular loop	$I = MR^2$
❖ Solid cylinder	$I = \frac{1}{2}MR^2$

The moment of inertia of a bodies depend on mass , radius and axis of rotation. Therefore , all have the same mass and radius. Hence The circular loop with this axis of rotation has the largest moment of inertia.

∴ ***Answer C***

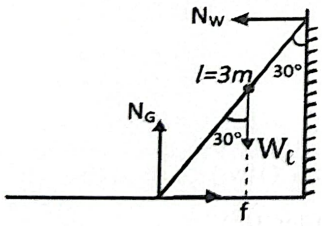
38.A ladder of length 3m and mass 20 kg, leans against a smooth , vertical wall so that the angle between the wall and the ladder is 30^0.Determine the magnitude of the normal forces at the contact point on the wall (N_w) and at the contact point on the ground (N_g)

A .$N_w = 57.7$ N ,$N_g = 200$ N
B .$N_w = 200$N , $N_g = 57.7$ N
C .$N_w = 0$N , $N_g = 200$ N
D .$N_w = 200$ N , $N_g = 0$ N

Given : $\ell = 3$m
m =20 kg
$w_\ell = mg = 200$N
$\theta = 30^0$ with the wall.

Required:
Normal force to the wall = N_w = ?
Normal forces to the ground = N_G = ?

Solution: By the first law of equilibrium:



$\sum F_i = 0 \Longrightarrow \sum F_{ix} = 0 \Longrightarrow f = N_w$
$\sum F_{iy} = 0 \Longrightarrow N_G - W_\ell = 200$ N

By the Second law of equilibrium:

$\sum \tau_O = 0$
$N_w \ell \sin 60^0 - W \frac{\ell}{2} \sin 30^0 = 0$
$N_w = \frac{W_\ell \times \ell}{\frac{4}{0.866\ell}}$
$N_w = \frac{W_\ell}{4(0.866)}$
$N_w = \frac{200N}{3.464}$
∴ $N_w = 57.7$ N

∴ ***Answer A***

39.Which statement is **NOT** among the assumptions made in the kinetic theory of gases?

A .There is no interaction between particles or between particles and the wall except during collision.
B .collisions between particles , and between particles and walls are perfectly elastic.
C .The volume occupied by the particle is negligible compared to the volume of the gas as a whole.
D .The distribution of velocities of particles is uniform.

Solution: Molecules are always in random motion. This implies that the velocity is not uniform.

∴ ***Answer D***

40.The loudness of sound changes from $L_1 = 30$ db to $L_2 = 60$ db. What is the ratio of intensities $\frac{I_2}{I_1}$ in the two cases?

A .2 B.4 C .600 D .1000

Solution:

Given:
$L_1 = \beta_1 = 30$ db
$L_2 = \beta_2 = 60$db
Required : $\frac{I_2}{I_1}$ = ?

$\beta = 10 \log_{10} \frac{I}{I_0}$
$\beta_1 = 10 \log_{10} \frac{I_1}{I_0}$
$\Longrightarrow 0.1\beta_1 = \log_{10} \frac{I_1}{I_0}$
$I_1 = 10^{0.1 \times 30} I_0$
∴ $I_1 = I_0 10^3$
$\beta_2 = 10 \log_{10} \frac{I_2}{I_0}$
$0.1\beta_2 = \log_{10} \frac{I_2}{I_0}$
$I_2 = 10^{0.1 \times 60} I_0$
∴ $I_2 = I_0 10^6$
$\frac{I_2}{I_1} = \frac{I_0 10^6}{I_0 10^3}$
$\frac{I_2}{I_1} = 1000$

∴ ***Answer D***

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41. What happens if the monochromatic light used in young's double slit experiment is replaced by white light?

A .No fringes are observed.

B .All bright fringes become white.

C .All bright fringes have colors between violet and red.

D .Only the central fringe is white, all other fringes are colored.

Solution: In Young's double slit experiment, if white light is used then observes the white central fringes surrounded by a few colored fringes on either sides

∴ ***Answer D***

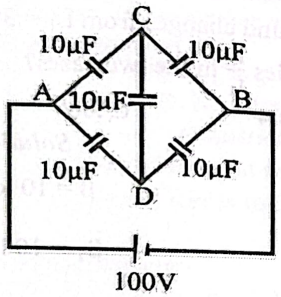
42. Five capacitors 10μF capacity each is connected to A ,D and C potential of 100 volt as shown in the next figure. What is the equivalent capacitance between A and B?

A .10μF

B .20μF

C .30μ F

D .60μF



Given: $C_1 = C_2 = C_3 = C_4 = C_5 = 10\mu F$

Solution: The bridge circuit shown in the figure at balance there is flow of charge across CD.

Case i: For ACB it is series then

$$\frac{1}{C_{eq\,1}} = \frac{1}{10\mu F} + \frac{1}{10\mu F}$$

$$\frac{1}{C_{eq\,1}} = \frac{2}{10\mu F} = \frac{1}{5\mu F}$$

$$\therefore C_{eq1} = 5\mu F$$

Case ii: For ADB also connected in series then

$$\frac{1}{C_{eq\,2}} = \frac{1}{10\mu F} + \frac{1}{10\mu F}$$

$$\frac{1}{C_{eq\,2}} = \frac{2}{10\mu F} = \frac{1}{5\mu F}$$

$$\therefore C_{eq2} = 5\mu F$$

Therefore, C_{eq1} and C_{eq2} are connected in parallel then the net capacitance between A and B is :

$\therefore C_{eq} = C_{Total} = 5\mu F + 5\mu F = 10\mu F$

∴ ***Answer A***

43. A galvanometer may be used as an ammeter by

A .Shunting the galvanometer with a high resistance.

B .Connecting a low resistance in parallel with the galvanometer.

C .Connecting a low resistance in series.

D .Connecting a high resistance in series.

Solution:

- **A galvanometer** may be used as an ammeter by connecting a low resistance (shunt) in parallel with the galvanometer.
- **A galvanometer** may be used as a voltmeter by connecting a high resistance (multiplier) in series with the galvanometer.

∴ ***Answer B***

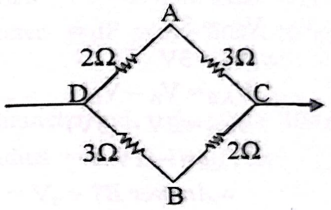
44. There is a current of 2A in the system of conductors shown in the figure below. The potential difference $V_A - V_B$ is

A .-2V

B .-1V

C .+1V

D .+2V



Given : I = 2A

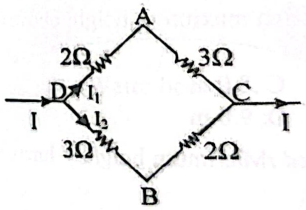
Required: $V_{A,B} = V_A - V_B$

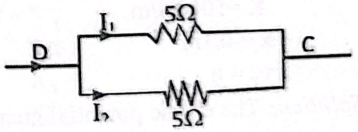
Solution:

$R_{T1} = 2\Omega + 3\Omega$

$\therefore R_{T1} = R_{T2} = 5\Omega$

$I = I_1 + I_2$





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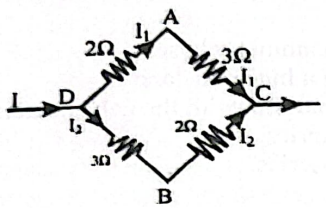


$I_1 = 5\Omega I_2$

$I_1 = I_2$

$I_1 + I_2 = I$

$2I_1 = I$

$I_1 = \frac{I}{2} = \frac{2A}{2} = 1A$

$\therefore I_1 = I_2 = 1A$

Current flows from higher potential to lower potential.

P.d between A and D is:

$V_{AD} = 1A \times 2\Omega = 2V$

$\therefore V_A = 2V$

P.d between Band D is:

$V_{BD} = 3\Omega \times 1A$

$V_{BD} = 3V$

$\therefore V_B = 3V$

$V_{AB} = V_A - V_B$

$V_{AB} = 2V - (3V)$

$\therefore V_{AB} = -1V$

∴ Answer B

45.A block of mass 0.25 kg is placed on top of a light vertical spring of force constant 5000 N/m and pushed down ward so that the spring is compressed by 0.10 m .After the block is released from rest it travels upward and then leaves the spring. To what maximum height above the point of release does it rise?

A .10.2m C .8.0 m

B. 9.2 m D. 9.0 m

Given : $m = \frac{1}{4}$ kg ***Required*** :Maximum height = hmax = ?

$K = 5000 N/m$

$x = 0.1m$

$v_y = 0$

Solution: The elastic potential energy of the system:

$E_{PE} = \frac{1}{2}Kx^2$

$E_{PE} = \frac{1}{2} \times 5000 \times (0.1)^2$

$\therefore E_{PE} = 25\ J$

The spring energy is changed in to gravitational potential energy.

$E_{PE} = mgh_{max}$

$h_{max} = \frac{EPE}{mg}$

$h_{max} = \frac{25\ J}{\frac{1}{4kg \times 10m/s^2}}$

$\therefore h_{max} = 10\ m$ or

$h_{max} = \frac{25\ J}{\frac{1}{4kg \times 9.8\ m/s^2}}$

$\therefore h_{max} = 10.2\ m$

∴ Answer A

46.Water flows through a fire hose of diameter 6.35 cm at a rate of $0.012 m^3/s$. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exists the nozzle?

A .23.4m/s B . 31.6m/s C. 63.2 m/s D. 46.8 m/s

Given: Fire hose of diameter $= dF = d_1 = 6.35 \times 10^{-2} m$

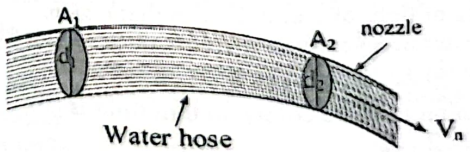
$\frac{V}{t} = 0.012\ m^3/s$

Nozzle of inner diameter $= d_n = d_2 = 2.2 \times 10^{-2} m$

Nozzle of inner radius $= r_n = 1.1 \times 10^{-2} m$

Required: Nozzle speed $= V_n = ?$

Solution:



$AV = constant$

$A_n V_n = \frac{V}{t}$

$V_n = \frac{V/t}{A_n}$

$V_n = \frac{V/t}{\pi r_n^2}$

$V_n = \frac{0.012 m^3/s}{3.14\ (1.1 \times 10^{-2} m)2}$

$\therefore V_n = 31.58 m/s$

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Therefore, the speed with which the water exists the nozzle is31.6m/s.

∴ ***Answer B***

47. A point source broad casts sound into a uniform medium. If the distance from the source is tripled, how does the intensity change?

A .It remains unchanged.
B .It becomes three times larger.
C .It becomes one – third as large.
D .It becomes one – ninth as large.

Given : $r_2 = 3r_1$ ***Required :*** $I_2 = ?$

Solution: $I \times r^2 = \text{constant}$

$$I_2 \times r_2^2 = I_1 \times r_1^2$$

$$I_2 = \frac{r_1^2}{r_2^2} \times I_1$$

$$I_2 = \frac{r_1^2}{(3r_1)^2} \times I_1$$

$$\therefore I_2 = \frac{1}{9} I_1$$

Therefore, the intensity change becomes one – ninth as large.

∴ ***Answer D***

48. When a soap bubble is in air, it exhibits different colours under sunlight. The absence of a particular colour in a region is due to

A .double refraction.
B . straight line propagation of light.
C . destructive interference of light of that colour in that region.
D . constructive interference of light of that colour in that region.

Solution: Constructive interference(maximum colors observed) in thin films is give by : $2nt = (m + \frac{1}{2})\lambda$..... $m = 0, 1, 2, 3....$

Destructive interference(minimum or no color) in thin films is give by : $2nt = m\lambda$..... $m = 0, 1, 2$ ....

Therefore , during destructive interference of light the particular colour **cannot** be seen

∴ ***Answer C***

49. A small object of mass m carries a charge q and is suspended by a thread between the vertical plates of a parallel- plate capacitor. The plate separation is d. If the thread makes an angle θ with the vertical, what is the potential difference between the plates?

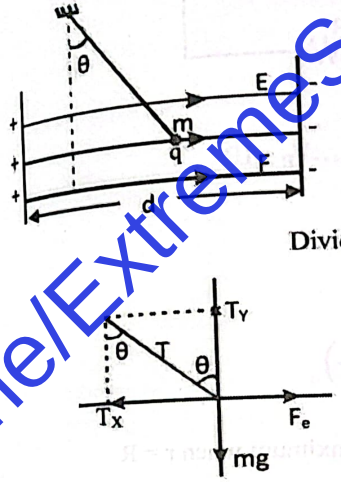
A . $\frac{mgd\tan\theta}{q}$ B . $\frac{md\tan\theta}{q}$ C. $\frac{mgd\sin\theta}{q}$ D. $\frac{mdsin\theta}{q}$

Given : mass = m
Charge = q
Plate separation = d = r

Angle = θ with the vertical

Required : Potential difference between the plates = V = ?

Solution:



The system is at equilibrium (rest)

$\sum F_i = 0$

$\Rightarrow \sum F_{ix} = \sum F_{iy} = 0$

$F_e - T_x = 0$.....$T_x = T\sin\theta$

$F_e - T\sin\theta = 0$

$F_e = T\sin\theta$ Eqution1

$T_y - mg = 0$ ..... $T_y = T\cos\theta$

$T\cos\theta - mg = 0$

$T\cos\theta = mg$....Eqution2

Dividing equation 1 by equation 2 we get :

$$\frac{T\sin\theta}{T\cos\theta} = \frac{F_e}{mg}$$

$$\tan\theta = \frac{F_e}{mg}$$

$F_e = \tan\theta \times mg$

$qE = mg\tan\theta$..... $F_e = qE$

$$q\frac{V}{d} = mg\tan\theta \text{.....} E = \frac{V}{d}$$

$$\therefore V = \frac{mgd\tan\theta}{q}$$

Therefore , the potential difference between the plates is $\frac{mgd\tan\theta}{q}$

∴ ***Answer B***

50. The maximum power delivered to the load resistance R occurs when the load resistance R and the internal resistance r have the following relation.

A. r = 2r B. r = R C. r = 3R D .r =7R

Given : A battery of emf = ε
Internal resistance = r
Load (total external resistance) = R

Required : What should be the relationship between the load resistance R and the internal resistance r for the power dissipated in to be R to maximum = ?

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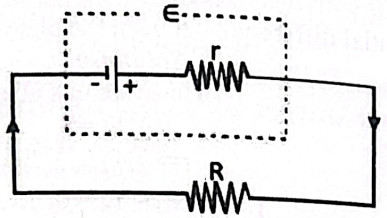
Solution:



$P = IV_R$

$P = (IR)I$.....$V_R = IR$

$P = I^2R$

$P = \left(\frac{\varepsilon}{r+R}\right)^2 R$

$P = \varepsilon^2 \left(\frac{R}{(r+R)^2}\right)$

$\therefore P = \frac{\varepsilon^2}{4r}$.....Maximum when r = R

∴ Answer B

1. Which of the following is correct about the meaning of photon?
A .Quantum of matter. C .Negatively charged particle.
B .Positively charged particle. D. Quantum of light.
Solution: The meaning of photon means quantum of light.
The units of energy are called **photons or quanta.**
∴ Answer D

2. Consider a series RC circuit for which R = 1.0 MΩ, C =5.0 μF and ε =30.0 V. What is the time constant of the circuit and the maximum charge on the capacitor after the switch is closed respectively?
A .5seconds, 150μC C . 10seconds , 160 μC
B .7seconds, 155 μC D . 9seconds, 135 μC
Given: R =1M Ω ***Required:***
C =5.0 μF The time constant = t = ?
ε =30.0 V Maximum charge = q_{max} = ?

Amount of charge stored after time t is given by :$Q = Q_0(1 - e^{\frac{-t}{RC}})$

The maximum charge is given by: $Q_{max} = Q_o$

$\therefore Q_{max} = \varepsilon C = 30V \times 5\mu F = 150\ \mu F$

The time constant is given by : $t = RC = 1 \times 10^6\ \Omega \times 5 \times 10^{-6}\ F = 5sec$

∴ Answer A

3 .What is the direction to which a fish must push the water with its fins in order to propel east ward?
A .East ward C. West ward
B .Upward D. Down ward.
Solution: **Newton III law states** that for every action force there is an equal and opposite reaction. i.e to move due eastward the fins must exist a force due west ward .
∴ Answer C

4. Suppose two molecules of different mass are kept at the same temperature. Which one of the following statements is **NOT** true?
A . Both remain at rest.
B .Both move with the same speed.
C .The lighter molecule moves faster.
D .The heavier molecule moves faster.
Solution: Every particles whose temperature is above absolute zero is in motion. The motion increases with temperature. Temperature is the measure of the average kinetic energy of the molecules.
Therefore , choice A , B and D are False.

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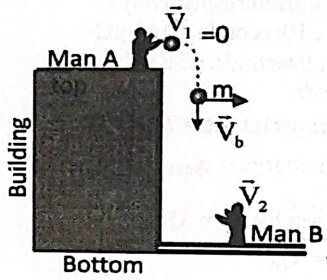
∴ ***Answer A ,B ,D***

5. One person drops a ball from the top of a building while another person at the bottom observes its motion. The two people **DISAGREE** on which of the following?
   A . The value of total energy of the ball.
   B .The change in potential energy of the ball.
   C . The value of the gravitational potential energy of the ball Earth system.
   D . The kinetic energy of the ball at some point in its motion if one of them is walking.



Solution:

$KE_A = \frac{1}{2} m_b v_1^2$ .... at time t.

$KE_B = \frac{1}{2} m v_2^2$

The velocity of the bottom relative to the man walking on the ground, $V_{b2} > v_{b1}$.

Therefore, they disagree on the kinetic Energy.

∴ ***Answer D***

6 .Two non-zero vectors $\vec{A}$ and $\vec{B}$ are related by $\vec{A} = c\vec{B}$,where c is a scalar. If the two vectors have opposite directions, then one of the following is true about c ?
   A .C is a positive number
   B .C is a negative number.
   C. C = 1
   D .C = 0

Solution: For any vector $\vec{B}$ and a scalar C their product is $\vec{A} = c\vec{B}$ the result is a vector. Direction of $\vec{A}$ is the same as $\vec{B}$ if C is positive number while opposite in $\vec{B}$ negative number.

∴ ***Answer B***

7 .On which of the following does the induced emf in a coil **NOT** depend when a magnet is inserted into it?
   A . Resistance of the coil.
   B . Number of turns in the coil.
   C. Speed of approach of the magnet
   D. Magnetic moment of the magnet

***Solution :*Faradays Second Law:** the strength of the induced emf (ε) is directly proportional to the time rate of change of the magnetic flux.

$$\varepsilon = \frac{-N\Delta(BA)}{\Delta t}$$

Therefore, induced emf in a coil depend on :

- number of turns in the coil
- speed of approach of the magnet
- magnetic moment of the magnet
- the magnetic field strength

But induced emf in a coil does not depend on the resistance of the coil.

∴ ***Answer A***

8 . A wave has a speed of 50 m/s and a wave length of 25,000 m. What is the frequency of the wave?
   A .2.0×10^{-3}Hz
   B .6.0×10^{1}Hz
   C. 5.0×10^{2}Hz
   D. 1.8×10^{6} Hz

Given:
v =50 m/s
λ = 25000 m

Required:
f =?

Solution: By the wave equation:

$V = \lambda f$

$f = \frac{V}{\lambda}$

$f = \frac{5 \times 10 m/s}{25 \times 10^3 m}$

∴ $f = 2 \times 10^{-3}$Hz

∴ ***Answer A***

9. An oscillating system is used to create a succession of waves on a water surface. The rate of wave generation is one every 0.33 seconds. What is the period of these waves?
   A .0.3 seconds
   B .1.0 seconds
   C.2. 0 seconds
   D. 3.0 seconds

***Solution*: Period (T)** of a vibrating system is the time required for the system to complete one full cycle of oscillation.
The rate of wave generation is one every 0.33 seconds implies one second .Therefore ,f =3 Hz.

$$T = \frac{1}{f} = \frac{1}{3Hz} = 0.33sec$$

∴ ***Answer A***

10. A vector that represents the position of an object in relation to another object is called _____.
   A .Unit vectors
   B .position vector
   C.-Coplanar vectors
   D. Collinear vectors.

***Solution*:** A vector that represents the position of an object in relation to another object is called **position vector.**

∴ ***Answer B***

11.In a reversible thermodynamic process, the system:
   A .is always close to equilibrium states.
   B .might never be close to any equilibrium state.
   C .is close to equilibrium states only at the beginning and end.

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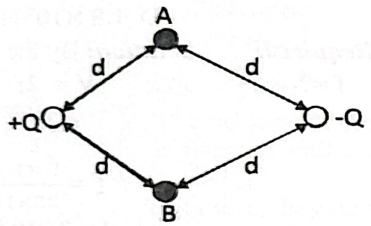
D .is close to equilibrium states throughout , except at the beginning and end.

Solution: A process which can be retraced in the opposite direction so that it passes through exactly the same states in all respects as in the direct process is called **a reversible process.**

In a reversible process , the system is very nearly in equilibrium all the time.

∴ Answer A

12.Points A and B are each a distance d from each of two equal and opposite charges as shown below.



What will be the work required to move a negative charge q from A to B?

A .Zero

B. $\frac{qQ}{4\pi\varepsilon_o d^2}$

C . $\frac{qQ}{4\pi\varepsilon_o d}$

D. $\frac{qQ}{4\pi\varepsilon_o(4d^2)}$

Required: $W_{A\rightarrow B} = ?$

Solution: The potential at point A is: $V_A = \frac{k(+Q)}{d} + \frac{k(-Q)}{d} = 0$

The potential at point B is: $V_B = \frac{k(+Q)}{d} + \frac{k(-Q)}{d} = 0$

$$V_{B.A} = V_B - V_A = 0$$

$$V_{B.A} = \frac{W_{A\rightarrow B}}{q}$$

$$\therefore W_{A\rightarrow B} = V_{B.A} \times q = 0$$

∴ Answer A

13.An air -filled parallel – plate capacitor has a capacitance of 1 PF. The plate separation is then doubled and a wax dielectric is inserted completely filling the space between the plates. As a result, the capacitance becomes 2pF .What is the dielectric constant of the wax?

A .8.0

B .4.0

C .2.0

D .0.5

Solution:

Given :

$K_1 = 1$

$C_o = 1pF$

$d_1 = d$

$C_2 = 2PF$

$d_2 = 2d_1$

Required: $K_2 = ?$

$$C_o = \frac{\varepsilon_o A}{d_1}$$

$$C_2 = K_2 \frac{\varepsilon_o A}{d_2}$$

$$C_2 = K_2 \frac{\varepsilon_o A}{2d_1} \ldots\ldots C_o = \frac{\varepsilon_o A}{d_1}$$

$$2C_2 = K_2 C_o$$

$$K_2 = \frac{2C_2}{C_o}$$

$$K_2 = 2 \times \frac{2pF}{1PF}$$

$$\therefore K_2 = 4$$

∴ Answer B

14. A step up transformer is used to _____.

A .increase the power.

B .decrease the power.

C. increase the voltage

D. decrease the voltage.

Solution: **A transformer** is a device to raise or lower the voltage in an ac circuit.

There are two types of transformer .These are :

i) Step up transformer and

ii) Step down transformer

Step up transformer is the number of turns in the primary coil(N_P) is less than the number of turns in the secondary coil (N_S) .

Step up transformer is the input voltage of the transformer (V_P) is smaller than the output voltage of the transformer (V_S).

Therefore , $\frac{V_p}{V_s} = \frac{N_p}{N_s}$

$$\Rightarrow V_S = \frac{N_S}{N_P} \times V_P$$

Therefore, a Step up transformer used to increase the voltage supplied.

∴ Answer C

15.A 220 – V power line is protected by a 15 – A fuse. What is the maximum number of "220 V , 500 W" light bulbs that can be operated at full brightness from this line?

A .0.2 B .0.4 C .0.6 D .0.8

Given : $\varepsilon = 220\ V$

$I = 15A$

Required : the maximum number of "220 V , 500 W" light bulbs

Solution: $P_{max} = \varepsilon . I = 220V \times 15A = 3300\ W$

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$$R = \frac{V^2}{P} = \frac{(220V)^2}{500W} = 96.8\Omega$$

$$n = \frac{3300W}{500W} = 6.6 \text{bulbs}$$

$$P = nI^2R$$

$$n = \frac{p}{I^2.R} = \frac{3300w}{(15)^2 \times 96.8\Omega} = 0.15 \approx 0.2$$

∴ ***Answer A***

16.If $\vec{A} = 8\hat{i} + \hat{j} - 2\hat{k}$, and $\vec{B} = 5\hat{i} - 3\hat{j} + \hat{k}$ then , what is the value of $\vec{B} \times \vec{A}$?

A $3\hat{i} + 4\hat{j} - 3\hat{k}$

B . $-3\hat{i} - 4\hat{j} + 3\hat{k}$

C. $\vec{A} = -5\hat{i} - 18\hat{j} - 29\hat{k}$

D . $\vec{A} = 5\hat{i} + 18\hat{j} + 29\hat{k}$

Given: $\vec{A} = 8\hat{i} + \hat{j} - 2\hat{k}$

$\vec{B} = 5\hat{i} - 3\hat{j} + \hat{k}$

Required : $\vec{B} \times \vec{A} = ?$

Solution:

$$\vec{B} \times \vec{A} = \begin{bmatrix} i & j & k \\ 5 & -3 & 1 \\ 8 & 1 & -2 \end{bmatrix}$$

$$\vec{B} \times \vec{A} = [(-3 \times -2) - (1 \times 1)]i + [(8 \times 1) - (5 \times -2)]j + [(5 \times 1) - (8 \times -3)]k$$

$$\therefore \vec{B} \times \vec{A} = 5\hat{i} + 18\hat{j} + 29\hat{k}$$

∴ ***Answer D***

17.The coordinate of a particle in meters is given by $x(t) = 25t - 3t^3$, where the time t is in seconds. At what value t will the particle become momentarily at rest?

A .2.78s B. 1.67 s C. 0.6 s D. 0.36 s

Given: V = 0

$X(t) = 25t - 3t^3$

Required: t = ?

Solution:

$$V = \frac{dx}{dt}$$

$$V = \frac{d}{dt}(25 - 3t^3)$$

$$V = 25 - 9t^2$$

$$25 - 9t^2 = 0$$

$$t = \sqrt{\frac{25}{9}}$$

$$\therefore t = 1.67\ S$$

∴ ***Answer B***

18.A woman driving a 2000 kg car along a level road at 30 m/s takes her foot off the gas to see how far her car will roll before it slows to a stop. She discovers that it takes 150 m. What is the average force of friction acting on the car?

A .9000N B .6000N C .3000N D .400N

Given: m =2000 kg

u = 30m/s

V =0

S = 150m

Required : Average friction force $= F_{av} = ?$

Solution: Work done by friction $= \Delta KE$

$$-W_f = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$-f.s = -\frac{1}{2}mu^2$$

$$f = \frac{mu^2}{2s}$$

$$f = \frac{2\times10^3 kg(30m/s)^2}{2\times150m}$$

$$\therefore f = 6000\ N$$

∴ ***Answer B***

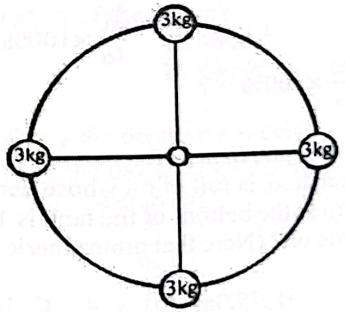
19.The radius of the wheel shown below is 0.75m.What is its moment of inertia?

A .2.25kgm^2

B .3.38kgm^2

C. 6.75kgm^2

D. 9.00kgm^2



Given : $m_1 = m_2 = m_3 = m_4 = 3kg$

$r_1 = r_2 = r_3 = r_4 = r = 0.75\ m$

Required: I =?

Solution: The total moment of inertia is the sum of the moment of inertia due to each mass

$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + m_4r_4^2$$

$$I = mr^2 + mr^2 + mr^2 + mr^2$$

$$I = 4mr^2$$

$$I = 4(3\ kg) \times (0.75m)^2$$

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$\therefore I = 6.75\ kg.m^2$

∴ Answer C

20. An ordinary refrigerator is turned on and its door is left open in a thermally insulated kitchen. What is the temperature of the room?
A .increases according to the first law of thermodynamics.
B .decreases according to the first law of thermodynamics.
C .remains constant according to the first law of thermodynamics.
D .remains constant according to the second law of thermodynamics.
Solution: As the room thermally insulated there is no transfer of heat energy between the room and the refrigerator. This is due to the first law of thermodynamics.

∴ Answer C

21. The length of an aluminum rod at a temperature of 20°C is 3m. What will be the ratio of change in the length of the rod to its original length if its temperature is raised to 120°C? The coefficient of linear expansion of aluminum is $2.31\times10^{-5}\ k^{-1}$.

A .23% B . 3.2% C. 3% D. 0.23%

Given: To = 20°C
T = 120°C
$\Delta T = 100°C = 100K$
$\ell_o = 3m$
$\propto = 2.31\times 10^{-5}K^{-1}$

Solution:
$\Delta\ell = \propto \ell_o \Delta T$
$\Delta\ell = 2.31\times 10^{-5}K^{-1}\times 3m\times 100K$
$\therefore \Delta\ell = 693\times 10^{-5}m$

Required: $\frac{\Delta\ell}{\ell_o}\times100\% = ?$

$\frac{\Delta\ell}{\ell_o}\times100\% = \frac{693\times10^{-5}m}{3m}\times100\%$
$= 231\times10^{-5}\times100\%$
$= 0.231\%$

∴ Answer D

22. A very large open tank is full of oil whose density is $800kg/m^3$. If the absolute pressure at the bottom of the tank is 1.81×10^5 pa, then what is the depth of the oil? (Note that atmospheric pressure is 1.01×10^5pa).

A .100.0m B .75.0m C .10.0m D .9.5m

Given: Density = $\rho_o = 800kg/m^3$
Absolute pressure = $P_{ab} = 1.81\times10^5$ Pa
Atmospheric pressure = $P_{atm} = 1.01\times10^5$pa

Required: depth = h = ?

Solution: $p_{fluid} + p_{atm} = p_{ab}$
$p_f = p_{ab} - p_{atm}$
$\rho gh = p_{ab} - p_{atm}$
$h = \frac{p_{ab}-p_{atm}}{\rho g}$

$h = \frac{1.81\times10^5 Pa - 1.01\times10^5 pa}{8\times10^2 kg/m^3\times10m/s^2}$

$\therefore h = 10m$

∴ Answer C

23. A 1.5 kg particle moves in the xy-plane with velocity $V = (4.2\hat{i} - 3.6\hat{j})$ m/s. What is its angular momentum when its position vector is $r = (1.5\hat{i} + 2.2\hat{j})$ m?

A . $-22\hat{k}$ kg m^2/s
B .$14.6\hat{k}$ kg m^2/s
C. $-14.6\hat{k}$ kg m^2/s
D. $22\hat{k}$ kg m^2/s

Given: m = 1.5 kg
$\vec{V} = (4.2\hat{i} - 3.6\hat{j})$ m/s
$\vec{r} = (1.5\hat{i} + 2.2\hat{j})$m

Required: $\vec{L} = ?$

Solution: $\vec{L} = \vec{r}\times\vec{p}$
$\vec{L} = \vec{r}\times(m\vec{V})$ ..... $\vec{P} = m\vec{V}$
$\vec{L} = m(\vec{r}\times\vec{v})$
$\vec{L} = 1.5kg\,(1.5\hat{i}+2.2\hat{j})\ m\ \times(4.2\hat{i} - 3.6\hat{j})m/s$
$\vec{L} = 1.5kg\,[6.3\,(\hat{i}\times\hat{i}) + 5.4\,(i\times i)] + 9.24\,(\hat{i}\times\hat{i}) + 7.92\,(\hat{j}\times\hat{j})$
$\therefore \vec{L} = -21.96\,\hat{k}\ kg.m^2$

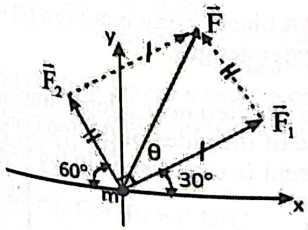
∴ Answer A

24. A 2 kg object experiences two concurrent forces of 100N at 30^0 above the positive x –axis and 80 N at 60^0 above the negative x –axis. What is the magnitude of acceleration of the object?

A .68m/s^2 C. 48m/s^2
B .64m/s^2 D. 46m/s^2

Given: m = 2kg
$\vec{F_1}$ = 100N at 30^0 above the positive x – axis.
$\vec{F_2}$ = 80N at 60^0 above the negative x – axis.

Required : a = ?



Solution:
$a = \frac{\vec{F}}{m}$
$F = \sqrt{F_1^2 + F_2^2}$
$F = \sqrt{(10^2N)^2 + (80N)^2}$
$\therefore F = 128N$
$a = \frac{\vec{F}}{m} = \frac{128N}{2kg}$
$\therefore a = 64m/sec^2$

∴ Answer B

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25. A driver moving at a constant speed of 20m/s sees an accident up a head and hits the brakes. If the car decelerates at a constant rate of $-5m/s^2$, how far does the car go before it comes to a stop?

A .120m B . 100m C .40 m D. 20 m

Given: V = 0 m/sec

u =20m/s

$a = -5m/s^2$

Required: S = ?

Solution: $2as + u^2 = v^2$

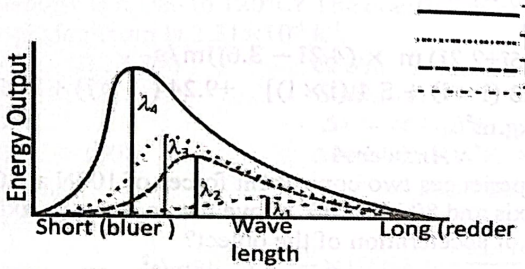
$$S = \frac{V^2 - U^2}{2a}$$

$$S = \frac{0 - (20m/s)^2}{2(-\frac{5m}{s^2})}$$

$\therefore$ **S = 40 m**

$\therefore$ Answer C

26. The energy spectrum emitted by a black body at different temperature is as shown in the figure below. Which of the following statements is **NOT** correct about the black body energy spectrum?



A .$T_1 < T_2 < T_3 < T_4$

B .$\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$

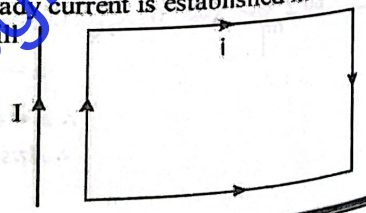
C . Black body radiate energy over the entire electromagnetic spectrum.

D .The wavelength corresponding to the maximum intensity increases with temperature.

Solution: The total energy emitted by a black body increases rapidly with increase in temperature for any wave length.

$\therefore$ Answer D

27. A rectangular loop carrying a current I situated near a long straight wire such that the wire is parallel to one of the sides of the loop and it is in a plane of the loop. If a steady current is established in the wire as shown in the figure, the loop will



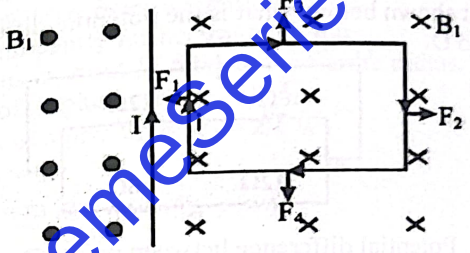
A .Remain stationary.

B .Move toward the wire.

C. Move away from the wire

D. Rotate about an axis parallel to the wire.

Solution:



The magnetic field B_1 produced by straight current carrying conduct to and the forces ($\vec{F_1}$ and $\vec{F_2}$) on the current carrying loop in B_1 are determined using right hand rule.

$F = BI\ell \sin\theta, \implies \vec{F} = I\vec{\ell} \times \vec{B}$

$$B_1 = \frac{\mu_o I_1}{r}$$

$F_3 = F_4$ ..... Balanced

$F_1 > F_2$ .... since $F_1 = \frac{\mu_o I}{r_1} \cdot \ell$ and $F_2 = \frac{\mu_o I \ell}{r_2}$..... and $r_1 < r_2$

Thus $F_1 - F_2 = F_{net}$ ..... to the left .

Thus the loop is attracted to the left.

$\therefore$ Answer B

28. Which device uses the principle of electromagnetic induction for its operation?

A .Ac generator C. Cathode ray tube.

B .Moving coil meter. D. Light emitting diode.

Solution: Ac generator uses the principle of electromagnetic induction.

$\therefore$ Answer A

29. An electron enters a region of uniform electric field of 5×10^3N/C. What is the force on the electron?

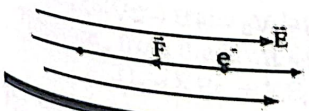
A .8×10^{-22}N C. 8×10^{-16}N

B .5.3×10^{-22}N D . 1.88×10^{22} N

Given: $E = 5\times 10^3$ N/C ***Required :*** $\vec{F_e} = ?$

$q = e^- = 1.6\times 10^{-19}$C

Solution:



$$E = \frac{F}{q}$$

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$F = qE$

$F = 1.6\times 10^{-19}\ C \times 5 \times 10^{3} N/C$

$\therefore F = 8.0 \times 10^{-16}\ N$

∴ Answer C

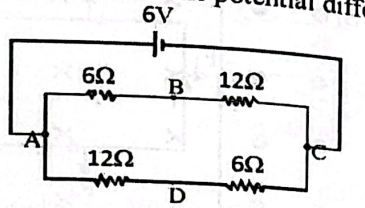
30.In the circuit shown below, what is the potential difference between points B and D.

A .1V

B .2 V

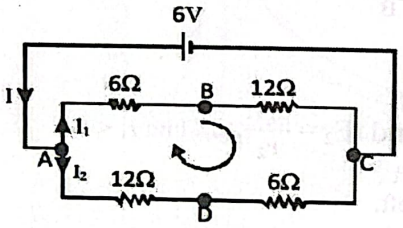
C. 3V

D .4 V



Required : Potential difference between points B and D $= V_{DB} = ?$

Solution:



$I = I_1 + I_2$ Equation1

$-6I_1 - 12I_1 + 6I_2 + 12I_2 = 0$ Equation2

$\Rightarrow I_1 = I_2$

$I = I_1 + I_2 \Rightarrow 2I_1 = I$

$\Rightarrow I_1 = I_2 = \frac{I}{2}$

$I = \frac{V}{R_T}$

$R_{T1} = 6\Omega + 12\Omega = 18\Omega$

$R_{T2} = 6\Omega + 12\Omega = 18\Omega$

$\frac{1}{R_T} = \frac{1}{R_{T1}} + \frac{1}{R_{T2}} = \frac{1}{18\Omega} + \frac{1}{18\Omega}$

$\therefore R_T = 9\Omega$

$I = \frac{V}{R_T} = \frac{6V}{9\Omega} = \frac{2}{3}\ A$

$I_1 = I_2 = \frac{I}{2} = \frac{\frac{2A}{3}}{2} = \frac{1A}{3}$

$V_D = 12\Omega \times I_2$

$V_D = 12\ \Omega \times \frac{1}{3}\ A = 4\ V$

$V_B = 6\Omega \times I_1$

$V_B = 6\Omega \times \frac{1}{3}A = 2V$

$V_{DB} = V_D - V_B$

$\therefore V_{DB} = V_D - V_B = 4V - 2V = 2V$

∴ Answer B

31.The planet Jupiter is about 318 times as massive as Earth, yet on its surface you would weigh only about 2.4 times your weight on earth. Why is this happened?

A. Your mass is 132.5 times less on Jupiter.

B .You are 132.5 times more weightless there.

C .Jupiter is significantly farther from the sun.

D .Jupiter's radius is about 11.5 times the Earth's radius.

Given: Mass of Jupiter $= M_J = 318M_E$

Weight of a person $= W_J = 2.4\ W_E$

Required: $W_E = ?$

Solution: $W_J = m.g_J$

$W_J = m.\frac{GM_J}{R_J^2}$

$2.4W_E = m\frac{GM_J}{R_J^2}$

$2.4m\frac{GM_E}{R_E^2} = mG\frac{318M_E}{R_J^2}$

$R_J = \sqrt{\frac{318}{2.4}} \times RE$

$\therefore R_J = 11.5\ R_E$

∴ Answer D

32.A uniform ball of radius r and mass m starts from rest and rolls down a frictionless incline of height h .When the ball reaches the ground how fast is it going ?

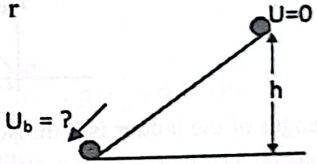
A. $\sqrt{\frac{10}{7}gh}$ B. $\sqrt{\frac{9}{10}gh}$ C. $\sqrt{\frac{10}{9}gh}$ D. $\sqrt{\frac{7}{10}gh}$

Given: Radius of the ball $= r_b = r$

$I = \frac{2}{5}MR^2$ ...Solid sphere

Mass of the ball $= m_b = m$

Velocity $= u = 0$ m/sec



Required : Velocity of the ball $= V_b = V_{bottom} = ?$

Solution: At the start , the sphere has gravitational potential energy $U_g = Mgh$. When it reaches the ground all of U_g will be changed to rotational(KEr) and translational (KE_t) kinetic energy.

Therefore , from the law of Conservation of Mechanical energy.

$U_g = KEr + KE_t$

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$$Mgh = \frac{1}{2}MV^2 + \frac{1}{2}I\,\omega^2$$

$$Mgh = \frac{1}{2}MV^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right) \times \left(\frac{V}{R}\right)^2 \quad \ldots\ldots \omega^2 = \frac{V^2}{R^2}$$

$$Mgh = \frac{1}{2}MV^2 + \frac{1}{5}MR^2 \times \frac{V^2}{R^2}$$

$$Mgh = \frac{1}{2}MV^2 + \frac{1}{5}MV^2$$

$$Mgh = \frac{7}{10}MV^2$$

$$\therefore V = \sqrt{\frac{10}{7}gh}$$

∴ ***Answer A***

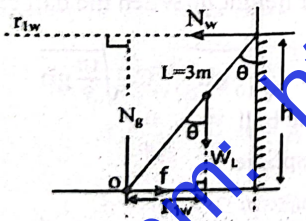
33. A ladder of length 3m and mass 20 kg leans against a smooth vertical wall so that the angle between the wall and the ladder is 30^0. What are the magnitudes of the normal forces at the contact point on the wall (N_w) and at the contact point on the ground (N_g)?

A. $N_w = 0$ N, $N_g = 200$ N
B. $N_w = 200$ N, $N_g = 0$N
C. $N_w = 57.7$ N, $N_g = 200$N
D. $N_w = 200$ N, $N_g = 57.7$ N

Given: $\ell = 3$m
$m = 20$ kg
$W_\ell = mg = 200$N
$\theta = 30^0$ with the wall.

Required:
$N_w = ?$
$N_g = ?$

Solution:



The length of the ladder is 3 m. So its center of mass is at 1.5 m from the ground. The ladder is in equilibrium, it satisfies the first and Second condition of equilibrium

first condition of equilibrium: $\sum \overrightarrow{F_\ell} = 0 \Rightarrow \sum \overrightarrow{F_{X\ell}} = 0$

$f - N_W = 0 \Rightarrow f = N_w$

$\sum F_{1y} = 0 \Rightarrow Ng - w_\ell = 0$

$N_g = w_\ell$

$\therefore N_g = 200$N

Second condition of equilibrium: $\sum \tau = 0$

$$\cos 30^0 = \frac{h}{\ell} = \frac{h}{3m}$$

$h = 3m \times 0.866$

$h = 2.598m = 2.6m$

$$r_{1\ell} = \frac{\ell}{2}\sin 30^0$$

$$\ell_{1\ell} = \frac{3}{4}\,m$$

$N_w \times h - W_\ell \times r_{1\ell} = 0$

$$N_w = \frac{w_\ell . r_{1\ell}}{h}$$

$$N_w = \frac{200N \times 3/4m}{2.6m}$$

$\therefore N_w = 57.7$N

∴ ***Answer C***

34. A container resting on the floor of an elevator contains an incompressible fluid of density ρ. When the elevator has an upward acceleration of magnitude a then what will be the pressure difference between two points in the fluid separated by a vertical distance Δh?

A. $\rho a \Delta h$
B. $\rho(g + a)\Delta h$
C. $\rho g \Delta h$
D. $\rho g a \Delta h$

Given: Fluid of density $= \rho_f = \rho$
Upward acceleration $= a_1 = a$
Separated by a vertical distance $= \Delta h$
Required: Pressure difference between two points $= \Delta p = ?$
Solution: The elevator accelerates upward the net force is given by:

Net force (ma) = Force of fluid (F) − Weight (mg)

$F - mg = ma$

$F = m(g + a)$

$Ap = m(g + a) \ldots\ldots p = \frac{F}{A} \Rightarrow F = PA$

$A\,\Delta p = \Delta m(g + a)$

$A\,\Delta p = \rho \Delta v(g + a) \ldots . \rho = \frac{\Delta m}{\Delta V} \Rightarrow \Delta m = \rho \Delta V$

$A\,\Delta p = \rho A \Delta h(g + a) \ldots . \Delta V = A\Delta h$

$\therefore \Delta p = \rho \Delta h(g + a)$

∴ ***Answer B***

35. If two particles have equal momenta, then which one of the following statements is TRUE about their kinetic energies?
A. Their kinetic energies are always equal.
B. Their kinetic energies are never equal.

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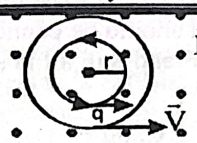

$F = F_C$

$B = \frac{mv^2}{r}$

$q = \frac{mv}{rB}$

Therefore, the charge of the particle is $\frac{mv}{rB}$.

∴ ***Answer D***

39. An alternating current is represented by I (t) = (10.0 A) sin (100π t).What are the amplitude and frequency of the alternating current?

A .100A , 75Hz
B .10.0 A , 100Hz
C. 10.0 A , 50 Hz
D. 5.0 A , 25Hz

Given: I (t) = (10.0) A sin (100 π t) ***Required:*** Io = ?and f = ?

Solution: I (t) = Io sin ωt.

I (t) = Io sin 2πf t

I (t) = (10.0 A) sin (100 π)t.

∴Io = 10 A

2π f = 100π t

∴f = 50 Hz

Therefore , the amplitude and the frequency of the alternating current are 10A and 50Hz respectively.

∴ ***Answer C***

40 . A series RLC circuit has a resistance of 45.0Ω and an impedance of 75.0 Ω.What average power is delivered to this circuit when ΔVrms = 210V ?

A 573W B .353 W C .255W D .247 W

Given: R = 45.0Ω ***Required:*** Average power = P_{av} = ?

Z = 75Ω

$\Delta V_{rms} = 210$ V

Solution: The average power delivered in the circuit is given by : $P_{av} = I^2_{rms} R$

$I_{rms} = \frac{V_{rms}}{Z}$

$I_{rms} = \frac{210\ V}{75\Omega} = 2.8$ A

$P_{av} = I^2_{rms} R$

$P_{av} = (2.8\ A)^2 \times 45.0\Omega$

∴ P_{av} = 353 W

∴ ***AnswerB***

41 .A thin film of oil (n = 1.25) is located on a smooth wet floor. When viewed perpendicular to the floor, the film reflects most strongly red light at 640 nm and reflects no blue light at 512nm. How thick is the oil film?

A .640 nm B . 612 nm C .548 nm D. 512nm

Given : n = 1.25 ***Required*** : Thickness t of the film = ?

$\lambda_1 = 640$ nm

$\lambda_2 = 512$ nm

Solution: Constructive interference in thin films is given by :

$2nt = (m + \frac{1}{2})\lambda$ ..... m = 0,1. 2,...

Destructive interference in thin films is given by :

$2nt = m\lambda$ ..... m= 0, 1 , 2 ...

The minimum film thickness for Constructive interference in the reflected light corresponds to m = 0.

$t = \frac{\lambda}{4n}$

$t = \frac{640\ nm}{4 \times 1.25} = \frac{640\ nm}{5} = 128$nm

∴ ***No Answer***

42 .Which one of the following changes to parallel plate capacitor would **Not** increase energy stored in the capacitor at a fixed voltage across the plates?

A .Increasing the area of the plates.
B .Increasing the dielectric constant.
C .Decreasing the charges on the plates.
D .Increasing the distance between the plates.

Solution: The energy stored in a capacitor is given by :$U = \frac{1}{2}Vq$

$U = \frac{1}{2}CV^2$ ..... Formula 1

From the given formula 1

U increases and C also increases .

For parallel plate capacitor is given by:

$C = \varepsilon_o \frac{A}{d}$..... Formula 2

From the given formula 2

C increases and A also increases.

1.When a dielectric constant increases ,the capacitance and the energy increases.
2. Increasing the distance between the plates , the capacitance and the energy decreases.
3.When the charge decreases the energy and the capacitance decreases.

∴ ***Answer D***

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43. Suppose that a voltage surge produces 240 V for a moment. By what percentage does the power output of 220 – V light bulb increase?

A .38% B .36% C .19% D .18%

Given: Vo = 220 V

V = 240 V

Required : Percentage increase of power output = ?

Solution: Percentage increase of power output =

$$= \left(\frac{p-p_o}{p_o}\right) \times 100\%$$

$$= \left(\frac{\frac{V^2}{R} - \frac{V_O^2}{R}}{\frac{V_O^2}{R}}\right) \times 100\%$$

$$= \left(\frac{V^2 - V_O^2}{V_O^2}\right) \times 100\%$$

$$= \frac{(240\ V)^2 - (210V)^2}{(210V)^2} \times 100\%$$

$$= 19.0\ \%$$

∴ Answer C

44. A wire carries a steady current of 2.4A . A straight section of the wire is 0.75 m long and lies along the x axis within a uniform magnetic field,

B = 1.6 T along + z axis. If the current is in the +x direction, what is the magnetic force on the section of wire?

A .2.88$\hat{j}$ N B. 1.62$\hat{j}$ N C. – 1.62 $\hat{j}$ N D. -2.88 $\hat{j}$N

Given: I = 2.4 A

$\vec{\ell} = \frac{3}{4} m\hat{\imath}$

$\vec{B} = 1.6\ T\hat{k}$

Required : $\vec{F_m}$ = ?

Solution: $I\vec{\ell} = 2.4 \times 0.75\ A.m\hat{\imath} = 1.8\ Am\hat{\imath}$

$\vec{F_m} = I\vec{\ell} \times \vec{B}$

$\vec{F_m} = 1.8Am\hat{\imath} \times 1.6\ T\hat{k}$

$\vec{F_m} = 2.88\ (\hat{\imath} \times \hat{k})$

$\therefore \vec{F_m} = -2.88\ \hat{j}N$

∴ Answer D

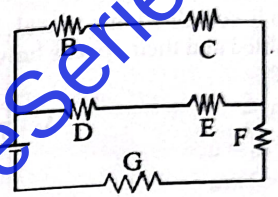
45. If all the resistors in the circuit shown in the figure have equal resistances, and the current flowing into resistor B is 4A, then what is the current flowing into resistor G?

A .16A

B .8A

C .2 A

D .1A

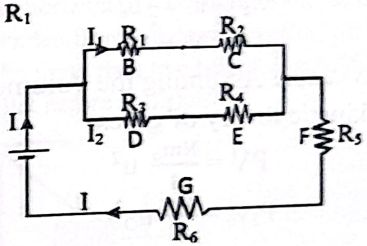


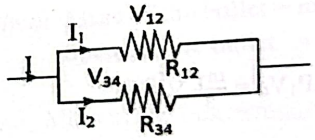
Given: Let $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R$

$I_B = I_1 = 4A$ through R_1

Required : $I_6 = I_G = I = ?$

Solution:





$I = I_1 + I_2$

$I_B = I_C = I_1 = 4A$

$R_{12} = R_1 + R_2 = 2R$

$R_{34} = R_3 + R_4 = 2R$

$V_{12} = V_{34}$ ..... why ?

$I_1R_{12} = I_2R_{34}$

$I_1.2R = I_2 2R$

$I_1 = I_2$

$I = I_1 + I_2$

$I = 2I_1$ ..... $I_1 = I_2$

$I = 2 \times 4A$

$\therefore$ $I = 8A$

∴ Answer B

46. Refraction of a wave does **NOT** involve change in its

A .speed C .frequency

B , wave length D. speed and wavelength.

Solution: Refraction of a wave change

- Speed of a wave

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- wave length
- But the frequency of a wave remains constant.

∴ ***Answer C***

47. Gas at a pressure p is contained in a vessel. If the masses of all the molecules are doubled and their speeds halved, then what will be the resulting pressure?

A .2P B .$\frac{P}{3}$ C .$\frac{P}{2}$ D .3P

Given: $P_1 = p_o =, u_1 = u_o$

$m_1 = m_T = Nm_o$

$v_1 = v_o$

$m_2 = 2m_T$

$v_2 = \frac{1}{2} u_1 = \frac{1}{2} u_o$

$V_2 = V_1 = Vo$

Required: $P_2 = ?$

Solution: Assuming the volume of the gas is constant .From kinetic theory of gases:

$$PV = \frac{Nm_o}{3} u^2$$

$$P_1 v_o = \frac{m_T}{3} u_o^2$$

$$P_2 V_O = \frac{m_2}{3} V_2^2$$

$$P_2 V_O = \frac{2}{3} m_T (\frac{1}{2} u_0)^2$$

$$P_2 V_O = \frac{1}{2} (\frac{m_T}{3} u_o^2)$$

$$P_2 V_O = \frac{1}{2} P_1 V_O \ldots\ldots P_1 V_O = \frac{m_T}{3} V_O$$

$$\therefore P_2 = \frac{1}{2} P_1$$

∴ ***Answer C***

48. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.012 m^3/s. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exists the nozzle?

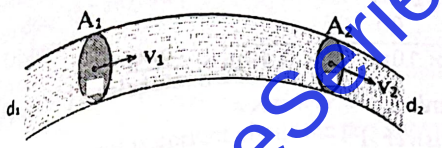
A .63.2m/s B . 46.8m/s C. 31.6m/s D. 23.4m/s

Given: $r_1 = \frac{d_1}{2} = 6.35 \times 10^{-2} m$

$\frac{V}{t} = 12 \times 10^{-3} m^3/s$

$r_2 = \frac{d_2}{2} = \frac{2.2 \times 10^{-2}}{2} = 1.1 \times 10^{-2} m$

Required : $u_2 = ?$



Solution: Au = constant

$$Au = \frac{V}{t} = \frac{\Delta V}{\Delta t}$$

$$A_n u_n = \frac{V}{t}$$

$$u_2 = \frac{V}{t}$$

$$u_2 = \frac{12 \times 10^{-3} m^3/s}{\pi r_2^2}$$

$$u_2 = \frac{12 \times 10^{-3} m^3/s}{3.14 (1.1 \times 10^{-2} m)^2}$$

$$\therefore u_2 = 31.58 m/s$$

∴ ***Answer C***

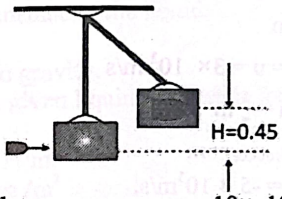
49. A bullet of mass 10 gm, moving horizontally, strikes and embeddes itself in a box of mass 1kg suspended from a light string as shown in the figure below. If the composite mass rises to a height, H = 0.45m , then what is the speed of the bullet before collision?

A .303m/s

B. 300m/s

C . 90.9m/s

D. 90m/s



Given: Mass of the bullet = $m_{bullet} = m_1 = 10 \times 10^{-3}$ kg

Speed of the bullet = $V_b = u_H$ – horizontal

$u_2 = 0$

Mass of the box = $m_{box} = m_2 = 1kg$ and speed of the box = $U_b = 0$

H =0.45 m

$m_3 = m_T = m_{bullet} + m_{box} = (1kg + 0.01kg) = 1.01kg$

Required: Speed of the bullet = $u_b = u_1 = ?$

Solution: Using conservation of momentum :$p_{before} = p_{after}$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$m_1u_1 = m_1v + m_2v \ldots\ldots\ldots v = v_1 = v_2$$

$$m_1u_1 = (m_1 + m_2)v$$

$$v_1 = \frac{m_1u_1}{m_1 + m_2} = \frac{m_1u_1}{m_T} \ldots\ldots\ldots \text{Equation1}$$

$$ME_1 = ME_2$$

$$\frac{1}{2} m_T v_1^2 = m_T g H$$

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$V_1^2 = 2gh$

$\left(\frac{m_1 u_1}{m_T}\right)^2 = 2gH$ .....From equation1

$u_1 = \sqrt{2gH} \times \frac{m_T}{m_1}$

$u_1 = \sqrt{2 \times 10m/S^2 \times 0.45m} \times \frac{1.01kg}{0.01kg}$

$\therefore u_1 = 303m/s$

∴ ***Answer A***

50.A rocket moving in free space has a speed of 3×10^3m/s relative to the Earth. Its engines are turned on , and fuel is ejected in a direction opposite the rocket's motion at a speed of 5×10^3m/s relative to the rocket. What is the speed of the rocket relative to the earth once the rocket's mass is reduced to half its mass before ignition?

A .7×10^3 m/s B . 5×10^3 m/s C. 4×10^3 m/s D . 2×10^3 m/s

Given: $m_T = m_1 + m_2 = m + \Delta m$

$m_T = m + \frac{1}{2}m$

$m_T = \frac{3}{2}m$

$u_1 = u_2 = u = 3\times 10^3$ m/s

$m_2 = \Delta m = \frac{1}{2}m$

$m_1 = m_{rocket} = m$

$V_2 = V_g = -5 \times 10^3$m/s.

Required: $V_{rocket} = V_1 = ?$

Solution: Using conservation of momentum : $p_{before} = p_{after}$

$m_1v_1 + m_2v_2 = (m + m_2)\, u$

$m_1v_1 + \Delta m v_2 = (m + \Delta m)u$

$mv_1 + \frac{1}{2}mv_2 = (m + \frac{1}{2}m)u$

$mv_1 = mu + \frac{1}{2}mu - \frac{1}{2}mv_2$

$mv_1 = m(u + \frac{1}{2}u) - m(\frac{1}{2}v_2)$

$v_1 = \frac{3}{2}(3\times 10^3 m/s) - \frac{1}{2}(-5 \times 10^3 m/s) = 7 \times 10^3 m/s$

∴ ***Answer A***

1 .What does the slope of a velocity – time graph represent?

A .Distance C. Displacement

B .Acceleration D. Speed

Solution: The slope of a velocity – time graph represents Acceleration.

∴ ***Answer B***

2 .Which statement is correct about the pressure under a fluid in a container?

A .It depends on the volume of the fluid.

B .It depends on the density of the fluid.

C .It depends on the shape of the container.

D .It is independent of the depth of the fluid.

Solution: Pressure under a fluid

i) Liquid pressure is independent of the size , shape and the base area of the container .

ii) **Liquid pressure depends only on the :**

a)dept (h) from the top surface of the liquid.

b) density of the liquid .

c) the acceleration due to gravity.

iii) The pressure exerted by a given liquid at a certain location is given by

$P = \rho gh$ where

P = pressure of liquid in N/m^2

ρ = density of liquid in kg/m^3

h = depth of the liquid.

∴ ***Answer B***

3 .A transverse sinusoidal wave is travelling on a string. Which statement is correct about a point on the string?

A . The point moves in the same direction as the wave.

B .The point moves in simple harmonic motion with a different frequency than that of the wave.

C .The point moves in simple harmonic motion with the same angular frequency as the wave.

D .The point moves in uniform circular motion with a different angular speed than the wave.

Solution: A Simple harmonicwave is sinusoidal (sine or cosine) in shape. Therefore, a transverse sinusoidal wave is travelling on a string , moves in simple harmonic motion with the same angular frequency as the wave.

∴ ***Answer C***

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4 .The distance between two point charges is doubled. What will happen to the force of interaction between the charges?

A .The force will be doubled.

B .The force will be reduced by half.

C .The force will be reduced by a factor of $\frac{1}{4}$.

D .The force will not be affected.

Given: $r_1 = r$ ***Required*** :$F_2 = ?$

$r_2 = 2r_1 = 2r$

$F_1 = F$

Solution: $F_1 = \frac{KQ_1Q_2}{r_1^2}$ Equation1

$F_2 = \frac{KQ_1Q_2}{r_2^2}$.....Equation2

Dividing Equation2 by Equation1 we get :

$$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2}$$

$$\Rightarrow F_2 = \left(\frac{r_1}{r_2}\right)^2 \times F_1$$

$$F_2 = \left(\frac{r_1}{2r_1}\right)^2 \times F_1$$

$$\therefore F_2 = \frac{1}{4} F_1$$

∴ Answer C

5 .Moving 2.5×10^{-6} coulomb of charge from point A to point B in an electric field requires 6.3×10^{-4} joule of work. The potential difference between points A and B is approximately.

A .4.0×10^{-3} V

B .2.5×10^{-2} V

C . 1.6×10^{-19} V

D. 1.0×10^{-14}V

Given: ***Required:*** ***Solution:***

$W_{A\rightarrow B} = 6.3\times 10^{-4}$J $V_{B.A} = ?$ $V_{B.A} = \frac{W_{A\rightarrow B}}{q}$

$q = 2.5\times 10^{-6}$ C $V_{BA} = \frac{6.3\times 10^{-4}J}{2.5\times 10^{-6}C}$

$\therefore V_{BA} = 2.52\times 10^{2}$V

∴ No Answer

6 .Two non zero vectors D and E have precisely equal magnitudes. For the magnitude of $\vec{D} + \vec{E}$ to be 3 times larger than the magnitude of $\vec{D} - \vec{E}$, what must be the angle between D and E?

A .30^0 B .37^0 C .53^0 D .60^0

Given: $|\vec{E}| = |\vec{D}|$

$|\vec{D} + \vec{E}| = 3\,(|\vec{D} - \vec{E}|)$

Required: $\theta = ?$

Solution: Using cosine law

$|\vec{D} + \vec{E}| = \sqrt{D^2 + E^2 + 2DE\cos\theta} = \sqrt{D^2(2 + 2\cos\theta)}$

$|\vec{D} - \vec{E}| = \sqrt{D^2 + E^2 - 2DE\cos\theta} = \sqrt{D^2(2 - 2\cos\theta)}$

$|\vec{D} + \vec{E}| = 3\,(|\vec{D} - \vec{E}|)$

$\sqrt{D^2(2 + 2\cos\theta)} = 3\sqrt{D^2(2 - 2\cos\theta)}$

$2 + 2\cos\theta = 9(2 - 2\cos\theta)$

$5\cos\theta = 4$

$\therefore \theta = \cos^{-1}(0.8) = 37^\circ$

∴ Answer B

7.Which one of the following statements is true about the motion of a particle in a circular path?

A .The centripetal acceleration is constant if the particle's speed is constant.

B .The tangential acceleration can be perpendicular to the velocity vector of the particle.

C .The centripetal acceleration is always in the direction perpendicular to the velocity vector of the particle.

D .The acceleration is always perpendicular to the velocity vector of the particle.

Solution: If a body moves in a circle of radius R with a tangential velocity V_C (constant in magnitude) its centripetal acceleration a_C is $a_c = \frac{V_t^2}{R}$.

∴ Answer C

8.Two men , A and B , are loading identical boxes onto a truck. A lifts his box straight up from the ground to the bed of the truck, where as B slides his box up a rough ramp to the truck .Which statement is correct about the change in the potential energy on the box – Earth system?

A .A and B change the potential energy by the same amount.

B .A changes more potential energy than B.

C .B changes more potential energy than A .

D .None of those statements is necessarily true because the force of friction and the angle of the incline is unknown.

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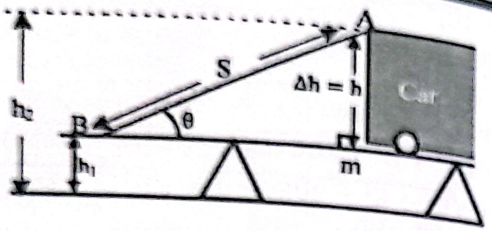
Solution:



Whenever work w is done energy transfers .Amount of energy transferred equals .The work done.

$W_A = F.S$

$W_A = m.g\,(h_2 - h_1) = mg\Delta h = \Delta PE$

$W_B = F.S$

$W_B = mg\sin\theta \times S$ ..... $F = mg\sin\theta$

$W_B = mg\frac{\Delta h}{s} \times s$ ..... $\frac{\Delta h}{s} = \sin\theta$

$W_B = mg\Delta h = \Delta PE$

Therefore , both men change the potential energy by equal amount.

∴ ***Answer A***

9 .A capacitor made of parallel plates is charged by connecting it to a battery. The distance between the plates is much less than the surface area of each plate. After fully charged if the plates are disconnected from the battery and then pulled apart a little, what will happen to the total stored energy , the electric field , and the potential difference between the plates?

A .The electric field between the plates increases.

B .The potential difference between the plates decreases.

C .The total energy stored in the capacitor increases.

D .The electric field, the potential difference and the stored energy remain the same.

Required :Identify the true statement about U_2 , E_2 and V_2 .

Solution: $C_1 = \varepsilon_0 \frac{A}{d_1}$

$U_1 = \frac{1}{2} V_1 Q_1$

$C_2 = \varepsilon_0 \frac{A}{d_2}$ ..... $C_2 < C_1$ and $d_2 > d_1$

$U_2 = \frac{1}{2} V_2 Q_2$ ..... $Q_2 = Q_1$

$Q_1 = C_1 V_1$

$Q_2 = C_2 V_2$

$V_2 C_2 = C_1 V_1$

$V_2 = \frac{C_1 V_1}{C_2}$ ..... $C_2 < C$ and $V_2 > V_1$

$\Rightarrow U_2 > U_1$

$V_2 = E_2 d_2$.....Equation1

$V_1 = E_1 d_1$.....Equation2

Dividing Equation 1 by Equation 2 we get:

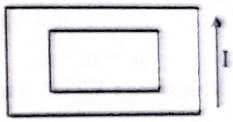
$\frac{E_2 d_2}{E_1 d_1} = \frac{V_2}{V_1}$

$E_2 = \frac{V_2}{V_1} \times \frac{d_1}{d_2} E_1 \Rightarrow V_2 \sim d_2$

$\therefore E_2 = E_1$

∴ ***Answer C***

10. Two rectangular loops of wire Lie in the plane of this paper as shown in the figure. If the current I in the outer loop is counter clock wise and increases with time, Which one of the following statements is correct about the current induced in the inner loop?

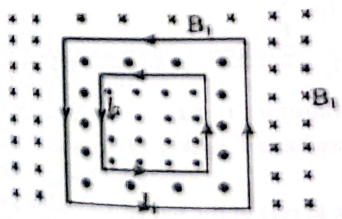


A .A current is induced in the inner loop in the clockwise direction.

B .A current is induced in the inner loop in the counter clock wise direction.

C .The induced current in the inner loop creates a magnetic field inside the area bounded by the inner loop in the same direction of the magnetic field that the outer loop produces in this same area.

D .It is zero.



Solution:

B_1 due to I_1 increases inside the inner loop, it induces current in the inner loop that produce B_2 has opposite direction to B_1 .Using right hand rule , the current is induced in the inner loop in the clock wise direction.

∴ ***Answer A***

11. The vector $2i + j + k$ is perpendicular to $i - 4j + \lambda k$ if λ is equal to

A .0 B .-1 C .2 D .3

Given: Vector $2i + j + k \perp i - 4j + \lambda k$

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$mr_B^2 \frac{v_B}{r_B} = mr^2_A \frac{v_A}{r_A}$ ..... $I = MR^2$

$v_B = \frac{r_A}{r_B} v_A$

$v_B = \frac{r_A}{8r_A} v_A$ ..... $r_B = 8r_A$

$v_B = \frac{1}{8} V_A$

$\therefore v_B = \frac{1}{8} \times 60{,}000 m/s = 7500 m/s$

∴ Answer A

15. A garden hose has an inside diameter of 16cm. The hose can fill a 10 liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm.

A .water exits with a speed twice the speed inside the hose.
B .a 10 litter bucked can be filled in 10 seconds.
C .a 20 litter bucket can be filled in 40 seconds.
D .water exits with a speed half the speed inside the hose.

Given: $r_1 = \frac{16}{2}$ c m $= 8 \times 10^{-2}$ m

$\frac{V}{t} = \frac{10 \text{ Liter}}{20 s} = 0.5 \ell/s$

$r_2 = \frac{d_2}{2} = \frac{8}{2} cm = 4 \times 10^{-2} m$

$\Rightarrow r_1 = 2r_2$

Required: Identify the correct statement

Solution: $\frac{V}{t} = Au = 0.5 \ell/s$

$A_1 = \pi . r_1^2 = \pi(2r_2)^2 = 4\pi r_2^2$

$A_2 = 4\pi r_2^2$

$A_1 = 4A_2$

$A_2 . V_2 = A_1 V_1$

$V_2 = \frac{A_1 V_1}{A_2}$

$V_2 = \frac{4A_2}{A_2} \times V_1$

$V_2 = 4V_1$

$= 0.5 \ell/S . t$

$V = 20\ell$

$\therefore t = 40 s$

∴ Answer C

16. A cup of coffee at a temperature of 80°C is placed on a table in a 20°C room. What happens to the thermal energy and molecular motion of the coffee when it is left on the table?

A .The thermal energy decreases and the molecules move more slowly.
B .The thermal energy increases and the molecules move more slowly.
C .The thermal energy decreases and the molecules move faster.
D .The thermal energy increases and the molecules move faster.

Solution: Heat flows from a body of higher temperature to a body of lower temperature until the bodies are in thermal equilibrium.

$\Rightarrow Q_{lost} = Q_{gained}$

$\Rightarrow$ The coffee decreases thermal energy.

The molecular motion of the particles decreases.

∴ Answer A

17. A mass suspended from the end of a spring vibrates up and down 24 times in 36 S .What are the frequency and period of the vibration?

A .1.5 Hz , 0.67 S
B .0.67 Hz , 1.5 S
C. 0.0012 Hz , 864S
D. 864 Hz , 0.0012S

Given: Nº of cycles completed = 24cycle

time taken t = 36s

Required: frequency (f) = ? and period (T) = ?

Solution: Frequency $= \frac{\text{Number of cycle}}{\text{Period}}$

$f = \frac{24 cycle}{36 s}$

$\therefore f = 0.67 Hz$

$T = \frac{1}{f} = \frac{1}{0.67}$

$\therefore T = 1.5 sec$

∴ Answer B

18 .What is the resistance of a copper cable if it has a cross- sectional area of $1cm^2$ and a length of 2 km? The resistivity of copper is $2 \times 10^{-8} \Omega m$

A .0.0004Ω
B .0.004Ω
C. 0.04Ω
D .0.4Ω

Given: $A = 1 \times (10^{-2} m)^2 = 10^{-4} m^2$ *Required:* R =?

$\ell = 2 \times 10^3 m = 2km$

$\rho = 2 \times 10^{-8} \Omega.m$

Solution: $R = \rho . \frac{\ell}{A}$

$R = \frac{2 \times 10^{-8} \Omega m \times 2 \times 10^3 m}{1 \times 10^{-4} m^2}$

$\therefore R = 0.4 \Omega$

∴ Answer D

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19.A small radio receiver uses a battery that delivers a constant current of 25mA for 3 hours. Calculate the total charge delivered by the battery.

A .1080000C
B .270000C
C. 540 C
D. 270 C

Given: $I = 25\times 10^{-3}A = 25\ mA$

$t = 3h = 3\times 3600s = 10800Sec$

Required : $Q = ?$

Solution: $I = \frac{Q}{t}$

$Q = I \times t$

$Q = 25\times 10^{-3}A \times 10800Sec$

$\therefore$ **$Q = 270\ C$**

$\therefore$ Answer D

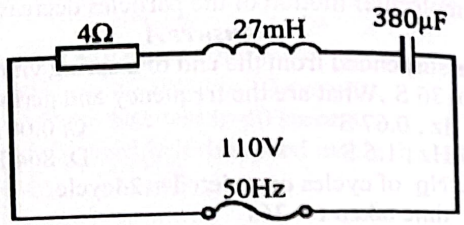
20.Determine the total impedance for the circuit shown below.

A . 0Ω

B .4Ω

C .27Ω

D .31Ω



Given: $R = 4\ \Omega$

$L = 27\times 10^{-3}H = 27\ mH$

$C = 380 \times 10^{-6}F = 380\ \mu F$

$V = 110V$

$f = 50\ Hz$

Required : $Z = ?$

Solution: $X_L = \omega L$

$X_L = 2\pi f_L = 2\pi \times 50Hz \times 27 \times 10^{-3}H$

$X_L = 8478\times 10^{-3} = 8.478\ \Omega$

$\therefore$ **$X_L = 8.5\Omega$**

$X_C = \frac{1}{\omega C}$

$X_C = \frac{1}{2\pi fC}$

$X_C = \frac{1}{2\pi\times 50Hz\times 380\times 10^{-6}F}$

$X_C = \frac{1}{2386.4\times 10^{-4}}$

$X_C = 8.378\ \Omega \approx 8.4\Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$Z = \sqrt{(4\Omega)^2 + (8.5\Omega - 8.4\Omega)^2}$

$Z = \sqrt{17.21}\ \Omega$

$Z = 4.14\Omega$

$\therefore$ Answer B

21. A body moving with constant acceleration covers the distance between two points 60 m apart in 5s.Its velocity as it passes the second point is 15m/s. What is the acceleration?

A .3m/s^2
B .2.4m/s^2
C . 1.8m/s^2
D. 1.2m/s^2

Given:

$S = 60m = S_{BA}$

$t = 5S$

$V_B = 15m/s$

Required: $a = ?$

Solution: $S_{B.A} = V_{B.t} - \frac{1}{2}at^2$

$a = \frac{2(V_{Bt} - S_{BA})}{t^2}$

$a = \frac{2(15\ m/s \times 5s - 60m)}{(5\ s)^2}$

$\therefore$ **$a = 1.2m/s^2$**

$\therefore$ Answer D

22.A placekicker must kick a football from a point which is at a distance of 36.0m from the goal. When kicked , the ball leaves the ground with a speed of 20.0m/s at an angle of 53.0^0 to the horizontal. If the ball hits the crossbar of the goal at a height h and bounces back what will be the height of the crossbar?

A .2.45m
B . 2.85m
C. 3.00m
D. 3.15m

Given: $U = 20m/s$

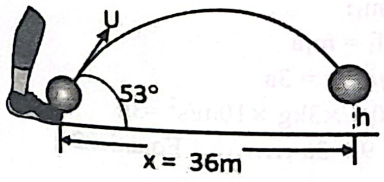
$\theta = 53^0$ with horizontal

$x = 36m$

Required: $h = ?$

Solution: $u_x = u\cos 53^0 = 20m/s \times 0.6 = 12m/s$

$u_y = u\sin 53^0 = 20m/s \times 0.6 = 16m/s$



$h = u_y t - \frac{1}{2}gt^2$

$U_x t = x$.....along x- aixs

$t = \frac{X}{U_X} = \frac{36m}{12m/s} = 3s$

$h = u_y t - \frac{1}{2}gt^2$

$h = 16m/s \times 3s - \frac{1}{2} \times 10m/s^2(3s)^2$

$\therefore$ **$h = 3m$**

$\therefore$ Answer C

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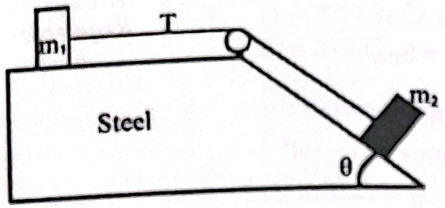
23. An aluminum block of mass $m_1 = 3kg$ and a copper block of mass $m_2 = 5kg$ are connected by a light string over a frictionless pulley. They sit on a steel surface as shown in the Figure ,Where $\theta = 53^0$.The coefficient of kinetic friction between the steel surface and each block is 0.3.If the blocks are released from rest what is the tension T in the string?

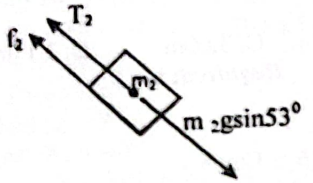
A .29.40 N
B .17.25 N
C .12.25 N
D .8.60 N



Given: $m_1 = 3kg$
$m_2 = 5kg$
$\theta = 53^0$
$\mu_k = 0.3$
$T_1 = T_2 = T$ ..... Frictionless pulley.

Required: T =?

Solution: The free – body diagram for m_2:



$\sum F_y = ma_y$ gives

$m_2 g\sin 53^0 - f_2 - T_2 = m_2 a$

$m_2 g \times 0.8 - \mu m_2 g\cos 53^0 - T = 5a$

$5kg \times 10m/s^2 \times 0.8 - 0.3 \times 5kg \times 10m/s^2 \times 0.6 - T = 5a$

$31 - T = 5a$ Equation1

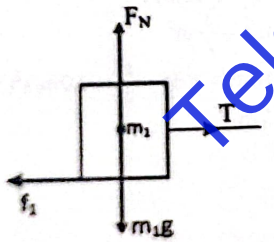
The free – body diagram for m_1:

$T - f_1 = m_1 a$

$T - \mu m_1 g = 3a$

$T - 0.3 \times 3kg \times 10m/s^2 = 3a$

$T - 9 = 3a$ Equation2



Adding Equation1 and Equation2 we get :

$\begin{cases} 31 - T = 5a \\ T - 9 = 3a \end{cases}$

$\therefore a = \frac{11}{4} m/s^2$

$T = 3a + 9$ .....Substituting the value of a in Equation2

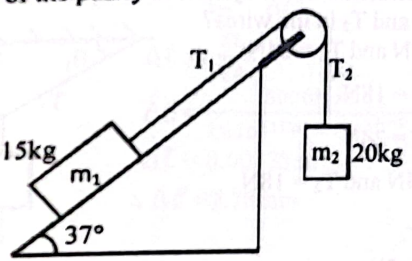
$T = 3 \times \frac{11}{4} N + 9N$

$\therefore T = 17.25 N$

$\therefore$ ***Answer B***

24. As shown in the Figure below ,two blocks are connected by a string of negligible mass passing over a pulley of radius r = 0.2 m and moment of inertia I. The block on the frictionless incline is moving with a constant acceleration $a = 2.0m/s^2$ up along the incline. What is the moment of inertia of the pulley?

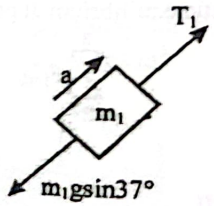
A .0.8kgm²
B .1.25kgm²
C .4kgm²
D .40kgm²



Given: $m_1 = 15$ kg
$\theta = 37^0$
$m_2 = 20kg$
$r = 0.2m$
$a = 2.0m/s^2$

Required: I = ?

Solution: The free – body diagram for m_1 :



$T_1 - m_1 g\sin 37^0 = m_1 a$

$T_1 = 15 \times 10 \times 0.6 + 15 \times 2$

$T_1 = 120$ N Equation1

The free – body diagram for m_2

$m_2 g - T_2 = m_2 a$

$T_2 = m_2 (g - a)$

$T_2 = 20kg\,(10m/s^2 - 2m/s^2)$

$\therefore T_2 = 160$ N

$\tau_{net} = \sum \tau_{cw} + \sum \tau_{ccw}$

$\tau_{net} = T_2 .r - T_1 .r$

$\tau_{net} = 0.2\,(160N - 120\,N)$

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$\therefore \tau_{net} = 8N.m$

$I = \frac{\tau_{net}}{\alpha}$

$I \cdot \alpha = \tau_{net}$

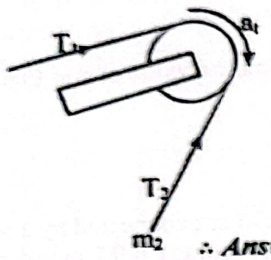
$\alpha = \frac{a_t}{r}$

$\alpha = \frac{2m/s^2}{2\times10^{-1}m}$

$\therefore \alpha = 10rad/s^2$

$I = \frac{8N.m}{10rad/S^2}$

$\therefore I = 0.8kg.m^2$



∴ *Answer A*

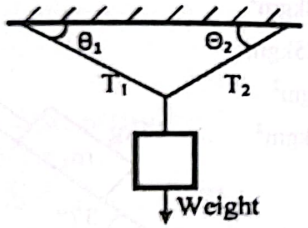
25. An object of mass 3 kg hangs from three light inextensible wires as shown in the Figure. Two of the wires make angle $\theta_1 = 53^0$ and $\theta_2 = 37^0$ with the horizontal. If the system is in equilibrium, what will be the tension T_1 and T_2 in the wires?

A. $T_1 = 18N$ and $T_2 = 24N$

B. $T_1 = T_2 = 18N$

C. $T_1 = T_2 = 24N$

D. $T_1 = 24N$ and $T_2 = 18N$



Given: $m = 3kg$

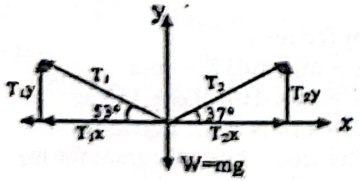
$\theta_1 = 53^0$

$W = mg = 30N$

$\theta_2 = 37$

Required : $T_1 = ?$

$T_2 = ?$



Solution: The system is at static equilibrium at point 0.

$\sum_{i=1}^{3} F_{i=0}$

$\sum F_{ix} = 0$

$\Rightarrow T_2X - T_1X = 0$

$\Rightarrow T_2 \cos 37^0 = T_1 \cos 53^0$

$T_2 = \frac{3}{4}T_1$ .....Equation 1

$\sum F_{iy} = 0 \Rightarrow T_1y + T_2y = W$

$T_1 \sin 53^0 + T_2 \sin 37^0 = W$

$T_1 \times 0.8 + T_2 \times 0.6 = 30N$ Equation2

Substituting Equation 1 in Equation2 we get:

$\therefore T_1 = 24\ N$ and $T_2 = 18N$

∴ Answer D

26. When a load of 500 kg is hanging from a steel wire of length 3m and cross sectional area 0.20 cm^2, the wire stretches beyond its no- load length. If the young's modulus for the wire made of steel is $Y = 2\times10^{11}pa$, what is the change in length of the wire beyond its no - load length?

A. 3.75mm B. 2.75mm C. 1.75mm D. 0.75mm

Given :

$m = 500\ kg$

$W = mg = 5000N$

$\ell = 3m$

$A = 2\times10^{-5}m^2$

$Y = 2\times10^{11}pa$

Required: $\Delta\ell = ?$

Solution : $Y = \frac{\text{stress}}{\text{strain}}$

$Y = \frac{\frac{F}{A}}{\frac{\Delta\ell}{\ell}} = \frac{F\ell}{A\Delta\ell}$

$\Delta\ell = \frac{F\ell}{YA}$

$\Delta\ell = \frac{5000N \times 3\ m}{2\times10^{11}Pa \times 2\times10^{-5}m^2}$

$\Delta\ell = 0.00375\ m$

$\therefore \Delta\ell = 3.75mm$

∴ Answer A

27. A table – tennis ball which is spherical in shape has a diameter of 6 cm and average density of $0.400g/cm^3$. What force is required to hold it completely submerged under water?

A. 0.68 N down ward.

B. 0.68N upward

C. 1.13N down ward

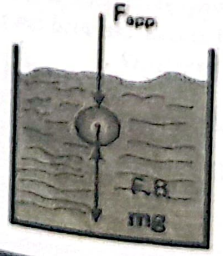
D. 1.13 N upward.

Given : $r = 3 \times 10^{-2}m$

Required : F_{app} =Force applied =?

$\rho_b = 0.4g/cm^3 = 400kg/m^3$

$\rho_w = 1g/cm^3$



Solution : B F − F_{app} − mg = 0

$F_{app} = BF - m_bg$

$= W_{dw} - m_bg$

$= m_{dw} \cdot g - m_bg$

$= V_{dw} \cdot \rho_w \cdot g - v_b \cdot \rho_b \cdot g$

$= v_b\, g\, (\rho_w - \rho_b)$..... $V_{dw} \cong V_b$

$= \frac{4}{3}\pi r_b^3 \cdot g\, (\rho_w - \rho_w)$

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$= \frac{4}{3} \times 3.14\,(3 \times 10^{-2}\text{m})^3 \times 10\text{m/s}^2\,(10^3 - 4 \times 10^2)\text{kg/m}^3$

$\therefore F_{app} = 0.68\ N$

∴ Answer A

28 .A horizontal pipe of 10 cm in diameter has a smooth reduction to a pipe of 5 cm in diameter. If the pressure of the water in the larger pipe is 80 kpa and the pressure in the smaller pipe is 60 kpa, at what rate does the water flow through the smaller pipe?

A .1.63m/ s B .6.53m/s C . 13.6m/s D. 42.67m/s

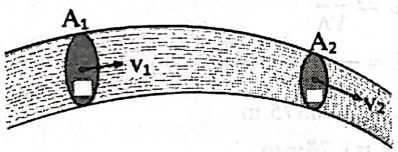
Given: $r_1 = 5 \times 10^{-2}$m ***Required:*** $V_2 = ?$

$p_1 = 80$kpa.

$r_2 = 2.5 \times 10^{-2}$m

$r_1 = 2r_2$

$P_2 = 60$kpa



Solution : $A_2V_2 = A_1V_1$

$A_1 = \pi r_1^2$

$A_2 = \pi r_2^2$

$A_1 = \pi r_1^2$

$A_1 = \pi(2r_2)^2$

$A_1 = 4\pi r_2^2$

$A_1 = 4A_2$

$A_2V_2 = 4A_2V_1 = V_2 = 4V_1$

$P_2 + \frac{1}{2}\rho_W V_2^2 = P_1 + \frac{1}{2}\rho_W V_1^2$

$v_2 = \sqrt{\frac{32}{15 \times \rho_w}\,(p_1 - p_2)}$

$v_2 = \sqrt{\frac{32}{15 \times 10^3\text{kg/m}^3}\,(80 - 60)10^3\text{pa}}$

$\therefore V_2 = 6.52$m/s

∴ Answer B

29 .An object of mass m is hung from a spring and set into oscillation. The frequency of the oscillation is measured and recorded as f .If the object of mass m is replaced with an object of mass 2m and set into oscillation, what will be the frequency of the motion?

A .$f/\sqrt{2}$ B .f C. $(\sqrt{2})f$ D. 2f

Given: $K_1 = k$ ***Required:*** $f_2 = ?$

$k_2 = k_1 = k$

$f_1 = f$

$m_1 = m$

$m_2 = 2m_1 = 2m$

Solution : $T = 2\pi\sqrt{\frac{m}{k}}$

$f_1 = \frac{1}{2\pi}\sqrt{\frac{K_1}{m_1}} = f_1 = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = f$

$f_2 = \frac{1}{2\pi}\sqrt{\frac{k_2}{m_2}} = \frac{1}{2\pi}\sqrt{\frac{k}{2m}}$

$f_2 = \frac{1}{\sqrt{2}}\left(\frac{1}{2\pi}\sqrt{\frac{k}{m}}\right) = \frac{1}{\sqrt{2}}f$ ..... $f = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$

$\therefore f_2 = \frac{f}{\sqrt{2}}$

∴ Answer A

30 .A viewing screen is separated from a double slit by 4.80m. The distance between the two slits is 0.030 mm. Monochromatic light is directed toward the double slit and forms an interference pattern on the screen. The first dark fringe is 4.50 cm from the center line on the screen .What is the wave length of the light?

A .1250nm B .1875nm C . 2500nm D. 5625nm

Given: D =4.8m ***Required:*** $\lambda = ?$

$d = 3 \times 10^{-5}$m

$y_o = 4.5 \times 10^{-2}$m

Solution : From Young's double slit experiment

$y_m = \frac{m\lambda D}{d}$ ..... Maximum and m = 0,1,2...

$y_m = \frac{(m+\frac{1}{2})\lambda D}{d}$ ..... Minimum and m = 0,1,2...

$y_o = \frac{(0+\frac{1}{2})\lambda D}{d}$ ..... 1st minimum (dark)

$\lambda = \frac{2dy_o}{D}$

$\lambda = \frac{2 \times 3 \times 10^{-5}\text{m} \times 4.5 \times 10^{-2}\text{m}}{4.8\text{m}}$

$\therefore \lambda = 562.5$nm

∴ No Answer

31. Wire B has twice the length and twice the radius of wire A .Both wires are made from the same material. If wire A has a resistance R , what is the resistance of wire B?

A .4R B .2R C .R/2 D .R/4

Given: $\rho_A = \rho_B = \rho$ ***Solution:***

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$\ell_B = 2\ell_A$

$r_B = 2r_A$

$A_B = \pi r_B^2$

$A_A = \pi r_A^2$

$A_B = \pi(2r_A)^2$

$A_B = 4A_A$

$R_A = \rho \frac{\ell_A}{A_A}$ Equation1

$R_B = \rho \frac{\ell_B}{A_B}$ Equation2

$R_B = \rho \frac{2\ell_A}{4A_A}$

$R_B = \frac{1}{2}(\rho \frac{\ell_A}{A_A})$

$R_B = \frac{1}{2} R_A$

$\therefore R_B = \frac{1}{2} R$ ..... $R_A = R$

Required: $R_B = ?$

∴ Answer C

32. If the terminals of a battery with zero internal resistance are connected across two identical resistors in series, the total power delivered by the battery is 8W. If the same battery is connected across the same resistors in parallel, what is the total power delivered by the battery?

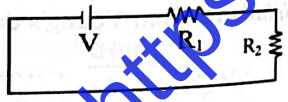
A .2W B .4W C .16W D .32W

Given: $R_1 = R_2 = R$ ***Required:*** $P_2 = ?$

$P_1 = 8W$

Internal resistance = $r = 0\Omega$

Solution: When R_1 and R_2 are connected in Series to the same battery.



$R_T = R_1 + R_2 = R + R$

$R_T = 2R$

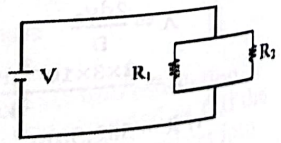
$P_1 = V.I$

$P_1 = \frac{V^2}{R_T} = \frac{V^2}{2R}$ Equation1

When R_1 and R_2 are connected in parallel to the same battery

$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R}$

$R_T = \frac{R}{2}$



$P_2 = \frac{V^2}{R_T} = \frac{V^2}{\frac{R}{2}}$

$P_2 = 2(\frac{V^2}{R})$ Equation2

Dividing Equation2 by Equation1 , we get:

$\frac{P_2}{P_1} = 4$

$P_2 = 4P_1$

$P_2 = 4 \times 8$ W

$\therefore P_2 = 32W$

∴ Answer D

33. A coil formed by wrapping 50 turns of wire in the shape of a square is positioned in a magnetic field so that the normal to the plane of the coil makes an angle of 37^0 with the direction of the field. When the magnetic field is increased uniformly from 200mT to 600 mT in 4.00 s, an emf of magnitude 160.0 mV is induced in the coil. What is the total length of the wire in the coil?

A .100m B . 200 m C. 300m D. 400m

Given : $N = 50$ ***Required :*** Total length = $\ell_T = ?$

$A = \ell \times \ell$

$B_o = 200mT = 0.2T$

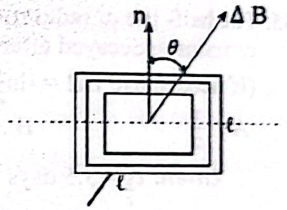
$B = 600mT = 0.6\ T$

$t = 4s$

$\Delta B = 600mT - 200mT = 0.4\ T$

$\varepsilon = 160mV = 0.16\ V$

$\theta = 37^0$



Solution:

The magnitude of induced EMF is obtained from Faradays Law :

$\mathcal{E} = \frac{-N\Delta\Phi}{\Delta t}$

$\varepsilon = \frac{N\Delta(AB\cos 37°)}{t}$ ..... $B = \frac{\Phi}{A}$

$A = \frac{\varepsilon.t}{0.8N\Delta B}$

$\ell^2 = \frac{0.16\ V \times 4S}{0.8 \times 50 \times 0.4T}$ ..... $\frac{0.64}{16}$

$\therefore \ell = 0.2m$

$\ell_T = 50(4\ell)$

$\ell_T = 50\ (4 \times 0.2)m$

$\therefore \ell_T = 40m$

∴ No Answer

34. A 6.00 V battery is connected across the primary coil of a transformer having 50 turns. If the secondary coil of the transformer has 100 turns, what voltage appears across the secondary?

A .24.0 V B .12.0V C .6.00V D .3.00V

Given: $V_1 = 6V$ ***Required:*** $V_2 = ?$

$N_1 = 50$ turns

$N_2 = 100$ turns

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Solution: By step up or step down transformer the ratio of V_2 to V_1 and N_2 to N_1 are equal.

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$$V_2 = \frac{N_2}{N_1} \times V_1$$

$$V_2 = \frac{100}{50} \times 6V$$

$$\therefore V_2 = 12V$$

∴ Answer B

35. The half- life of radium – 224 is 3.5 days. What fraction of the sample remains undecayed after fourteen days?

(If needed use $\ln 2 = -\ln\frac{1}{2} \approx 0.693$)

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{8}$ D. $\frac{1}{16}$

Given: $t_{\frac{1}{2}} = 3.5$ days, $t = 14$ days ***Required:*** $\frac{N}{N_O} = ?$

Solution: λ = decay constant or disintegration

N_o = number of undecayed radioactive nuclei

N = number of decayed radioactive nuclei

$N = N_o e^{-\lambda t}$ ..... Radioactive Formula

The half – life of radioactive substance is the time it takes half of a given number of radioactive nuclei to decay.

Setting $N = \frac{N_o}{2}$ and $t = t_{\frac{1}{2}}$

$\frac{N_o}{2} = N_o e^{-\lambda t_{\frac{1}{2}}}$ ..... Half- life of radioactive

$e^{\lambda t_{\frac{1}{2}}} = 2$ and taking the natural logarithm of both sides, we have : $t_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$

$3.5 \text{ day} = \frac{0.693}{\lambda}$

$\lambda = 0.198$

$N = N_o e^{-\lambda t}$

$\frac{N}{N_O} = e^{-2.772}$

$\therefore \frac{N}{N_O} = \frac{1}{16}$

∴ Answer D

36. Which of the following combinations of properties would be most desirable for a heat sink in a machine?

A .High specific heat and low conductivity.

B .Low specific heat and high conductivity.

C .High specific heat and high conductivity.

D .Low specific heat and low conductivity.

Solution: A heat engine operates between the heat reservoir of higher temperature T_h and cold reservoir (heat sink) at low temperature Tc. The efficiency of the Carnot engine is given by :

$$\eta = \frac{(T_h - T_c)}{T_h} \times 100\%$$

$$\eta = \left(1 - \frac{T_C}{T_h}\right) \times 100\%$$

η is large means $T_h - T_c$ is large or $\frac{T_C}{T_h}$ very small for this to hold the material that is used as heat sink must be high Specific heat capacity and low conductivity like water.

∴ Answer A

37. The change in entropy is zero for:

A .reversible isobaric processes.

B .reversible adiabatic processes.

C .reversible isothermal processes.

D .reversible processes during which no work is done.

Given : $\Delta S = 0$

Required: In which reversible processes ?

Solution: $\frac{\Delta T}{T} = \Delta S$

$\frac{\Delta Q}{T} = 0$

$\Delta Q = 0$

Therefore , during adiabatic processes , no heat is added or removed from the body .i.e adiabatic reversible process.

∴ Answer B

38. Two separate coherent sources produce waves whose wavelengths are 0.10m .Which statement is correct about points that are 3.2 m away from one source?

A .Constructive interference occurs if the points are 3.0 m from the other source.

B .Constructive interference occurs if the points are 3.05m from the other source.

C .Constructive interference occurs if the points are 2.95m from the other source.

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D .Destructive interference occurs if the points are 3.2 m from the other source.

Given: $\lambda = 0.1m$

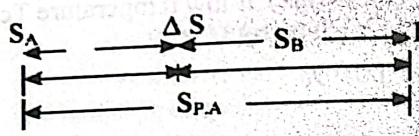
Two coherent sources A and B

$S_B = 3.2m$

Solution:

$\Delta S = m\lambda$ ... Constructive m = 0,1,2 ...

$\Delta S = (m+\frac{1}{2})\lambda$ ..... destructive



$\Delta S = S_B - S_A$

$m\lambda = 3.2m - 3.0\ m$

$\therefore m = \frac{0.2m}{0.1m} = 2$..... constructive

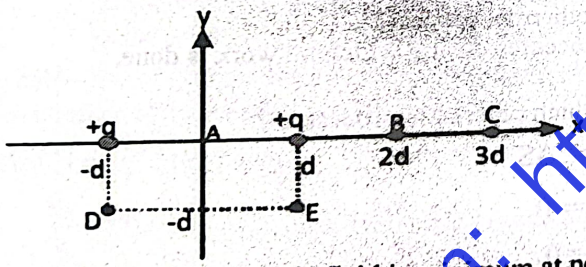
$m\lambda = S_B - S_A$ if $S_A = S_B$

$m = \frac{0}{0.1} = 0$ .....Constructive

$\therefore$ m= 0,1 ,2 ,3 ... Constructive

$\therefore$ Answer A

39 .Two positive charges of magnitude q are each a distance d from the origin A of a coordinate system as shown below.



The magnitude of the electric field is minimum at point

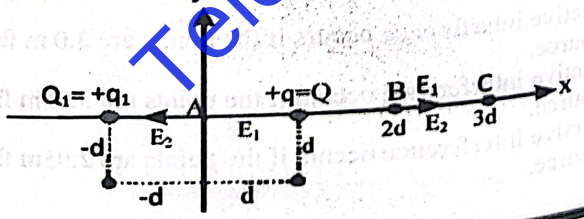
A .A B .B C .C D .D

Given: $q_1 = q_2 = +q$

Located at –d and d from the origin.

Required : Magnitude of the electric field is minimum at point =?

Solution:



$E = \frac{KQ}{r^2}$ ....due to at a point charge

The direction is defined by placing the positive charge at the reference point.

1 .At point A : $E = E_1 - E_2$

$E = \frac{Kq_1}{d^2} - \frac{Kq_1}{d^2}$

$\therefore E = 0$..... $q_1 = q_2$ and $d_1 = d_2$

2 .At point B: $E = E_1 + E_2$

$E = \frac{kq_1}{(3d)^2} + \frac{kq}{d^2} = E$

Therefore , E = 0 at A and is minimum

$\therefore$ Answer A

40.Two particles with masses 2m and 3m are moving toward each other along the x axis with the same initial speeds V. Particle 2m is travelling to the left, and particle 3m is travelling to the right. They undergo an elastic glancing collision such that particle 2m is moving in the negative y direction after the collision. What are the x component of the final velocity of particle 3m and the kinetic energy of particle 2m, respectively?

A .0.33V and 0.7 mV^2 C. 0.67V and 0.70 mV^2

B .0.33V and 1.40 mV^2 D. 0.67 V and 1.4 mV^2

Given: $m_1 = 2m$ ***Required:*** $u_{2x} = ?$

$u_1 = u_{1x} = -v$ $KE_{2f} = ?$

$u_{1Y} = 0$

$m_2 = 3m$

$u_2 = u_{2X} = v$

$u_{2Y} = 0$

$v_1 = v_{1Y}$

$v_{1X} = 0$

Solution: $m_1v_{1x} + m_2v_{2x} = m_1u_{1x} + m_2u_{2x}$

$v_{2x} = \frac{2m(-v)+3mv}{3m}$

$v_{2x} = \frac{1}{3}U$

$m_1v_{1y} + m_2v_{2y} = m_1u_{1y} + m_2u_{2y}$

$2v_{1y} + 3mu_{2y} = 0$

$v_{1y} = -\frac{3}{2}v_{2y} = v_1$

$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2$

$2u_1^2 + 3u_2^2 = 5u^2$

$V_1^2 = \frac{9}{4}v_2^2 - \frac{1}{4}u^2$

$5u^2 = 2(\frac{9}{4}v_2^2 - \frac{1}{4}v^2) + 3v_2^2$

$\frac{11}{2}u^2 = \frac{15}{2}v_2^2$

$V_2^2 = \frac{11}{15}u^2$

$KE_2 = \frac{1}{2}m_2v_2^2$

$KE_2 = \frac{3}{2}m(\frac{11}{15}u^2)$

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$v_2^2 = v_{2x}^2 + v_{2y}^2$

$v_2^2 = (\frac{1}{3}u)^2 + (\frac{2}{3}v_1)^2$

$v_2^2 - \frac{1}{9}u^2{}_2 = \frac{4}{9}v_1^2$

$\therefore KE_2 = 1.1mu^2$

Answer B

41. A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 2000N/m. The spring is compressed 2.0 cm and is then released from rest. At what speed will the block pass through the equilibrium position if a constant friction force of 10.0N retards its motion from the moment it is released?

A .70m/s B. 50m/s C. 70cm/s D.50cm/s

Given: u = 0 m/s ***Required*** :V = ?

m =1.6 kg

K = 2000N/m

x =2× 10^{-2}m = 2cm

F = 10N

Solution: PE =KE +W_F

KE =PE –W_F

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 - f.x$

$V = \sqrt{\frac{kx^2 - 2fx}{m}}$

$V = \sqrt{\frac{2\times10^3 N/m(2\times10^{-2}m)^2 - 2\times10N\times2\times10^{-2}m}{1.6kg}}$

$\therefore V = \frac{1}{2}$ **m/s = 50 cm/s**

∴ Answer D

42. A 10.0 m uniform ladder weighing 200 N rests against a frictionless wall. The ladder makes 53^0 angle with the horizontal. If the ladder is just on the verge of slipping when a 600.0N fire fighter is 6.00 m long the ladder from the bottom, what is the coefficient of static friction between the ladder and ground?

A .0.3240 B. 0.4313 C. 0.5750 D. 0.7667

Given: ℓ = 10m

weighing ladder = W_ℓ = 200N

$\theta = 53^0$

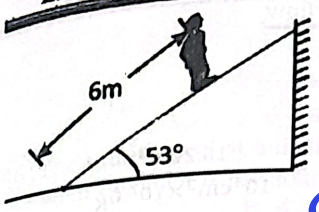
If a 600 N man can Clump the ladder 6m before if slips under him or

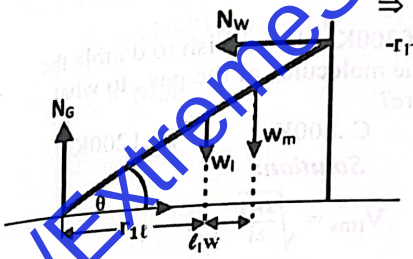
W_m = 600 N.

Required: μ_s = ?

Solution: $\mu_s N_G = f \Rightarrow \mu_s = \frac{f}{N_G}$

$N_G - W_m - W\ell = 0 \Rightarrow N_G = 600N + 200N$





$\therefore N_G = 800$ N

$\frac{r_{1\ell}}{5m} = \cos 53^0 \Rightarrow r_1\ell = 3m$

$\frac{r_{1m}}{6m} = \cos 53^0 \Rightarrow r_{1m} = 3.6m$

$\frac{r_{1Nm}}{10m} = \sin 53^0 \Rightarrow r_{1}Nm = 8m$

$f - N_W = 0 \Rightarrow f = N_W$

$\sum \tau_o = 0$

$\Rightarrow \sum \tau_{cw} + \sum \tau_{ccw} = 0$

$-r_1\ell \times w_\ell - r_{1m} \times W_m + r_{1NG} \times Nw = 0$

$N_w = \frac{r_{1\ell}.w_\ell + r_{1m} \times w_m}{r_{1NG}}$

$N_w = \frac{3m \times 200m + 3.6m \times 600N}{8m}$

$N_w = f = 345$ N

$\mu_S = \frac{f}{N_G}$

$\mu_S = \frac{345N}{800N}$

$\therefore \mu_S = 0.4313$

∴ Answer B

43. A glass flask whose volume is exactly 1000cm^3 at 0°C is totally filled with 1000cm^3 of mercury at this temperature. When flask and mercury are heated to 100°C ,15.2cm^3 of mercury overflow.If the coefficient of volume expansion of mercury is 1.82 ×10^{-4} per Celsius degree, what should be the coefficient of volume expansion of the glass?

A .1.52× $10^{-4}/C^0$ C. 3.00 ×$10^{-5}/C^0$

B .1.82 ×10^{-4} /C^0 D. 1.00× $10^{-5}/C^0$

Given :Volume of mercury = V_m =10^3 cm^3 at T_o = 0°C

Volume of glass = V_g = 10^3 cm^3 at T_o = 0°C

Temperature of mercury = T_{om} =0°C

Temperature of glass = T_{og} = 0°C

Over flow = 15.2 cm^3 at $T_{mercury} = T_{glass}$ = 100°C

$\gamma_m = 1.82\times 10^{-4}$ K^{-1}

$\Delta T_m = \Delta T_g = 100°C - 0°C = 100$ °C

Required: γ_g = ?

Solution: Over flow = $\Delta V_m - \Delta V_g$

$\Delta V_g = \Delta V_m$ – over flow

$\gamma_g V_{og} \Delta T_g = \gamma_m v_{om} \Delta T_m$ – over flow

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$$\gamma_g = \frac{\gamma_m v_{om} \Delta T_m - \text{overflow}}{v_{og} \Delta T_g}$$

$$\gamma_g = \gamma_m - \frac{\text{over flow}}{v_{og} \Delta T_g}$$

$$\gamma_g = 1.82\times 10^{-4}\ K^{-1} - \frac{15.2cm^3}{10^3 cm^3 \times 10^2\ 0_K}$$

$$\therefore \gamma_g = 3\times 10^{-5}\frac{1}{k} = 3\times 10^{-5}/°C$$

∴ ***Answer C***

44. An ideal gas is at a temperature of 300K. . If we wish to double the root mean square (rms) speed of the molecules of the gas , to what value must we raise its temperature?

A .450K B .600K C .800K D .1200K

Given :

$V_{rms} = V_1$ at $T_1 = 300k$

$V_{rms} = 2V_1$

Required: $T_2 = ?$

Solution:

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$V_1 = \sqrt{\frac{3RT_1}{M}} \quad \text{..... Equation 1}$$

$$V_2 = \sqrt{\frac{3RT_2}{M}}$$

$$2V_1 = \sqrt{\frac{3RT_2}{M}} \quad \text{..... Equation 2}$$

Equating Equation 1 and 2 we get

$$2\sqrt{\frac{3RT_1}{M}} = \sqrt{\frac{3RT_2}{M}}$$

$$\Longrightarrow T_2 = 4T_1$$

$$T_2 = 4\times 300K$$

$$\therefore T_2 = 1200K$$

∴ ***Answer D***

45. A heat engine is being designed to have a Carnot efficiency of 65.0% when operating between two energy reservoirs. If the temperature of the cold reservoir is 20.0°C, what must be the temperature of the hot reservoir?

A 57°C B .310°C C .564°C D. 807°C

Given: $\eta = 65\%$

$T_c = 20.0\ °C = 293k$

Required : $T_h = ?$

Solution: $\eta = (1 - \frac{T_C}{T_h}) \times 100\%$

$$0.65 = 1 - \frac{T_c}{T_h}$$

$$\therefore T_h = 837k = 564\ °C$$

∴ ***Answer C***

46. The sound level at a distance of 4.00 m from a source is 120dB .At what distance is the sound level 100 dB?

A .0.40m B. 4.80m C. 40 m D. 48 m

Given: $I_o = 10^{-12} W/m^2$

$\beta_1 = 120dB$ at $r_1 = 4m$

$\beta_2 = 100dB$

Required: $r_2 = ?$

Solution: Ir^2 = constant

$$\beta_1 dB = 10\log_{10}\frac{I}{I_o}$$

$$\frac{I}{I_o} = 10^{\frac{\beta_1}{10}}$$

$$I_1 = I_O \times 10^{\frac{120}{10}}$$

$$I_1 = 10^{-12} w/m^2 \times 10^{12}$$

$$\therefore I_1 = 1W/m^2$$

$$\beta_2 dB = 10\log_{10}\frac{I}{I_o}$$

$$\frac{I}{I_o} = 10^{10}$$

$$I_2 = 10^{-12} w/m^2 \times 10^{10}$$

$$\therefore I_2 = 10^{-2} w/m^2$$

$$I_2 r_2^2 = I_1 r_1^2$$

$$r_2 = \sqrt{\frac{I_1}{I_2}} . r_1$$

$$r_2 = \sqrt{\frac{1\frac{W}{m^2}}{10^{-2}\frac{W}{m^2}}} \times 4m$$

$$\therefore r_2 = \frac{4}{10^{-1}m} = 40m$$

∴ ***Answer C***

47. Suppose that the index of refraction of a soap- bubble film is 1.25 , what is the minimum thickness of the soap – bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is λ= 600nm?

A .120n m B. 150 nm C. 300n m D. 450nm

Given: n = 1.25

$\lambda = 600nm$

Required: thickness = t = ?

Solution: Constructive interference of thin films is given by:

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$$\lambda\,(m+\tfrac{1}{2}) = 2nt \ldots\ldots \; m = 0, 1, 2\ldots$$

$$t = \frac{\lambda}{4n}$$

$$t = \frac{600nm}{4\times 1.25}$$

$$\therefore\ t = 120\ nm$$

Therefore ,the minimum thickness of the film is **120 nm.**

∴ ***Answer A***

48 .Two different capacitors of capacitances $C_1 = 2\mu F$ and $C_2 = 3\mu F$ are connected in series across a 120 V supply line.The charged capacitors are disconnected from the line and from each other, and reconnected with terminals of like sign together. What will be the final charges on C_1 and C_2, respectively?

A . 57.6 μC and 86.4μC

B . 115.2μC and 127.8μC

C .144μC across each

D .288 μC across each

Given : $C_1 = 2\mu F$

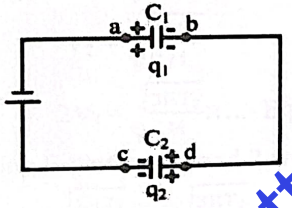
$C_2 = 3\mu F$

V= 120 V

Required: $q_1 = ?$

$q_2 = ?$

Solution: **Series connected:**



$$\frac{1}{C_{T1}} = \frac{1}{C_1} + \frac{1}{C_2}$$

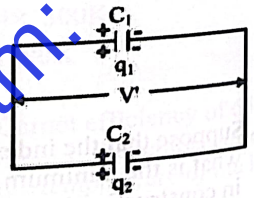
$$\frac{1}{C_T} = \frac{1}{2\mu F} + \frac{1}{3\mu F}$$

$$\therefore\ C_T = \frac{6}{5}\mu F$$

$$q_1 = q_2 = V_T \,.C_T = \frac{6}{5}\mu F \times 120V$$

$$\therefore\ q_1 = q_2 = 144\mu C$$

Parallel connected:



$$V' = V'_1 = V'_2$$

$$V' = \frac{q'_1}{c_1} = \frac{q'_2}{c_2}$$

$$\Rightarrow q'_2 = \frac{c_2}{c_1} q'_1 = \frac{3\mu F}{2\mu F} q'_1$$

$$q'_2 = 1.5 q'_1$$

$$q'_1 + q'_2 = q$$

$$q'_1 + q'_2 = 144\mu c$$

$$q'_1 + 1.5q_1 = 144\mu c$$

$$2.5q_1 = 144\mu c$$

$$q_1 = \frac{144}{2.5}\mu c$$

$$\therefore\ q'_1 = 57.6\mu c$$

$$\therefore\ q'_2 = 86.4\mu c$$

∴ ***Answer A***

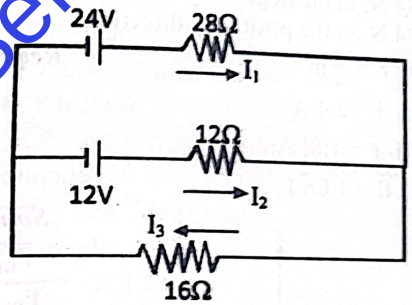
49. In the Figure , the current in the 16 Ω resistor is :

A .0.64A

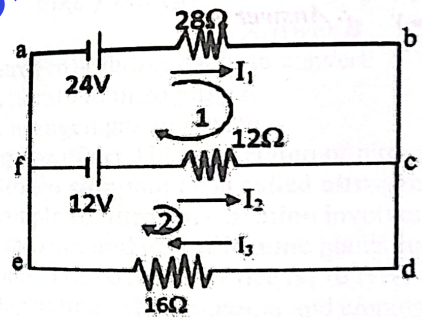
B .0.15 A

C .0.49 A

D .1.28 A



Required : $I_3 = ?$



Solution:

a) At junction C

$I_1 + I_2 = I_3$ Equation1

b) loop abc f a

$24v - 28I_1 + 12I_2 - 12 = 0$

$12 - 28\,I_1 + 12I_2 = 0$ ... Equation2

c) loop fcdef

$12 - 12I_2 + 16I_3 = 0$ Equation3

Substituting Equation1 in Equation2 we get

$12 - 28\,(I_3 - I_2) + 12I_2 = 0$ Equation4

$12 + 40I_2 - 28I_3 = 0$..... Equation5

$12 - 12\,I_2 + 16\,I_3 = 0$..... Equation 6

Solve equation 5and 6 simultaneously we get :

$$\begin{cases} 3 - 3I_2 + 4I_3 = 0 \\ 3 + 10I_2 - 7I_3 = 0 \end{cases}$$

$$\Rightarrow 61I_3 = 39$$

$$\therefore I_3 = 0.639A \approx 0.64\ A$$

∴ ***Answer A***

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50. A wire carries a steady current of 2.4A .A straight section of the wire is 0.75m long and lies in xy –plane along the x axis within a uniform magnetic field. B = 1.6KT. If the current is in the positive x direction, what is the magnetic force on the section of wire?

A .2.88 N , in the negative y direction.
B .2.88 N ,in the positive y direction
C .3.84 N, in the negative y direction.
D .3.84 N ,in the positive y direction.

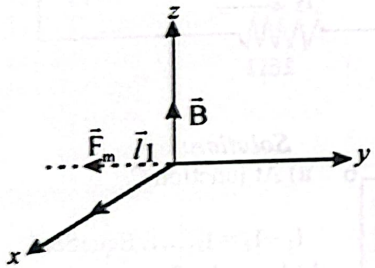
Given: $\ell = \frac{3}{4}$ m

$I = 2.4$ A

$\vec{\ell} = 1.8\hat{i}$ Am

$\vec{B} = 1.6\hat{K}$T.

Required : $\vec{F} = ?$



Solution:

$\overrightarrow{F_m} = I\vec{\ell} \times \vec{B}$

$\overrightarrow{F_m} = 1.8\text{Am} \cdot 1.6\text{T}\,(\hat{i} \times \hat{k})$

$\therefore \overrightarrow{F_m} = -2.88\,\hat{j}$

∴ Answer A

1. What is the balanced equation for the cell notation of the following galvanic cell?

$Pb(s) \mid Pb^{2+}(aq) \parallel Cl_2(g) \mid Cl^-(aq) \mid C(s)$

A . $Pb(s) + 2\,Cl^-(aq) \longrightarrow Pb^{2+}(aq) + Cl_2(g)$

B . $Pb(s) + Cl_2(g) \longrightarrow Pb^{2+}(aq) + 2Cl^-(aq)$

C . $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow Pb(s) + Cl_2(g)$

D . $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow PbCl_2(s)$

Explanation:

i) Oxidation reaction at the anode :

$Pb(s) \longrightarrow Pb^{2+}(aq) + 2e$

ii) Reduction reaction at the cathode:

$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$

iii) Overall reaction (the sum of the half –reactions) is

$Pb(s) + Cl_2(g) \longrightarrow Pb^{2+}(aq) + 2Cl^-(aq)$

∴ Answer B

2. Nitrogen – fixing bacteria convert:

A. nitrate to nitrogen gas
B. nitrogen gas to nitrate
C. nitrate to ammonia
D. ammonia to nitrate

Explanation: The conversion of nitrogen from the free element to nitrogen compounds is called **nitrogen –fixation**. The best known example of nitrogen –fixation involves the Rhizobium bacteria found in the root nodules of legume plants such as peas, clover alfalfa and beans. The bacteria reduce N_2 to NH_3 and NH_4^+ salts at room temperature .The ammonia and ammonium salts are oxidized to nitrate by the action of soil bacteria .

NH_3 or $NH_4^+ \longrightarrow NO_2^- \longrightarrow NO_3^-$

∴ Answer B

3. What is the standard entropy change for the reaction:

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Given the following data : (S^o of $SO_2(g) = 248.1$ J/(mol.K) , S^o of $O_2(g) = 205.0$ J/(mol.K) ; S^o of $SO_3(g) = 256.6$ J/(mol.K)

A .-188.0 J/K
B .+17.0 J/K
C. -196.5 J/K
D. +60.1 J/K

Explanation : The **entropy** change, ΔS , for a chemical reaction is equal to the sum of the entropies of the products minus the sum of the

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entropies of the reactants : $\Delta S_{rxn} =$

$m\sum S_{(products)} - n\sum S_{(reactants)}$

"m" and "n" are coefficients

$\Delta S_{reaction} = 2(256.6) - [2(248.1) + 205]$

$\Delta S_{rxn} = 513.2 - 701.2$

$\therefore \Delta S_{rxn} = -188$ J/K

∴ Answer A

4.How many moles are there in 159 g of alanine , $C_3H_7NO_2$?

A .0.560 B .0.992 C .1.78 D .3.31

Explanation: **The number of moles (n) of a substance** is equal to thegiven mass (m) divided by the molar mass (M) of the substance.

The molar mass of alanine = (3× 12) +(7×1) +14 + (2× 16)

= 89

$$n = \frac{m}{M} \Rightarrow n = \frac{159\,g}{89\,g/mole} = 1.7865 \text{ mole}$$

∴ Answer C

5.Which one of the following is **NOT** true of metallic bonding?

A .It gives rise to excellent electrical conductivity.

B .Electrons are free to move throughout the structure.

C .The strength of metallic bonds increases down a group.

D .The strength of metallic bonding affects the boiling point of metals.

***Explanation*: Metallic bond** is the strong electrostatic force of attraction between positively charged nuclei and the negatively charged valence electrons. Therefore , metallic bonding:

✓ high melting point, boiling point and high density.

✓ the high thermal and electrical conductivity of metals is accounted for by the mobility of the free valence electrons.

∴ Answer C

6.Synthetic fibers are:

A .naturally produced

B .manufactured through the use of chemical substances.

C .produced from animal hair fibers

D. produced from plant substances

***Explanation* : Fibers** are polymers that can be modified and shaped into fine threads or ropes. They can be natural or synthetic.

i) **Natural fibers** : thread or rope like products obtained from naturally existing materials.

Example : cotton fibers.

ii) **Synthetic fibers** :are modifications of the natural fibers and are manufactured through the use of chemical substances.

Examples : Cellulosic fibers , Nylon 6 , Nylon 66 ,Polyester and polyacrylics.

∴ Answer B

7. An amphiprotic species is a molecule or ion that can _____.

A .accept protons.

B .donate more than one protons.

C .both accept and donate protons.

D .be formed into a double ion.

Explanation : An amphiprotic species is a molecule or ion that can either **donate or accept** a proton. For example a water molecule forms

i) H_3O^+ by accepting a proton or

ii) A hydroxide ion (OH^-) by donating a proton

∴ Answer C

8.Which one is true for a triprotic acid ,such as phosphoric acid H_3PO_4?

A .$K_{a1} > K_{a2} > K_{a3}$ C. $K_{a1} > K_{a2} = K_{a3}$

B .$K_{a3} > K_{a2} > K_{a1}$ D. $K_{a1} = K_{a2} = K_{a3}$

Explanation : Phosphoric acid (H_3PO_4) dissociates in 3 steps.

i) $H_3PO_4 + H_2O \rightleftharpoons H_2PO_4^- + H_3O^+$

$$K_{a1} = \frac{[H_2PO_4^-][H_3O^+]}{[H_3PO_4]} = 7.1\times 10^{-3}$$

ii) $H_2PO_4^- + H_2O \rightleftharpoons HPO_4^{2-} + H_3O^+$

$$K_{a2} = \frac{[HPO_4^{2-}][H_3O^+]}{[H_2PO_4^-]} = 6.3\times 10^{-8}$$

iii) $HPO_4^{2-} + H_2O \rightleftharpoons PO_4^{3-} + H_3O^+$

In the first step a proton separates from $H_2PO_4^-$ .In step two a proton separates from an anion of -2 charge HPO_4^{2-} and this is a more difficult separation. Dissociation is even more difficult in the third step.

$\therefore ka1 > k_{a2} > k_{a3}$

∴ Answer A

9.Which group of elements are the most reactive of all the metallic elements?

A .Alkali metals C. Transition metals

B .Alkaline earth metals D. Group 2B Metals.

***Explanation*: Group IA metals** also called the alkali metals constitute the most reactive group of metals. They react with oxygen and water vapor in the air spontaneously. Because of their high reactivity , they are stored under inert substances like kerosene or other liquid alkanes .

∴ Answer A

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10.All of these are characteristics of most ionic compounds in the solid phase **EXCEPT:**

A .high melting point.
B .solubility in water
C. high electrical conductivity.
D. insolubility in organic solvents.

Explanation: The major characteristics of most ionic compounds in the solid state (phase) are listed below:

- ✓ have high melting points.
- ✓ are soluble in polar solvents like water but insoluble in organic solvents like ethanol, hexane and benzene.
- ✓ in the solid phase, they do not conduct electricity. But in the molten form or in aqueous solution ionic compounds can conduct electricity.

∴ Answer C

11. Consider the following :

$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)\ \Delta H = -78.38\ KJ$

What conditions of temperature and pressure will produce the highest yield of NOCl at equilibrium?

A .High temperature ,High pressure
B .High temperature, Low pressure
C .Low temperature ,High pressure
D .Low temperature, Low pressure

Explanation:

i) Since it is an exothermic reaction lowering the temperature gives higher yield of product.

ii) Increasing the pressure shifts the equilibrium position towards the side with fewer number of moles of gaseous reactants or products. This direction is to the right and it produces higher yield of product. Therefore, low temperature and high pressure results in higher yield of product.

∴ Answer C

12.In the process known as "roasting " a(n) _____ is chemically converted to a (n) _____.

A .Carbonate /oxide
B .Oxide /sulfate
C. Sulfide /oxide
D. Phosphate / phosphide

Explanation: **Roasting:** Burning a sulphide ore in air to give the corresponding metal oxide and sulfur dioxide. For example, galena (Pbs) , the sulfide ore from which lead metal is extracted is roasted and reduced to lead metal as follows

i) **Roasting:** $2PbS(s) + 3O_2 \longrightarrow 2PbO(s) + 2SO_2$

ii)**Smelting:** Reduction of the oxide

$PbO + CO \longrightarrow Pb(\ell) + CO_2$

∴ Answer C

13.Which of the following statement is **NOT** true according to 1st law of thermodynamics?

A .Energy can neither be created nor destroyed although it can be changed from one from to another.
B .The total energy of the universe is constant.
C .The change in the internal energy of a closed system is equal to the sum of heat absorbed by the system and work done by the system.
D .The change in internal energy of a system is zero in all the processes.

Explanation: Some points on the first law of thermodynamics .The total energy of the universe is a constant.

$E_{syst} + E_{surr}$ = constant . Implies that energy is neither created nor destroyed.

- ✓ The change in internal energy (ΔE) of a system is equal to the sum of the heat (Q) absorbed and the work (W) done on the system. i.e $\Delta E = Q + W$. This change in internal energy can be zero for some processes but not for all processes.
- ✓ In a closed system , there is no exchange of matter .But there is exchange of energy in the form of heat (Q) or work done (W) .

∴ Answer D

14.How many chloride ions are in 1.0 mole of $CaCl_2$?

A .3.01×10^{23} Cl^- ions
B .6.02×10^{23} Cl^- ions
C. 1.81×10^{24} Cl^- ions
D. 1.20×10^{24} Cl^- ions

Explanation: The dissociation equation for $CaCl_2$ is given as

$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$ .For every mole of calcium chloride, there are two moles of chloride ions. Multiply the number of particles in one mole by two or $2\times 6.02\times 10^{23} = 1.204\times 10^{24}$ chloride ions.

∴ Answer D

15.Which of the following types of solutions are possible?

I . Solid dissolved in a liquid.
II. Gas dissolved in a liquid.
III Gas dissolved in a gas.
IV Solid dissolved in a solid.

A .I and II
B .I, II, III and IV
C. I ,II and IV
D. I

Explanation: List the types of solutions to answer this questions

Types of solutions		***Example:***
❖ Gas –Gas	III	Air ,Natural gas
❖ Gas –Liquid	II	CO_2 in water (Ambo)
❖ Liquid – Liquid		Ethanol in water
❖ Solid – Liquid	I	Sugar in water

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- Liquid – Solid — Dental Amalgum (Hg /Ag)
- Solid –solid — IV — Bronze (Sn in Cu)

All the four types of solutions listed are possible

∴ ***Answer B***

16. What is the normality of 1.0 M Solution of Na_2CO_3?

A .1 N B .0.5N C .2 N D .3N

Explanation:

Normality = Number of equivalent × Molarity

$\Rightarrow N = n \times M$

For Na_2CO_3: $n = 2$

$\therefore N = 2 \times 1.0 = 2.0\ N$

∴ ***Answer C***

17. What is the type of monomer unit in a natural protein polymer?

A .Amino acids C. ∝ - amino acids

B .Essential amino acids D. Nucleic acids

Explanation: The type of monomer units in a natural protein are **∝ - amino acids.**

∴ ***Answer C***

18. Which of the following substances could be amphiprotic?

I .NO_3^- III. NH_3

II.H_2O IV .HCO_3^-

A.I ,II and III C.II and IV

B.I and II D.II , III and IV

Explanation: The species in II, III and IV can accept or donate a proton. Therefore , they are amphiprotic .But NO_3^- does not have a proton to donate but can accept a proton.

∴ ***Answer D***

19. Which one of the following does NOT form hydroxide ions when placed in water?

A .Ionic hydrides C. Nonmetal oxides

B .Ionic metal oxides D. Ionic nitrides.

Explanation :

i) Metal hydrides react with water to yield hydrogen gas

$H^-(aq) + H_2O(\ell) \longrightarrow H_2(g) + OH^-(aq)$

ii) Ionic metal oxides react with water to form hydroxide ions,

$M_2O + H_2O \longrightarrow 2M^+ + 2OH^-$ where M represents a metal.

iii) **Non metal oxides** form acidic solutions not hydroxide ions.

∴ ***Answer C***

20. Which of the following is true for a 0.10M solution of a weak base ,HB?

A . $[H^+] = 0.10$ M C. $[HB] > [H^+]$

B .$[B^-] = 0.10$ M D .PH= 1.0

Explanation : Choices A ,B and D will be true only for a strong acid. Weak acids like HB given here ,dissociate only slightly .The concentration of HB is always greater than $[H^+]$or $[B^-]$. When [HB] = 0.10M , then $[H^+] < 0.1$ and $[B^-] < 0.1$ .Therefore , $[HB] > [H^+]$

∴ ***Answer C***

21. What is the bond order of O_2^+?

A .2 B .2.5 C .1.5 D .3

Explanation: The molecular ion O_2^+ contains 15 electrons. For more than 14 electrons, the following order is used:

$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_x^2 \begin{cases} \pi 2P_y^2 \\ \pi 2P_z^2 \end{cases} \begin{cases} \pi^* 2P_y^1 \\ \pi 2P_z \end{cases}$

Bond order $= \dfrac{\text{Number of electrons in bonding Mo} - \text{Number of electrons in antibonding Mo}}{2}$

$$\text{Bond order} = \frac{10 - 5}{2} = 2.5$$

∴ ***Answer B***

22. Which of the following is characteristic of an antibonding molecular orbital?

A .It is a molecular orbital with a high probability of finding electrons parallel to the region between bonded atoms.

B .It has no electrons

C .It is a molecular orbital with a high probability of finding electrons away from the region between bonded atoms.

D .It is a molecular orbital with a high probability of finding electrons in the region between bonded atoms.

Explanation: **Anti –bonding molecular Orbital's** have a region of zero electron density (anode) between the nuclei. It places high electron charge density away from the region between the nuclei. They posses energy higher than the energy of the starting atomic orbital's

∴ ***Answer C***

23. What is the molarity of a solution obtained by dissolving 0.01 moles of NaCl in 500 mL of solution?

A .0.01M B. 0.005 M C .0.02M D .0.10M

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Explanation : Molarity = $\frac{\text{Number of moles of solute}}{\text{volume of solution}}$

$$C = \frac{n}{v} = \frac{0.01 \text{ moles}}{0.5 \text{ L}} = 0.02 \text{ M}$$

Therefore ,the molarity of the solution is **0.02M.**

∴ Answer C

24. Which of the following does **NOT** affect the rate of a chemical reaction?

A .Enthalpy of the reaction. C. Surface area

B .Concentration of reactants D. Temperature

***Explanation*:** Factors that affect the rate of a reaction .

i) **concentration of reactants** is increasing the concentration of reactants results in higher number of collisions per unit time and the reaction rate is correspondingly higher.

ii) **Temperature :** As the temperature of the reacting species is raised, the average kinetic energy of the reacting particles increases considerably. This causes the particles of the reactants to collide more frequently with greater energy. This results in an increase in reaction rate.

iii) As **the surface area of the reactants** increases (powder form) there is more area of contact between the reactants. The rate increases as the surface area increases , that is , as the size of the particles is minimized.

iv) The **enthalpy of a reaction** is equal to the heat absorbed or given off. It does not affect the rate of a reaction

∴ Answer A

25. Which of the following is **NOT** a valid expression for the rate of the reaction given below?

$4NH_3 + 7O_2 \longrightarrow 4NO_2 + 6H_2O$

A . $-\frac{1}{7}\frac{\Delta[O_2]}{\Delta t}$

B. $\frac{1}{4}\frac{\Delta[NO_2]}{\Delta t}$

C. $-\frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$

D . $-\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t}$

***Explanation*:** For the general reaction $aA + bB \longrightarrow cC + dD$, the rate of the reaction can be written as follows:

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

For the reaction in the question:

$$\text{Rate} = -\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t} = -\frac{1}{7}\frac{\Delta[O_2]}{\Delta t} = \frac{1}{4}\frac{\Delta[NO_2]}{\Delta t} = \frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$$

∴ Answer C

26. Each of the choices below gives a reaction and the corresponding rate law. Of these choices ,which one could be an elementary process or individual step in a chemical reaction?

A .$2A \longrightarrow P$ Rate = k [A]

B .$A + B \longrightarrow P$ Rate = k[A][B]

C .$A + 2B \longrightarrow P$ Rate = k $[A]^2$

D .$A + B + C \longrightarrow P$ Rate = K [A][C]

***Explanation*:** For an elementary process, the order of the reaction is equal to the molecularity of the elementary steps.

∴ Answer B

27. Consider the reaction in which nitricoxide is oxidized to nitrogen dioxide: $2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$

For which the rate law is rate = $k[NO]^2[O_2]$.If this reaction takes place in a sealed vessel and the partial pressure of nitric oxide is doubled ,what effect would this have on the rate of reaction?

A .The reaction rate would increase by a factor of four.

B .The reaction rate would increase by a factor of three.

C .The reaction rate would increase by a factor of eight.

D .The reaction rate would increase by a factor of two.

***Explanation*:** The rate law can be written in terms of partial pressure.

Rate = $K(P_{NO})^2 (P_{O_2})$

Doubling the partial pressure of NO yields the following rate law.

Rate = $K(2P_{NO})^2 (P_{O_2})$

Rate = $4K(P_{NO})^2 (P_{O_2})$

The reaction rate increased by a factor of four.

∴ Answer A

28. Which one of the following will change the value of an equilibrium constant?

A .Changing the temperature.

B .Adding other substances that do not react with any of the species involved in the equilibrium.

C .Varying the initial concentrations of reactants.

D .Varying the initial concentrations of products.

***Explanation*:** Changing the temperature or the form of the equation can change the value of the equilibrium constant.

∴ Answer A

29. The equilibrium constant for reaction (1) is K. What is the equilibrium constant for reaction (2)?

$SO_2(g) + \frac{1}{2}O_2(g) = SO_3(g)$ (1)

$2SO_3(g) = 2SO_2(g) + O_2(g)$ (2)

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A .K^2 B .2K C .$\frac{1}{2}K$ D .$\frac{1}{K^2}$

Explanation : $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$..... K

Reverse this equation

$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$.....K' and $K' = \frac{1}{k}$

Multiply this equation by 2

$2SO_3(g) \longrightarrow 2SO_2(g) + O_2(g)$.....K" and $K'' = \left(\frac{1}{k}\right)^2$

$$K'' = \frac{1}{K^2}$$

∴ Answer D

30.Which of the following are carbohydrates?

I .Nucleic acids IV . Aldoses

II. Polyhydroxyketones V . Triglycerides

III.Disaccharides.

A .II,III and IV C .I,III , and V

B .I and V D .III and IV

Explanation :

i) Carbohydrates are defined as polyhydroxy aldehydes (aldoses) and polyhydroxy ketones (ketoses).The dimers of simple sugars are known as **disaccharides.**

ii) Triglycerides are fats and oils.

iii) **Nucleic acids**: Major constituents of the chromosomes within the cell nucleus and deoxyribose carries the genetic code in chromosomes.

∴ Only II, III, and IV are carbohydrates.

∴ Answer A

31.Which of the following is true concerning this acid –base reaction?

$HS^- + CH_3Cl \longrightarrow CH_3SH + Cl^-$

A .CH_3Cl is a Bronsted –Lowry acid

B .HS^- is a Bronsted – Lowry acid

C .HS^- is a Lewis acid

D .HS^- is a Lewis base

Explanation: A Lewis base is an electron pair donor and HS^- has donated a pair of electrons to the CH_3 which has a vacant orbital after giving up the chloride ion.

∴ Answer D

32. For the stepwise dissociation of aqueous H_3PO_4, which of the following is **NOT** a conjugate acid –base pair?

A .HPO_4^{2-} and PO_4^{3-} C .$H_2PO_4^-$ and HPO_4^{2-}

B .H_3PO_4 and $H_2PO_4^-$ D. H_2PO_4 and PO_4^{3-}

Explanation: An acid and its conjugate base or a base and its conjugate acid differ only by one hydrogen. That is the case in choices A,B and C. But in choice D , the acid has two more hydrogen than the base.

∴ Answer D

33.Which solution below has the highest concentration of hydroxide ions?

A .PH =3.21 C. PH =7.93

B .PH =12.59 D .PH =9.82

Explanation: Acids have PH values of less than 7 .For bases the PH values greater than 7. The higher the PH the stronger the base. Therefore , the solution with the largest PH ,has the highest concentration of hydroxide ions .

∴ Answer B

34.The standard reduction potential of zn^{2+}/ Zn is (-0.76)V and that of Cu^{2+}/Cu is +0.34 V. What would be the electromotive force (emf) of the cell constructed between these two electrodes?

A .1.10V B .0.42V C .-1.10V D .-0.42V

***Explanation*: oxidation – reductions** occur in electrochemical cells. In the zinc – copper cell.

i) **Oxidation** occurs at the anode (at the zinc electrode)

$E^o_{oxidation}$ = 0.76V.In magnitude, this is equal to the standard reduction potential for zinc but opposite in sign,– 0.76 V.

ii) **Reduction** – occurs at the cathode (at the copper electrode). The standard reduction potential for copper is 0.34V.

E^o_{cell} (emf) = oxidation potential + Reduction potential.

$Emf = E^o_{oxid} + E^o_{red}$

$Emf = 0.76\ V + 0.34\ V$

∴Emf = 1.10 V

∴ Answer A

35.At constant T and P ,which one of the following statements about the enthalpy (ΔH) and internal energy (ΔU) changes is correct for the reaction?

$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$

A .$\Delta H = \Delta U$

B .$\Delta H < \Delta U$

C.$\Delta H > \Delta U$

D. ΔH and ΔU are independent of physical states of the reactants.

***Explanation*: The enthalpy change** is equal to the heat absorbed under constant pressure, when the only form of work is pressure – volume work.

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Under conditions of constant T and P volume is also constant. For many processes ΔH and ΔE (ΔU) are roughly the same in magnitude when V is constant .Therefore, change in enthalpy (ΔH) is equal to the change in internal energy (ΔU).

∴ ***Answer A***

36. Which one of the following conditions regarding a chemical process ensures its spontaneity at all temperatures?

A . $\Delta H < 0$, $\Delta S < 0$ C. $\Delta H < 0$, $\Delta S > 0$

B . $\Delta H > 0$, $\Delta S < 0$ D. $\Delta H > 0$, $\Delta S > 0$

Explanation : A chemical reaction is spontaneous whenever $\Delta G < 0$. The condition that makes ΔG negative at all temperatures in the Gibbs free energy equation $\Delta G = \Delta H - T\Delta S$ is only when $\Delta H < 0$ and $\Delta S > 0$.

∴ ***Answer C***

37. Why are silicate minerals **NOT** commonly used as sources of metals?

A .They are rare minerals

B .They usually do not contain important metals.

C .They are difficult to reduce and concentrate.

D .They are only found at excessive depths in the oceans.

Explanation: It is hard to isolate metals from silicate minerals. Because it is highly expensive to reduce and concentrate silicate minerals.

∴ ***Answer C***

38. Which of the following metals is the best conductor of heat and electricity?

A .Copper C. Gold

B .Silver D. Tungsten

Explanation: silver is a soft metal with the highest known electrical and thermal conductivity

∴ ***Answer B***

39. HNO_3 is used as:

I. a hydrogenating agent.

II. an acid

III. an oxidizing agent.

IV . an ammoniating agent.

A .II and III C .I ,II, III and IV

B .I and II D .II, III and IV

Explanation :HNO_3

(i) is a strong acid

(ii) is a powerful oxidizing agent when concentrated.

∴ ***Answer A***

40. Which equation represents an oxidation –reduction reaction ?

A .$H_2SO_4 + 2NH_3 \longrightarrow (NH_4)_2 SO_4$

B .$H_2SO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + H_2O + CO_2$

C .$2K_2CrO_4 + H_2SO_4 \longrightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$

D .$2H_2SO_4 + Cu \longrightarrow CuSO_4 + 2H_2O + SO_2$

Explanation: A reaction such as the one of choice D in which a free element appears is always an **oxidation reduction reaction** .

∴ ***Answer D***

41. Which set of quantum numbers (n, ℓ, m_ℓ, m_s) is **NOT** permitted by the rules of quantum mechanics?

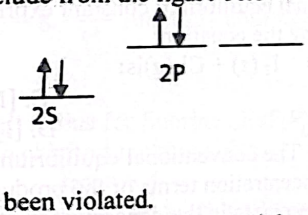
A .1 ,0 ,0 ,$\frac{1}{2}$ C .3, 3, 1, $\frac{-1}{2}$

B .2, 1, 1, $\frac{-1}{2}$ D .4, 3, 2 ,$\frac{1}{2}$

Explanation : Each set of the quantum numbers in choices A,B and D are permissible. But the set in choice C is not permissible. Because "n" and ℓ must not be equal according to the rule of quantum mechanics.The maximum value of ℓ is n -1 and $\ell \neq n$.

∴ ***Answer C***

42. What can you conclude from the figure below?



A .Hund's rule has been violated.

B .The pauli exclusion principle has been violated.

C .The Aufbau principle has been violated.

D .This is a valid orbital diagram

Explanation: According to **Hund's rule** p, d or f orbital's are singly occupied before pairing . This rule is violated by pairing only one of the 2P orbital's in A leaving the other 2p orbital's empty.

∴ ***Answer A***

43. Which set contains only covalently bonded molecules?

A .BCl_3 ,$SiCl_4$, PCl_3 C. I_2 ,H_2S ,NaI

B .NH_4Br ,N_2H_4,HBr D. Al, O_3 ,As_4

Explanation:

i) Only choice A contains covalently bonded molecules.

ii) In choice B, the bond between NH^+_4 and Br^- is **ionic bond**. In the other two molecules the atoms are covalently bonded.

iii) The bond between Na^+ and I^- ions is **ionic** in choice C. H_2S and I_2 are **covalent molecules.**

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iv) The bond in aluminum metal (Al) of choice D is **metallic** .Ozone contains covalent bond.

∴ ***Answer A***

44. How many grams of $Ca(OH)_2$ are contained in 1500 mL of 0.0250 M $Ca(OH)_2$ solution?

A .1.85g B .2.78g C. 3.17 g D. 4.25 g

Explanation :

(i) Calculate the number of moles of Ca $(OH)_2$ present in the solution using the relation : $\mathbf{n = C \times V}$

$n = 0.025\ mol/L \times 1.5\ L$

$\therefore \mathbf{n = 0.0375\ mole}$

ii) Use the relation : $n = \frac{m}{M}$ to convert the number of **moles** to mass(m)

$m = n \times m$

$\Longrightarrow m = 0.0375\ mole \times 74g/mol.$

$\therefore \mathbf{m = 2.775\ g \approx 2.78\ g}$

∴The mass in grams of Ca $(OH)_2$ present in the solution is 2.78g.

∴ ***Answer B***

45.The conventional equilibrium constant expression (K_C) for the system as described by the equation:

$2ICl\ (s) \rightleftharpoons I_2\ (s) + Cl_2(g)$is:

A .$[Cl_2]$ C. $[I_2][Cl_2]/[ICl]^2$

B . $[Cl_2]/[ICl]^2$ D. $[I_2][Cl_2]/2[ICl]$

Explanation: The conventional equilibrium constant expression places the concentration terms of the products in the numerator and those of the reactants in the denominator as in choice C. But the position of such a heterogeneous equilibrium does not depend on the amounts of pure solids or liquids present. Therefore, the amounts of $I_2(S)$ and ICl (S) should not appear in the expression.

∴ ***Answer A***

46.What is the conjugate base of HSO_4^-?

A .H^+ B .H_2SO_4 C .OH^- D .SO_4^{2-}

Explanation: When an acid loses (donates) a proton, it is changed to its conjugate base.

Example : HSO_4^- $\rightleftharpoons$ H^+ + SO_4^{2-}

(Acid) (proton) (Conjugate base)

∴ ***Answer D***

47. An electrochemical cell constructed for the reaction:

$Cu^{2+}(aq) + M(s) \longrightarrow Cu(s) + M^{2+}(aq)$

has an $E^O_{cell} = 0.75$ V .The standard reduction E^O_{red} potential for Cu^{2+} (aq) is 0.34 V. What is the standard reduction potential for M^{2+} (aq)?

A .−1.09V B .−0.410 V C .0.410V D .1.09 V

Explanation:

(i). Cu^{2+} is reduced indicating that copper is the cathode since reduction occurs at the cathode.

$Cu^{2+} + 2e \longrightarrow Cu(S)$

ii) M(S) is oxidized and M is the anode .

Anode reaction : $M(S) \longrightarrow M^{2+}(aq) + 2e^-$

$E^O_{cell} = 0.75$ V, $E^O_{red} = +0.34$ V for Cu.

E^O_{cell} = Oxidation potential + Reduction potential

Substituting the given two quantities in the above equation gives us 0.41 V as the oxidation potential of M. Therefore ,the standard reduction potential for M^{2+} is the opposite of the oxidation potential + 0.41V and that is − **0.41 V.**

∴ ***Answer B***

48. Which of the following compounds would be expected to have the highest melting point?

A .BaF_2 B .$BaCl_2$ C. $BaBr_2$ D. BaI_2

Explanation: The ionic barium halide with the largest lattice energy has the highest melting point .

Columbic force $(F) = K\frac{q^+q^-}{r^2}$ (multiply both sides by r)

$\mathbf{F.\ r = \frac{q^+q^-}{r}}$

Because of the small radius for fluorine ,BaF_2 has the largest lattice energy and hence the highest melting point.

∴ ***Answer A***

49.Which of the following is true about chlorofluorocarbons?

A .React directly with stratospheric ozone to destroy it.

B .Interact with UV energy and become free radicals which destroy ozone.

C .Become free radicals that react with oxygen to create ozone.

D .React with free radicals to remove carbon dioxide.

***Explanation*: Chlorofluoro** carbons undergo photodecomposition with UV energy to release chlorine atoms. Species with unpaired electron like Cl atom is known as **a free radical** .The chlorine atoms readily react with ozone and result in the depletion of the ozone layer. This damages the ozone shield which is the blanket of our planet earth.

∴ ***Answer B***

50.Which metal can be found as the free element?

A .Cr B .Fe C. Mn D. Pt

Explanation: The two triads below the iron triad in the periodic table that include the six metals of the **platinum family** are very noble, that is they are unreactive and occur in nature in the uncombined state.

∴ ***Answer D***

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51. The enthalpy change for which reaction represents the standard enthalpy of formation for hydrogen cyanide, HCN ?

A .H(g) + C(graphite) + N (g) ⟶ HCN (g)

B . $\frac{1}{2}H_2(g)$ + C(graphite) + $\frac{1}{2}N_2(g)$ ⟶ HCN(g)

C .HCN (g) ⟶ $\frac{1}{2}H_2(g)$ + C(graphite) + $\frac{1}{2}N_2(g)$

D .$H_2(g)$ + 2C(graphite) + $N_2(g)$ ⟶ 2HCN(g)

***Explanation*:** The standard enthalpy of formation, ΔH^{O}_{f}, is the change in enthalpy of a reaction when 1.0 mole of substance is formed from the elements in their standard states. The standard enthalpy of formation for hydrogen cyanide, HCN is the enthalpy change when one mole of HCN is formed.

$\frac{1}{2}N_2 + \frac{1}{2}H_2 + C \longrightarrow 1\ HCN$

∴ Answer B

52. Which one of the compounds below is most likely to be ionic?

A.CCl_4 B. NO_2 C .$ScCl_3$ D. ClO_2

***Explanation*:** The compounds of choices A,B and D are all **covalent**. The bond in each one of them is the bond between common non-metals. But for the compound of choice **C** it is a chemical bond between metallic **Scandium (Sc)** and non-metallic **chlorine (Cl)** and it is ionic.

∴ Answer C

53. What is the correct order when the substances O_2, H_2O, OF_2, and H_2O_2 are arranged in order of increasing oxidation number for oxygen?

A .O_2, H_2O, OF_2, H_2O_2 C .H_2O_2, O_2, H_2O, OF_2

B ,H_2O, H_2O_2, O_2, OF_2 D .OF_2, O_2, H_2O_2, H_2O

Explanation : The oxidation number of oxygen in each of the substance mentioned in this question is given as follows:

Substance	H_2O	H_2O_2	O_2	OF_2
Oxidation number of oxygen	-2	-1	0	+2

The order listed in choice B agrees with the table.

∴ Answer B

54. A monoatomic ion that has 20 protons and a+2 charge

A .has 16 protons C. has 18 neutrons

B .has the symbol Ar^{2+} D. is isoelectronic with Ar

Explanation : The atomic number of this particular element is 20 since it contains 20 protons. A neutral atom of this element has 20 electrons. To attain a charge of +2 , it loses 2 electrons and remains

with 18 electrons. This monoatomic ion is iso- electronic with argon (Ar), i.e ,2,8,8

∴ Answer D

55. What is the molality of a solution that contains 51.2 g of naphthalene, $C_{10}H_8$, in 500 mL of carbon tetrachloride? Given the density of CCl_4 = 1.60g/mL

A .0.25m B. 0.50m C. 0.75m D. 0.84m

***Explanation*: Molality** : is the number of moles of solute per kilogram of solvent. It can be calculated for the solution in the question through the following three steps

i) Calculate the number of moles of naphthalene ($C_{10}H_8$) present in 51.2g of naphthalene using the relation $n = \frac{m}{M}$

Molar mass of $C_{10}H_8$ = 120 + 8 = 128 g/mol

$$\Longrightarrow n = \frac{m}{M} = \frac{51.2\ g}{128\ g/mol^{-1}} = 0.4\ mole$$

ii) Calculate the mass of the solvent from the given volume and density.

$$d = \frac{mass}{volume}$$

$$\Longrightarrow mass = volume \times density$$

Mass = 500mL ×1.60g/moL

∴ **Mass = 800g = 0.8 kg**

iii) $$Molality = \frac{number\ of\ moles\ of\ solute}{kilogram\ of\ solvent}$$

$$Molality = \frac{0.4\ mole}{0.8\ kg}$$

∴ **Molality = 0.5 molal**

∴ Answer B

56. A homogeneous liquid reaction mixture is often heated to increase the rate of reaction. This is best explained by the fact that raising the temperature:

A .increases the heat of reaction.

B .decreases the energy of activation.

C .increases the vapor pressure of the liquid.

D .increases the average kinetic energy of the reactants.

***Explanation*:** As the temperature of a homogenous reaction mixture increases,

i) the average kinetic energy of the reactants increases.

ii) the number of collisions per unit time (collision frequency) increases.

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iii) the rate of the reaction increases.

∴ ***Answer D***

57. Which of the following titrations will have an equivalence point at a PH > 7.00?

A .Weak acid with a strong base
B .Strong acid with a strong base.
C .Weak acid with a weak base.
D .Strong acid with a weak base.

Explanation: For acid –base titration , the PH at the equivalence point depends on the relative strength of the acid or the base.
Construct a table for the various combinations.

Acid –base combinations (titrations)	
❖ Strong acid – strong base	PH = 7
❖ Strong acid –weak base	PH < 7
❖ Weak acid – strong base	PH > 7
❖ Weak acid – weak base	PH may be 7 or less or more

For **weak acid –weak base titration** , the position of the equivalence point depends on the degree of hydrolysis of the cation or the anion.

∴ ***Answer A***

58. A 0.10 M solution of a weak acid, HX, is 0.059% ionized. What is the dissociation constant for the acid ?

A.3.8×10^{-9} B.3.5×10^{-8} C.6.5×10^{-7} D.4.2×10^{-6}

Explanation: Calculate the concentration of the ions from per cent ionization. $HX \rightleftharpoons H^+ + X^-$

$$\frac{[X^-]}{[HX]} = 0.059\% = \frac{0.059}{100}$$

$$\frac{[X^-]}{0.1} = \frac{0.059}{100} \Rightarrow [X^-] = \frac{0.059}{100} \times 0.1 = [H^+]$$

$$\Rightarrow [X^-] = 5.9 \times 10^{-5} = [H^+]$$

$$K_a = \frac{[H^+][X^-]}{[HX]}$$

$$K_a = \frac{(5.9\times10^{-5})\times(5.9\times10^{-5})}{0.1}$$

$$\therefore K_a = 3.5\times 10^{-8}$$

∴ ***Answer B***

59. Consider the standard voltaic (or galvanic) cell: Fe, Fe^{2+} versus Au, Au^{3+} .Which answer identifies the cathode and gives the E°_{red} for the cell? Given $E^\circ (Au^{3+}/Au) = 1.50V, E^\circ(Fe^{2+}/Fe) = -0.44V$.

A. Fe, -0.44V B. Au, 1.06V C. Au,1.94V D.Fe,1.94V

Explanation : In a galvanic cell, the species with a more positive standard reduction potential is the cathode and it is Au in this case. The more active metal iron (Fe) in this case serves as the anode and oxidation occurs here.

E^O_{cell} = Oxidation potential + Reduction potential

E^O_{cell} = 0.44V + 1.50 V = 1.94 V

∴ E°_{cell} = **1.94 V** and Au is the cathode since it is an inert (**unreactive**) metal with a relatively more positive standard reduction potential .

∴ ***Answer C***

60. What is the enthalpy for the reaction ,

$C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$?

Given the following information:

$2C(s) + 3H_2(g) \longrightarrow C_2H_6(g)$ $\Delta Hf^\circ = -84.7 KJ.mol^{-1}$

$C(s) + O_2(g) \longrightarrow CO_2(g)$ $\Delta Hf^\circ = -393.5 KJ.mol^{-1}$

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$ $\Delta Hf^\circ = -285.8 KJ.mol^{-1}$

A.−764 KJ B. −1560 KJ C. −1664 KJ D. −3120 KJ

Explanation : The **enthalpy of a reaction** is equal to the sum of the enthalpies of formation of products minus the sum of the enthalpies of formation of reactants.

$\Delta H^O_{rxn} = \sum n\Delta H_f^\circ (prod) - \sum m\Delta H_f^O (react)$, where m and n are the coefficient in the balanced chemical equation.

$\Delta H^O_{rxn} = 3(-285.8) + 2(-393.5) - (-84.7 + \frac{7}{2} \times 0)$

$\Delta H^O_{rxn} = -857.4 - 787 + 84.7 + 0$

$\Delta H^O_{rxn} = -1559.7 KJ \approx -1560 KJ$

∴ ***Answer B***

61. How many atoms are present in 64 g of SO_2 ?

A.6.02×10^{23} B.$2\times 6.02\times 10^{23}$ C.$3\times 6.02\times 10^{23}$ D.$4\times 6.02\times 10^{23}$

Explanation:

i) In one mole of a substance, there is an Avogadro number of particles.
ii) 64g of SO_2 is 1.0 mole of SO_2 .In one mole of SO_2 , there is 1.0 mole of S atoms and 2 moles of oxygen atoms.
iii) In 64g of SO_2 there are 3.0 moles of atoms.
iv) The number of atoms present in 64g of SO_2 is $3.0\times 6.02\times 10^{23}$ atoms

∴ ***Answer C***

62. What is the correct order of the atomic radius in the elements below ?

A Mg <Al < Cl C. Mg > Al > Cl
B. Al < Mg < Cl D. Cl >Al > Mg

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Explanation : **Mg , Al and Cl** are elements in the third period of the periodic table. In moving across the period from left to right that is from Mg to Cl , the atomic radius decreases. Mg has the largest atomic radius and Cl the smallest .

∴ **Mg > Al > Cl or Cl < Al < Mg**

∴ ***Answer C***

63. The outermost electron(s) of which of the following experiences is the greatest effective nuclear charge ?

A.K B.Ca C.Br D.P

Explanation : **Effective nuclear charge** is equal to the total nuclear charge minus the number of inner electrons that contribute to the shielding effect.

Element	Z	Number of inner electrons	Effective nuclear charge
K	19	18	+1
Ca	20	18	+2
Br	35	28	+7
P	15	10	+5

∴ ***Answer C***

64. Which of the following is **NOT** probable shape of a molecule ?

A.A tetrahedral $GeCl_4$ C.A trigonal bipyramidal PCl_5

B.A bent HCN D.A T shaped BrF_3

Explanation:

Molecule	No of sets	Bond pairs	Lone pairs	Shape
$GeCl_4$	4	4	0	Tetrahedral
HCN	2	4	0	Linear
PCl_5	5	5	0	Trigonal bipyramid
BrF_3	5	3	2	T -shape

∴ ***Answer B***

65. Four elements A ,B ,C ,D have atomic numbers Z−1 ,Z ,Z+1 and Z+ 2 respectively. If Z is 9 , then the bond between which pair of elements will be ionic ?

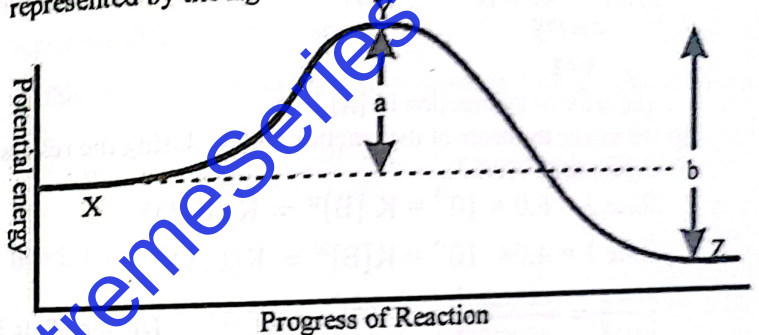
A.A and C B. C and D C. B and D D. B and C

Explanation:

Elements	A	B	C	D
Atomic Number	8	9	10	11
	Z−1	Z	Z+1	Z+2

A and D or B and D can form **ionic bond**. The ionic bond between B (halogen) and D (an alkali metal) is stronger.

66. Which of the following statements is INCORRECT for the reaction represented by the figure below ?



A. Y represents the transition state of the reaction.

B. The enthalpy of the reaction is given by b – a.

C. The forward reaction is endothermic.

D. b is the activation energy of the forward reaction.

Explanation:

i) **Y** : represents the transition state or the activated complex.

ii) **ΔH +a = b** .This indicates that the enthalpy change ΔH for the reaction is equal to **b – a** .

iii) **a** is the activation energy for the forward reaction and b is the activation energy for the reverse reaction.

iv) Since the energy of the product Z is less than the energy content of the reactant X, the forward reaction is exothermic and the reverse reaction is **endothermic.**

∴ ***Answer C and D***

67. Given the initial - rate data below for the reaction :

$aA + bB \longrightarrow$ Products, what is the correct rate law ?

Experiment	Concentration of A(M)	Concentration of B(M)	Initial Rate M/s
1	0.100	0.100	4.0×10^{-5}
2	0.200	0.100	1.6×10^{-4}
3	0.100	0.200	8.0×10^{-5}

A. Rate $= K[A][B]$ C. Rate $= K[A]^2[B]^2$

B. Rate $= K[A]^2[B]$ D. Rate $= K[A][B]^2$

Explanation :

i) To determine the order in [A] use the values of experiments 1 and 2

Rate 2 $= 1.6\times10^{-4} = k[A]^x = K(0.2)^x$

Rate 1 $= 4.0\times10^{-5} = k[A]^x = K(0.1)^x$

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$\frac{\text{Rate 2}}{\text{Rate 1}} = \frac{1.6 \times 10^{-4}}{4.0 \times 10^{-5}} = \left(\frac{0.2}{0.1}\right)^x$ .....Dividing Rate 2 by Rate1

$4 = (2)^x$

$\therefore x = 2$

The order of the reaction in [A] is 2 .

ii) Determine the order of the reaction in [B] .Using the results of experiments 1 and 3 .

Rate 3 $= 8.0 \times 10^{-5} = K\,[B]^x = K\,(0.2)^x$

Rate 1 $= 4.0 \times 10^{-5} = K[B]^X = K(0.1)^X$

$\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{8.0 \times 10^{-5}}{4.0 \times 10^{-5}} = \frac{k(0.2)^x}{k(0.1)^x} = \left(\frac{0.2}{0.1}\right)^x$ .. Dividing Rate 3 by Rate1

$2 = 2^x$

$\therefore$ **x= 1**

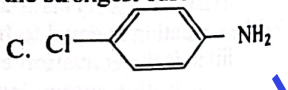
The order of the reaction in [B] is equal to one , that is , first –order

∴ The rate law for this reaction will be **Rate = K[A]^2[B].**

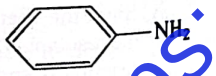
∴ Answer B

68.Which of the following compounds is the strongest base ?

A.NH_2Cl

C. 

B.NH_3

D. 

***Explanation*:**

i) **Alkyl or Aryl amines** are generally slightly stronger bases than ammonia.

ii) The substitution of electronegative atom like Cl for hydrogen as in choice C withdraws electrons from the amine group making the base Stronger.

∴ Answer C

69.For preparing a buffer solution of PH 6.0 by mixing sodium acetate and acetic acid , what would be the ratio of the concentration of salt and acid ($K_a = 10^{-5}$) ?

A.1:10 B.10:1 C.100:1 D.1:100

***Explanation*: Buffers** as solutions that resist a change in PH upon addition of a small amount of an acid or a base. **Buffered solutions** contain both members of a conjugate weak acid-base pair. The PH of

a buffer solution can be calculated using the Henderson –Hassel Balch equation: $PH = P^{K_a} + \log \frac{[Base]}{[Acid]}$

In this buffer question the acid is acetic acid and the base is acetate ion.

$PH = P^{K_a} + \log \frac{[CH_3COO\,Na]}{[CH_3COOH]}$ .....PH = 6 and $P^{K_a} = 5$

$6 = 5 + \log \frac{[CH_3cooNa]}{[CH_3COOH]}$

$\Rightarrow 6-5 = \log \frac{[CH_3COONa]}{[CH_3COOH]} = 1 \Longrightarrow \frac{[CH_3COO\,Na]}{[CH_3COOH]} = 10^1 = 10 = \frac{10}{1}$

The ratio of the concentration of the salt to the acid is 10:1

∴ Answer B

70. Given the following data at 25°C:

$2NOCl(g) \longrightarrow 2NO(g) + Cl_2(g)$ $\Delta H^O = +75$ KJ

$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$ $\Delta H^O = -113.05$ KJ

$2NO_2(g) \longrightarrow N_2O_4(g)$ $\Delta H^O = -58.03$ KJ

Calculate ΔH^Oat 25°C for the reaction:

$N_2O_4(g) + Cl_2(g) \longrightarrow 2NOCl(g) + O_2(g)$

A .+96.08 KJ C. 186.8KJ

B .+299 KJ D .85.52 KJ

***Explanation*: Hess's Law** is applied to solve this problem. If different processes are used to bring about the same change, the changes in enthalpies during these processes are the same. To come to the target equation, reverse all the three equations and change the sign of ΔH. Adding the reversed equations gives the overall target equation.

$\Delta H_{rxn} = \Delta H_1 + \Delta H_2 + \Delta H_3$

$\Delta H_{rxn} = -75 + 113.05 + 58.03$

$\therefore \Delta H_{rxn} = 96.08$ **KJ**

∴ Answer A

71. How many moles of $KBrO_3$ are required to prepare 0.0700 moles of Br_2 according to the reaction:

$KBrO_3 + KBr + HNO_3 \longrightarrow KNO_3 + Br_2 + H_2O$

A .0.0233 C .0.210

B .0.0732 D .0.220

***Explanation*: First** of all balance the equation using the change in oxidation number. **Bromine** is oxidized and reduced in this reaction.

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Such a reaction in which the same element is oxidized and reduced is known as **disproportionation reaction.**

$$\overset{+5}{KBrO_3} + \overset{-1}{KBr} + HNO_3 \longrightarrow KNO_3 + \overset{0}{Br_2} + H_2O$$

Reduced by gaining 5 electrons

Oxidized by losing 1 electrons

✓ Multiply the oxidation half- reaction by 5 to equalize the number of electrons **lost and gained.** Balance the remaining atoms by inspection.

$KBrO_3 + 5KBr + 6HNO_3 \longrightarrow 6KNO_3 + 3Br_2 + 3H_2O$

1mole $KBrO_3$ .....Produces 3 mole Br_2

X mole $KBrO_3$.....produces 0.07 mole Br_2

$\frac{x}{1} = \frac{0.07}{3} \Rightarrow \mathbf{x = 0.0233\ mole}$

∴ 0.0233 moles of $KBrO_3$ is required to produce 0.0700 moles of Br_2

∴ Answer A

72. A possible mechanism for the reaction , $2A + B \longrightarrow C + D$,is:

1 . $A + A \rightleftharpoons A_2$ fast ,equilibrium

2 .$A_2 + A \longrightarrow A_3$ Slow

3 .$A_3 + B \longrightarrow A + C + D$ fast

According to the mechanism , the rate law will be:

A . Rate = k [A] C. Rate =K $[A]^3$

B .Rate = k $[A]^2$ D . Rate = k $[A]^2$ [B]

Explanation : The slowest step in a mechanism is the rate determining step. Therefore, the rate law for the overall reaction must be the same as the rate law for step (2) , that is , rate = K $[A_2]$ [A]

This rate law contains the concentration of the intermediate $[A_2]$ and it must be removed from the rate law expression by applying the following assumption. The rates of the forward and backward reactions in the first step are equal. $A + A \underset{K_b}{\overset{K_f}{\rightleftharpoons}} A_2$

K_f and K_b are the rate constants for the forward and reverse reactions respectively .i.e Forward rate = Reverse rate.

$K_f [A][A] = K_b [A_2]$

$[A_2] = \frac{K_f}{k_b} [A][A]$

Substituting the quantity on the right hand side for $[A_2]$ in the rate determining step give the rate law **Rate = $K[A]^3$**

∴ Answer C

73. What is the $[H_3O^+]$ in a 0.050M solution of Ba $(OH)_2$?

A .1.0×10^{-13}M C .1.0×10^{-5}M

B .5.0×10^{-10}M D .5.0×10^{-2}M

***Explanation*:** Assume the complete dissociation of Ba $(OH)_2$

$Ba(OH)_2\ (aq) \longrightarrow Ba^{2+}(aq) + 2OH^-(aq)$

0.05 mol/L 0.05mol/L + 2 (0.05) mol/L= (0.1 mol/L)

The $[H_3O^+]$ can be calculated using the ion product of water , Kw.

$Kw = [H_3O^+][OH^-]$

$$[H_3O^+] = \frac{kw}{[OH^-]} = \frac{1.0\times10^{-14}}{0.1} = 1.0\times10^{-13}$$

∴ Answer A

74. Which species has the largest F—A –F bond angle where A is the central atom?

A .BF_3 B .CF_4 C .NF_3 D .OF_2

***Explanation*:** Using the following table Simplifies the solution:

Molecule	Number of electron sets	Number of bonding pairs	Number of lone pairs	Bond angle
BF_3	3	3	0	120^0
CF_4	4	4	0	109.5^0
NF_3	4	3	1	107^0
OF_2	4	2	2	104.5^0

✓ In NF_3 , the **bonding pair – lone pair** repulsion reduces the bond angle from 109.5^0 to 107^0 .

✓ The **bonding pair –bonding pair** repulsion in OF_2 is even stronger and has the smallest bond angle.**BF_3** has the largest bond angle.

∴ Answer A

75. When the atoms Li, Be, B, Na are arranged in order of increasing atomic radius, which is the correct order?

A .Li , Be ,B ,Na C. Na ,Li , Be , B

B .Li ,Na ,B ,Be D. B, Be ,Li ,Na

***Explanation*:** The order of **increasing atomic radius** can be deduced from the following periodic property generalization. Trend in atomic radius

Decrease →

Increase ↓

	Be	B	C
Li			
Na			
K			

For the atoms mentioned in the questions, **Na** has the largest and **B** the smallest atomic radius. Therefore ,the order is , **B, Be , Li , Na.**

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∴ *Answer D*

76. One of the steps in the manufacture of nitric acid is the oxidation of ammonia shown in this equation:

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$$

If gaseous water appears at a rate of 0.025 $mol.min^{-1}$, at what rate does ammonia disappear?

A .0.0040 $mol.min^{-1}$
B .0.017$mol.min^{-1}$
C. 0.038 $mol.min^{-1}$
D. 0.150$mol.min^{-1}$

Explanation: $\text{Rate} = +\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t} = +\frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$

$$\frac{\Delta[NH_3]}{\Delta t} = \frac{4}{6}(0.025)\frac{mol}{min} = \mathbf{0.017}\frac{mol}{min}$$

∴ *Answer B*

77. What is the half –life of the irreversible first order reaction, A ⟶ B if 75% of A is converted to B in 60 minutes?

A .30 minutes
B . 45 minutes
C. 60 minutes
D. 80 minutes

Explanation: If 75% of the reactant is used up in 60 minutes, 25 % remains unreacted.

$$\frac{\text{Amount that remained [A]}}{\text{The initial amount } [A]_o} = 25\%$$

$$\frac{[A]}{[A]_o} = \frac{25}{100} \Rightarrow \frac{[A]_o}{[A]} = \frac{100}{25} = 4$$

The integrated rate law for a first –order reaction

$$\text{Log}\frac{[A]_o}{[A]} = \frac{kt}{2.303} = \log 4 \text{ since } 4 = \frac{[A]_o}{[A]}$$

$$\frac{kt}{2.303} = \log 4$$

$$Kt = \log 4 \times 2.303$$

$$K = \frac{\log 4 \times 2.303}{60\ min} = \frac{0.602 \times 2.303}{60\ min} = \frac{0.0231}{min}$$

$$\therefore K = 0.0231\ min^{-1}$$

For an irreversible first order reaction, the half –life ($t_{½} = \frac{0.693}{K}$). Calculated k = 0.0231 min^{-1} substituting the value of the rate constant K in this equation gives half- life of 30 minutes.

∴ *Answer A*

78. At a certain temperature for the reaction:

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ the equilibrium concentrations are: [NO] = 0.52 M ; $[O_2]$ = 0.24 M; $[NO_2]$ = 0.18 M. What is the value of K_C at this temperature?

A .0.063
B .0.50
C.1.4
D .2.0

Explanation : $K_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.18)^2}{(0.52)^2(0.24)} = 0.499 \approx 0.5$

$\therefore K_C = 0.5$

∴ *Answer B*

79. A 25.00 mL sample of 0.1050 M H_2SO_4 is titrated with a NaOH Solution of unknown concentration. The phenolphthalein endpoint was reached when 17.23 mL of the NaOH Solution had been added. What is the concentration of the NaOH?

A .0.07617M
B .0.1447M
C .0.1524M
D .0.3047M

Explanation: Convert the molarity of H_2SO_4 using the relation

$N = n \times M$:Where n is the number of equivalents of the acid Per mole and it is equal to 2 for H_2SO_4

$N = 2 \times 0.105M = \mathbf{0.21\ N\ for\ H_2SO_4}$.

At the equivalent point, the number of equivalents of the acid (N_AV_A) is equal to the number of equivalents of the base (N_BV_B).

That is :$N_BV_B = N_AV_A$

$$\Rightarrow N_B = \frac{N_AV_A}{V_B}$$

$$\Rightarrow N_B = \frac{0.21N \times 25mL}{17.23\ mL} = \frac{5.25}{17.23}\ N = 0.3047N$$

For NaOH the normality and molarity is the same.0.3047 N NaOH = 0.3047 M NaOH

∴ *Answer D*

80. During the electrolysis of an aqueous solution of $AgNO_3$, what would happen to the mass of silver metal deposited if the current is doubled and the electrolysis time is decreased to $\frac{1}{2}$of its initial value?

A .It would stay the same.
B .It would increase to twice its initial value.
C .It would decrease to $\frac{1}{4}$ of its initial value.
D .It would decrease to $\frac{1}{2}$ of its initial value.

∴ *Answer A*

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1.Which of the following is **NOT** a state function?

A .Work C. Internal energy

B .Enthalpy D. Pressure

Explanation :

i) **State function** is the property of the state of a system that depends on the initial and final states and is independent of the path way leading from one state to the other.

Example : **Potential energy , Internal energy , Entropy, Enthalpy , volume , pressure**

ii. **Path functions:** Are quantities that depend on the path way or on how the state is prepared.

Heat (q) and work (w) depend on the path way between the system and its surroundings.

Heat and work are therefore path functions but the change in the internal energy ΔE depends on the initial and final states and not on the path.

Example : Heat ,Work

∴ Answer A

2.Which of the following metals has the highest positive oxidation state?

A .Ti B. Cr C. Mn D. Cu

Explanation :The maximum oxidation number in most cases is equal to the group number.

Element	Group number	Maximum oxidation number
Ti	4B	+4
Cr	6B	+6
Mn	7B	+7
Cu	IB	+2

∴ Answer C

3.Which of the following is **NOT** a natural polymer?

A .Proteins B. Starch C. Cellulose D. Teflon

Explanation : **Starch and cellulose** are naturally occurring carbohydrate polymers. **Proteins** are also natural polymers. **Teflon** is trade name for the synthetic polymer **polytetrafluoroethylene (PTFE).**

∴ Answer D

4.The coefficients required to correctly balance the reaction below are:

$$Al(NO_3)_3 + Na_2S \longrightarrow Al_2S_3 + NaNO_3$$

A .2,1,3,2 B .2,3,1,6 C .1,1,1,1 D .4,6,3,2

Explanation : **The LCM** method is convenient to balance double displacement reactions. Place the total valence of each compound above its formula.

$$\overset{3}{Al(NO_3)_3} + \overset{2}{Na_2S} \longrightarrow \overset{6}{Al_2S_3} + \overset{1}{NaNO_3}$$

The LCM of 1,2,3 and 6 is 6. Divide the LCM by each valency to get the coefficients. The coefficients are thus $\frac{6}{3}$, $\frac{6}{2}$, $\frac{6}{6}$ and $\frac{6}{1}$, that is **2,3,1 and 6.**

∴ Answer B

5.One mole of $(NH_4)_2HPO_4$ contains _____ moles of hydrogen atoms.

A .One B. Five C. Six D. Nine

Explanation : One formula unit of the compound $(NH_4)_2HPO_4$ contains **9 H –atoms.** One mole of the compound contains 9.0 moles of hydrogen atoms.

∴ Answer D

6 .A 500 g toothpaste sample has 0.2 g fluoride concentration .What is the concentration of fluoride in terms of ppm ?

A .200 B .250 C .400 D .1000

Explanation : **PPm** means parts per million. Determine the number of parts of fluoride ions in one million parts of the toothpaste.

0.2 g in 500g.

Xg in 10^6 g and determine X .

$$\frac{x}{0.2} = \frac{10^6 g}{500g} \Longrightarrow \mathbf{X = 400ppm}$$

In one million parts of the toothpastes , 400 parts is the fluoride ion.

∴ Answer C

7. Which solution below has the highest concentration of hydroxide ion?

A .PH = 3.21 C. pH = 9.82

B .pH = 7.93 D. pH = 12.59

Explanation : Bases have pH greater than 7.The higher the pH , the more basic is the solution and the higher is the hydroxide ion concentration. The solution with the highest pH , has the largest concentration of hydroxide ion.

∴ Answer D

8. What substance is reduced in the following reaction?

$$Cr_2O_7^{2-} + 6S_2O_3^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 3S_4O_6^{2-} + 7H_2O$$

A .$Cr_2O_7^{2-}$ B .$6S_2O_3^{2-}$ C .H^+ D. Cr^{3+}

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Explanation : In a redox reaction, the species whose oxidation number has decreased is the substance that is reduced. In the reaction above, the oxidation number of Cr has decreased from +6 in $Cr_2O_7^{2-}$ to +3 in Cr^{3+}

∴ Answer A

9 .What percent is gold in 18 karat gold?

A .18 B .89 C .25 D .75

Explanation : The quality of gold is defined by the number of parts of gold in 24 parts of gold alloy. The standard for this part is given in terms of the unit karat. Thus 24 karat gold is pure gold. 18karat gold has 18 parts gold and 6 parts other metals like copper. 21 karat gold contains 21 parts gold and 3 parts other metals. The percentage of gold in 18 karat gold is equal to $\frac{18}{24} \times 100\% = 75\%$

∴ Answer D

10. Which one of the following is a colligative property of solvent in a solution?

A .Lowering of vapor pressure C. Boiling point

B .Osmotic pressure D. Change in entropy.

Explanation :The colligative property of a solvent in a solution is the change that arises due to the addition of a solutes to the solvent.

✓ **Colligative properties** refer to the changes in vapor pressure , boiling point , freezing point and osmotic pressure due to the addition of a solutes to the solvent.

∴ Answer A

11.Which one of the following is the criterion for a spontaneous change using the thermodynamic parameter entropy (S) ?

A .$\Delta S_{system} < 0$ C. $\Delta S_{universe} \leq 0$

B . $\Delta S_{system} \leq \Delta S surr$ D. $\Delta S_{universe} > 0$

Explanation :In any **spontaneous change** , there is always an increase in the entropy of the universe, that is , $\Delta S_{universe} > 0$.

∴ Answer D

12.Which element corresponds to the electron configuration $[Ne]3s^23p^1$?

A .Mg B. Al C. Si D. Ga

Explanation : The electronic configuration $[Ne]3S^2\ 3P^1$ is for an element whose atomic number is equal to 13 and that is aluminium.

∴ Answer B

13.Which bond between the following groups of elements will be the most polar?

A .S – F B .S – Cl C .S- Br D. S – I

Explanation : The **halogen** that forms the most polar bond with sulfur is fluorine . Since **fluorine** is the most electro negative element. The

electro negativity difference is maximum for S – F bond and minimum for S – I bond.

∴ Answer A

14.What is the unit for the rate of a chemical reaction if the concentration is expressed in moles per liter and the time in seconds?

A .$Lmol^{-1}s^{-1}$ C. s^{-1}

B .$molL^{-1}S^{-1}$ D. $molL^{-1}S^{-2}$

Explanation :The unit for the rate of a chemical reaction can be derived from the equation of rate.

$$Rate = \frac{\text{CHANGE IN CONCENTRATION (mol/L)}}{\text{CHANGE IN TIME (sec)}}$$

✓ Unit of rate is equal to **mol L^{-1} S^{-1}**

∴ Answer B

15. What is the most important type of solute – Solvent interaction in a solution of $CaCl_2$ in water?

A .Dipole – dipole C. Hydrogen bonding

B .Dispersion forces D. Ion – dipole

Explanation : In the dissolution process of $CaCl_2$ in water.

i) The partial negative end of water attracts Ca^{2+} ions.

ii) The partial positive end of water molecule attracts the Cl^- ions.

iii) There is interaction between the dipole water and the ions of the salt.

The solute – solvent interaction is called , **ion – dipole.**

∴ Answer D

16. A bleaching solution contains 5.2 mass percent sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2500g of bleaching solution?

A .90 B .130 C .480 D .520

Explanation : The solution is 5.2% NaOCl by mass . That means there are 5.2g NaOCl in 100g of the bleaching solution.

5.2g ..... NaOCl in 100g solution

Xg..... NaOCl in 2500 g solution

$$\frac{x}{5.2} = \frac{2500}{100} \Rightarrow x = 130\ g$$

∴ Answer B

17. What is the correct order of increasing acid strength in the following species: HBrO , $HBrO_2$,$HBrO_3$, and $HClO_3$?

A .$HBrO < HBrO_2 < HBrO_3 < HClO_3$

B .$HBrO < HBrO_2 < HClO_3 < HBrO_3$

C .$HClO_3 < HBrO_3 < HBrO_2 < HBrO$

D .$HBrO_3 < HBrO_2 < HBrO < HClO_3$

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Explanation : The strength of the oxyacids of the halogens increases:
i) as the oxidation number increases.
ii) as the electro negativity increases.
iii) as the number of oxygen atoms increases.
Based on this guide line, we get the following order. Therefore , $HBrO < HBrO_2 < HBrO_3 < HClO_3$ since chlorine is more electronegative.

∴ Answer A

18. A 50.0 mL sample of 0.50M acetic acid (CH_3COOH, $K_a = 1.8\times10^{-5}$) is titrated with 0.150 M NaOH solution. What is the pH of the solution when no NaOH is added?

A .1.6 B .2.28 C .2.58 D .3.17

Explanation : If no NaOH is added, we consider the dissociation of the weak acid which determines the PH of the solution.

$$CH_3COOH\ (aq) \rightleftharpoons H^+\ (aq) + CH_3COO^-\ (aq)$$

	CH_3COOH	H^+	CH_3COO^-
Initial	0.5M	0	0
Change	- x	+x	+x
Equil	0.5 – x	x	x

$$K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$$

$$1.8\times 10^{-5} = \frac{(x)(x)}{0.5-x} \quad \text{..... (Assume } 0.5 - x \approx 0.5)$$

$$1.8\times 10^{-5} = \frac{x^2}{0.5} \Longrightarrow x = 3.0\times10^{-3}M = [H^+]$$

$$PH = -\log [H^+]$$

$$PH = -\log (3.0\times 10^{-3}) \Longrightarrow pH = 3 - \log 3 = 2.58$$

∴ Answer C

19. The enthalpy change for the conversion of white phosphorus to red phosphorus is $-17.7\ kJ\ mol^{-1}$. Which of the following statement is true?
A .One mole of red phosphorus will give off more heat when burn in air than one mole of white phosphorus.
B .One mole of white phosphorus will give off more heat when burned in air than one mole of red phosphorus.
C .The heat generated by both types is the same.
D .The enthalpy change for the conversion of red phosphorus to white phosphorus will be $-35.4\ kJmol^{-1}$.

Explanation : Phosphorus (white) ⟶ phosphorus (red) $\Delta H = -17.7$ kJ

White Phosphorus spontaneously changes to the more stable red phosphorus. Systems become stable when their potential energy is minimized. Initially the energy content of the white phosphorus is

larger. Therefore , one mole of white phosphorus will give off more heat when burned in air than one mole of red phosphorus.

∴ Answer B

20. Which of the following is NOT a spontaneous process?
A . The rusting of iron.
B . The dissolution of salt in water.
C .A clear solution becomes a uniform color after a few drops of dye are added.
D .Cleaning a room.

Explanation : **A spontaneous process** is a process that occurs without an external influence. Once started it does not require any external influence in order to continue. The process in A , B and C are spontaneous. But in cleaning a room force must be applied continuously by some body.

∴ Answer D

21. What are the expected products of the electrolysis of an aqueous solution of $CuCl_2$?
A .Cu and H_2 at the cathode and Cl_2 and O_2 at the anode.
B .Cu at the cathode and Cl_2 at the anode.
C .Cu and H_2 at the cathode and only Cl_2 at the anode.
D .Cu at the cathode and Cl_2 and O_2 at the anode.

Explanation : For the electrolysis of aqueous solution of $CuCl_2$:
i) the substances available for oxidation at the anode are Cl^-, H_2O or OH^-. For higher concentration of $CuCl_2$, Cl_2 (g) is produced and O_2 (g) for dilute solution.
ii) the substances available for reduction at the cathode are Cu^{2+}, H^+ and H_2O . Cu^{2+} is preferentially discharged to Cu metal.

∴ Answer B

22. Which is the second most abundant element in the earth's crust?
A .Aluminum C. Oxygen
B .Silicon D. Iron.

Explanation : Relative abundance of the elements in the earth's crust for the elements in the options is given as:

Element	Oxygen	Silicon	Aluminum	Iron
Percentage	47%	26%	7.5%	4.6%

∴ Answer B

23. The reaction of a dialcohol with diprotic carboxylic acid is an example of ;
A .a condensation polymerization.
B .a copolymer.
C .an addition polymerization.
D .an thermoset polymer.

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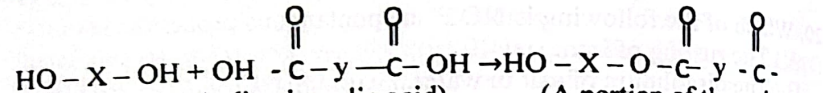
Explanation : A **dialcohol and a dicarboxylic acid** can polymerize by the expulsion of water molecules. Such kind of polymerization is known as **condensation polymerization.**



(dialcohol) (a dicarboxylic acid) (A portion of the polymer)

Where X and Y are methylene groups.

∴ Answer A

24 .Which of the following is **NOT** a carbohydrate?

A .Glucose
B .Cellulose
C. Glycine
D. Starch.

Explanation :

- **Glucose** is a monosaccharide carbohydrate.
- **Starch and cellulose** are polysaccharide carbohydrates.
- **Glycine** is not a carbohydrate. It is an amino acid one of the many small units of proteins.

∴ Answer C

25. The exothermic step in the dissolution of a salt in water is

A .separation of the salt into ions in gaseous state.
B .separation of the water molecules.
C .hydration (solvation) of the ions from the salt.
D . separation of the salt into ions in gaseous state and separation of the water.

Explanation : In the dissolution process of a salt in water:

i) the **ion- dipole forces** attract the positive end of water to the negative ion of the salt.
ii) the **ion – dipole forces** attract the negative end of water to the positive ion of the salt.
As a result of (i) and (ii) water molecules surround the ions of the salt. This condition is known as **hydration** and it is **exothermic.**

∴ Answer C

26. Which of the following is **NOT** correct?

A .Addition of a strongly solvated solute decreases the solubility of a gas in liquid.
B .For solid dissolved in liquid that is exothermic, increase in temperature increases the solubility.
C .The solubility of a substance is its concentration in the saturated solution.
D .For solution to occur solvent – solute attractions must overcome solute – solute and solvent – solvent attraction.

Explanation : For an equilibrium exothermic process, increase in temperature decreases the solubility.

∴ Answer B

27. How many milligrams are in 0.2 decigrams?

A .20 milligrams
B .2000 milligrams
C. 0.002 milligrams
D. 0.00002 milligrams.

Explanation : Use the relevant conversion factor from the following equalities to answer the question.

i) One gram = 10 decigram
ii) One decigram = 10 centigram
iii) One centigram = 10 milligram
iv) From (ii) and (iii) one decigram = 100 milligram

If one decigram is equal to 100 milligram, then

$$0.2 \text{ decigram} \times \frac{100 \text{milligram}}{1 \text{ decigram}} = \mathbf{20 \text{ milligrams}}$$

∴ Answer A

28. How many grams of $CaCO_3$ will be formed by passing excess CO_2 gas through 100 mL of 0.25M $Ca(OH)_2$ solution?

A .10 grams B. 2 grams C. 2.5 grams D. 5 grams

Explanation : i) Determine the number of moles of $Ca(OH)_2$ in 100mL of 0.25M solution, using the relation: $C = \frac{n}{V}$

$n = C \times V$
$n = 0.25M \times 100\ mL$
$n = 0.25\ mol/L \times 0.1L$
$\therefore$ **n = 0.025 moles of $Ca(OH)_2$**

ii) Write a balanced equation for the reaction.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

The mole ratio 0.025 mole 0.025 mole

iii) Convert 0.025 mole of $CaCO_3$ to gram using the equation:

$$n = \frac{m}{M}$$

$\Rightarrow$ mass $m = n \times M$.

Mass (m) = 0.025 mole × 100g/mole (molar mass of $CaCO_3$)

$\therefore$ **Mass of $CaCO_3$ produced is 2.5 grams.**

∴ Answer C

29 .Which one of the following electromagnetic radiation has the longest wavelength?

A .Gamma rays
B .Microwaves
C. Radio waves
D. X –rays

Explanation : In wave Length:

gamma ray < x- rays < uv < Visible < IR < Micro wave <TV< Radio wave.

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❖ **Radio waves** – have the longest wavelength.
❖ **Gamma rays** – have the shortest wavelength (highest energy)

∴ ***Answer C***

30 .What are the possible values of the azimuthal quantum number for an electron in the 4f subshell?

A . 0,1,2,3 B .1,2,3 C. 3 D. 4

Explanation :If n = 4 , the possible values of the azimuthal quantum number and the corresponding subshells are given below.

Azimuthal quantum number (ℓ)	0	1	2	3
Type of subshell	4s	4p	4d	4f

Therefore, the value of the azimuthal quantum number (ℓ) for a 4f electron is equal to 3.

∴ ***Answer C***

31.Which one of the following is the correct electronic configuration of Ni?

A . $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$ C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 5s^2$

B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^1$

C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 5s^2$

D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2 4p^2$

Explanation :The atomic number of **nickel** is given as 28. For an atom of nickel, the number of electrons is 28 having the following configuration. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$.

∴ ***Answer A***

32.What type of hybridization is used by carbon in C_2H_2 (acetylene)

A . sp B. sp^3 C. sp^2 D. sp^2d

Explanation :To determine the type of hybridization used by carbon in **acetylene (C_2H_2)** work out the structure as **H – C ≡ C – H.** In this structure there are two sets of electrons around each carbon atom that require two hybrid orbitals. To accomplish this, an **S – orbital** and a **P – orbital** mix up and hybridize to give two equivalent **sp hybrid** orbitals. These orbitals are oriented linearly on the opposite side of the carbon nucleus.

∴ ***Answer A***

33. Which of the following rate relationships is valid for the reaction:

$2NO_2(g) \longrightarrow 2NO(g) + O_2(g)$?

A . $\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$ C. $-\frac{1}{2}\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$

B . $\frac{-\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$ D. $\frac{\Delta[O_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$

Explanation :The rate expression for the reaction can be given in terms of the variation of concentration of each reactant or product with time.

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[NO_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[NO]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$$

∴ ***Answer C***

34.Which of the following statement is NOT correct regarding equilibrium constant? The value of the equilibrium constant does not depend upon the _____.

A . initial concentration of the reactants.
B .initial concentration of the products.
C .nature of the reactants.
D .Presence of a catalyst.

Explanation :Starting with the reactant or product equilibrium can be reached (attained).

i) therefore , the initial concentrations of reactants or products can affect the direction of shift of equilibrium.
ii) nature of the reactants is not a factor .
iii) presence of catalyst does not affect the equilibrium position.
iv) the value of the equilibrium constant does not depend upon: **concentration , pressure , presence of catalyst.**
vi) the value of the equilibrium constant depends upon:

- changes in temperature
- the form of the equation.
- the nature of the reactant

∴ ***Answer C***

35. Consider the reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ in a closed container at equilibrium at a constant temperature. What will be the effect of adding more PCl_5 on the equilibrium concentration of $Cl_2(g)$?It

A .decreases
B .cannot be predicted without the value of Kp.
C .remains unaffected.
D .increases.

Explanation :According to Le chatelier's principle:

i) addition of more $PCl_5(g)$ is a stress on the equilibrium system.
ii) the direction that relives the stress is the direction that consumes the PCl_5 (g).
iii) $PCl_5(g)$ is consumed in the forward reaction ,and the equilibrium shifts to the right **increasing** the concentration of Cl_2 (g) and $PCl_3(g)$

∴ ***Answer D***

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36. What is the mole fraction of a solute in one molal aqueous solution?
A .0.009 B. 0.018 C .0.027 D. 0.036

Explanation : To determine the mole fraction of a component in a solution, the number of moles of the components must be known (calculated).In a 1.0 molal aqueous solution:

(i) the number of moles of solute is 1.0.

(ii) the number of moles of solvent 1kg/molar mass of water since it is an aqueous solution.

$$n_{water} = \frac{1kg}{18g/mole} = \frac{1000g}{18g/mole}$$

Moles of solvent (water) = 55.6 mole

Moles of solute = 1.0 mole

$$X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{1.0}{1.0+55.6} = 0.0176$$

∴ The mole fraction of solute is 0.0176 which is approximately equal to 0.018.

∴ ***Answer B***

37. Which of the following aqueous solutions will have the lowest freezing point? 0.1 molal solution of _____.
A .urea C. sodium chloride
B . acetic acid D. calcium chloride.

Explanation :

i) **NaCl and $CaCl_2$** dissolve in water and ionize giving rise to a large number of ions. They have greater **colligative effect** on the freezing point of the solution than acetic acid which ionizes only slightly.

ii) **Urea** is a white crystalline solid that is highly soluble in water due to hydrogen bond. It forms molecules of urea not ions. Being a non-electrolyte 0.01 molal solution of urea has the lowest freezing point.

∴ ***Answer D***

38. Which one of the following pairs **CANNOT** be mixed together to form a buffer solution?
A .NH_3 and NH_4Cl C. RbOH and HBr
B .$NaC_2H_3O_2$ and $HC_2H_3O_2$ D. $H_3PO_4$and KH_2PO_4

Explanation :**A buffer solution** contains:

i) a weak acid and its conjugate base as the pairs in Choices B and D or

ii) a weak base and its conjugate acid as the pair given in Choices A.

iii)choice C contains a strong base and a strong acid and the mixture does not form a buffer.

∴ ***Answer C***

39. Which one of the following aqueous solutions will be basic?
A .NaCl C .Na_2CO_3
B .Na_2SO_4 D .$FeCl_3$

Explanation :

i) The salts in A, B and D are derived from strong acids and strong bases. They are **neutral salts** since their ions do not react with water to produce an acidic or basic solution.

ii) But **Na_2CO_3** of choice C is a salt derived from the strong base NaOH and the weak acid H_2CO_3. It is a basic salt since its anion the CO_3^{2-} hydrolyzes to give a basic solution as follows.

$CO_3^{2-}(aq) + H_2O(\ell) \rightleftharpoons HCO_3^-(aq) + OH^-(aq)$

∴ ***Answer C***

40. Consider the reaction:
$CH_4(g) + 4Cl_2(g) \longrightarrow CCl_4(g) + 4HCl(g)$, $\Delta H = -434KJ$
What energy change occurs when 1.2 moles of methane reacts?
A .5.2×10^5 J are released C. 3.4×10^5 J are released
B .5.2×10^5 J are absorbed D. 3.6×10^5 J are absorbed

Explanation :When

1.0 mole of CH_4 reacts ⟶ 434 kJ is given off.

1.2 mole of CH_4 reacts ⟶ X kJ is given off.

$$\frac{X}{434} = \frac{1.2\ mole}{1.0\ mole} \Longrightarrow x = 520.8KJ$$

$$\text{Or } x = 520.8\ KJ \times \frac{1000J}{KJ}$$

∴**x = 520800.Joules** (released)

$x = 5.2 \times 10^5$ Joules of heat energy is given off.

∴ ***Answer A***

41. When electricity is passed through aqueous solution of aluminium chloride, 13.5 g of Al are deposited. What must be the number of Faradays of electricity passed?
A .2.0 B. 1.5 C .1.0 D .0.5

Explanation: **Redox reaction** for the deposition of Al at the cathode.

$Al^{3+} + 3e^- \longrightarrow Al$

i) From the equation 3.0 moles of electrons (3F) deposit 1.0 mole of Al (27g of Al).

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ii) to deposit half mole of Al which is 13.5 g, 1.5 moles of electrons (1.5F) of electricity is required.

∴ ***Answer B***

42 .Which of the following is the anodic reaction in the discharge of a lead – acid storage battery?

A .$Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(S) + 2e^-$

B .$PbSO_4(S) + 2e^- \longrightarrow Pb(s) + SO_4^{2-}(aq)$

C .$PbO_2(s) + SO_4^{2-}(aq) + 4H^+ + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(\ell)$

D .$PbSO_4(s) + 2H_2O(\ell) \longrightarrow PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^-$

Explanation: The target of the equation is to identify anodic reaction.

i) The reactions of choices B and C are not anodic reaction since there is reduction by gain of electrons.

ii) The reactions of choices A and D are anodic reaction (oxidation reaction) since there is lose of electrons.

When the battery is **discharged:**

i) PbO_2 is reduced (**cathodic**).

ii) Pb (s) is oxidized (**Anodic**).

∴ ***Answer A***

43.With which one of the following is gold complexed to recover it from its ore?

A .NO_3^- B .CO_3^{2-} C .SO_4^{2-} D.CN^-

Explanation: A process called **cyanidation** treats gold ore with an aqueous cynanide (CN^-) solution in the presence of oxygen. The process dissolves the gold by forming the complex ion $Au(CN)_2^-$. Gold is recovered from this complex using zinc powder as a reducing agent.

∴ ***Answer D***

44. For the reaction , $3O_2(g) \longrightarrow 2O_3(g)$; $\Delta H^\circ = 284KJ$. What are the signs of ΔG and ΔS respectively?

A .positive and negative C. negative and negative

B .positive and positive D. negative and positive

Explanation: For this reaction to be non - spontaneous:

i) ΔG must be positive in $\Delta G = \Delta H - T\Delta S$

ii) ΔS must be Negative

The combined effect of (i) and (ii) makes the reaction non-spontaneous at higher temperature.

∴ ***Answer A***

45.Which of the following is **NOT** true about the anode of a voltaic cell?

A .Reduction occurs at it.

B .It is assigned as the negative pole.

C .Electrons flow from it.

D .Cations in solution move away from the surface of the electrode as the redox reaction occurs.

Explanation: choices B ,C and D are true. Choice A is not true. Because it is oxidation that occurs at the anode and not reduction.

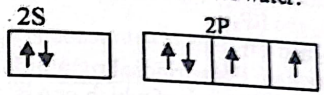
∴ ***Answer A***

46. What is the orbital hybridization of the oxygen atom in water?

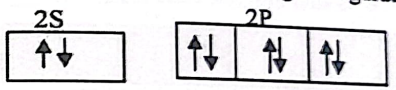
A .sp B. sp^2 C. sp^3 D. sp^3d

Explanation: Orbital hybridization of oxygen atom in water:

Ground state :



After two pairs of electrons are shared, the following configuration is obtained.



✓ **Two electrons** from hydrogen and **six electrons** from oxygen give rise to 4 pairs (sets) of electrons around the central oxygen atom corresponding to 4 hybrid orbital's. S+P+P+ P mix and transform into four equivalent sp^3 hybrid orbital's.

∴ ***Answer C***

47. Within a specified period, an increase in atomic number is usually accompanied by:

A .an increase in atomic radius and an increase in electro negativity.

B .a decrease in atomic radius and an increase in electro negativity.

C .an increase in atomic radius and a decrease in electro negativity.

D .a decrease in atomic radius and a decrease in electro negativity.

Explanation: The trend in some periodic properties across a specific period with increase in atomic number.

i) Atomic radius (Atomic size) decreases.

ii) Electro negativity increases.

iii)Ionization energy increases.

∴ An increase in atomic number is accompanied by a decrease in atomic radius and increase in electronegativity.

∴ ***Answer B***

48. Which of the following sets of quantum numbers is NOT allowed in the hydrogen atom?

A .$n=2, \ell=1, m_\ell = -1$ C. $n=8, \ell=7, m_\ell = -6$

B .$n=1, \ell=0, m_\ell = 0$ D. $n=2, \ell=0, m_\ell = 2$

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Explanation: The value of the azimuthal quantum number(ℓ) determines the value of the magnetic quantum number(m_ℓ).As an example if $\ell = 0$ the value of m_ℓ will be only zero and cannot be equal to 2 as in option D. There fore 2 is not allowed for m_ℓ when $\ell = 0$.

∴ ***Answer D***

49.NF_3,BF_3 and ClF_3 all have molecular formulae of the type XF_3.Which molecules (s) has (have) trigonal pyramidal molecular geometry?

A .NF_3 and BF_3 C. NF_3 and ClF_3

B .BF_3 and ClF_3 D. NF_3

Explanation: For a trigonal pyramidal molecular geometry there must be 4 sets of electrons of which one is a lone pair around the central atom. The following table simplifies the problem.

Molecule	C			
NF_3	N	3	1	Trigonal pyramid
BF_3	B	3	0	Trigonal planar
ClF_3	Cl	3	2	T - shape

∴ ***Answer D***

50 .Which of the following molecules **CANNOT** be formed?

A .NCl_3 C .ONF_3

B .OF_6 D .PCl_3

Explanation: Oxygen being a second period element can accommodate a maximum of 8 electrons. This forms 4 pairs of electrons corresponding to 4 covalent bonds. Therefore, oxygen can form a maximum of 4 bonds. Six bonds as in OF_6 is impossible.

∴ ***Answer B***

51.Of Br_2 , Ne , HCl and N_2 , which is likely to have the largest inter molecular dispersion forces?

A .Br_2 B. Ne C. HCl D.N_2

Explanation: Dispersion forces are predominantely associated to non-polar molecules. The strength of dispersion forces increases with increasing molecular size. This is because larger molecules polarize easily to form induced dipoles. Here , it is Br_2 that is the larger molecule mass and has the largest intermolecular dispersion forces.

∴ ***Answer A***

52 .Thiosulfate ion is oxidized by iodine according to the following reaction:

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

If , in a certain experiment, 0.006 mol of $S_2O_3^{2-}$ is consumed in 1.0L of solution each second, at what rates are $S_4O_6^{2-}$ and I^- produced in this solution?

A .0.012 $molL^{-1}S^{-1}(S_4O_6^{2-})$ and 0.006 $molL^{-1}s^{-1}(I^-)$.

B .0.006$molL^{-1}S^{-1}(S_4O_6^{2-})$ and 0.006 $molL^{-1}S^{-1}(I^-)$.

C .0.003$molL^{-1}S^{-1}(S_4O_6^{2-})$ and 0.006$molL^{-1}S^{-1}(I^-)$.

D .0.012$molL^{-1}S^{-1}(S_4O_6^{2-})$ and 0.012$molL^{-1}S^{-1}(I^-)$.

Explanation: The following facts are obtained from the balanced chemical equation.

i) The rate at which $S_2O_3^{2-}$ is consumed is twice as much as the rate at which $S_4O_6^{2-}$ is produced . $\frac{\Delta[S_2O_3]^{2-}}{\Delta t} = 2\frac{\Delta[S_4O_5]^{2-}}{\Delta t}$

$$\frac{0.006}{2} = \frac{\Delta[S_4O_6]^{2-}}{\Delta t} \Longrightarrow \frac{\Delta[S_4O_6]^{2-}}{\Delta t} = 0.003 \text{ mol } L^{-1}S^{-1}$$

ii) The rate at which $S_2O_3^{2-}$ is consumed is the same as the rate at which I^- is produced $\frac{\Delta[S_2O_3]^{2-}}{\Delta t} = \frac{\Delta[I^-]}{\Delta t} = 0.006 \text{mol } L^{-1}S^{-1}$

∴ ***Answer C***

53.A particular reaction was found to depend on the concentration of the hydrogen ion, $[H^+]$.The initial rate changed as a function of $[H^+]$ as follows:

$[H^+]$ (M)	0.005	0.01	0.02
Initial rate (M/s)	3.4×10^{-7}	1.7×10^{-7}	0.85×10^{-7}

What is the order of the reaction in $[H^+]$?

A .-1 B .1 C .2 D .0

Explanation: Determine the order of the reaction from the first two sets of measurements.

Rate 2 = k (0.01)^x Equation1

Rate 1 = k (0.005)^x Equation2

Divide equation 1 by equation 2 we get:

$$\frac{\text{Rate2}}{\text{Rate1}} = \frac{k(0.01)^x}{k(0.005)^x}$$

$$\frac{1.7\times10^{-7}}{3.4\times10^{-7}} = \left(\frac{0.01}{0.005}\right)^x$$

$$\frac{1}{2} = \left(\frac{2}{1}\right)^x \Longrightarrow 2^{-1} = 2^x$$

⇒ x = -1. Therefore ,the order of the reaction is $[H^+]$ is -1.

∴ ***Answer A***

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54. The conversion of ozone into O_2 proceeds in two elementary steps as follows:

$O_3(g) \longrightarrow O_2(g) + O(g)$

$O_3(g) + O(g) \longrightarrow 2O_2(g)$

What is the equation for the overall reaction?

A. $2O_3(g) \longrightarrow 3O_2(g)$

B. $2O_3(g) + O(g) \longrightarrow 3O_2(g) + O(g)$

C. $O_3(g) + O(g) \longrightarrow O_2(g) + 2O(g)$

D. $2O_3(g) + O(g) \longrightarrow 3O_2(g)$

Explanation: The equation for the overall reaction is obtained from the sum of the elementary steps. In doing so cancel out the species that appear on both sides.

$$O_3(g) \longrightarrow O_2(g) + O(g)$$
$$O_3(g) + O(g) \longrightarrow 2O_2(g)$$
$$2O_3(g) + O(g) \longrightarrow 3O_2(g) + O(g)$$

Cancelling O(g) from both sides gives

$$2O_3(g) \longrightarrow 3O_2(g)$$

∴ Answer A

55. Which of the following factors does **NOT** affect the rate of a chemical reaction that is of zero order?

A. Concentration of reactants.
C. Temperature
B. Surface area of reactants.
D. Presence of a catalyst.

Explanation: Factors B,C and D affect the rate of a zero- order reaction. A reaction that is zero –order overall is independent of any **concentration of reactants.**

∴ Answer A

56. How would you prepare 0.90 L of 1.2×10^{-2}M NaCl aqueous solution?

A. Weigh 0.012 g of NaCl, dissolve in a small amount of water and dilute to 0.90L.

B. Weigh 0.63 g of NaCl , dissolve in a small amount of water and dilute to 0.90L.

C. Weigh 1.2g of NaCl, dissolve in a small amount of water and dilute to 0.90L.

D. Weigh 1.4g of NaCl, dissolve in a small amount of water and dilute to 0.90L.

Explanation: From the given information :

i) determine the number of moles of sodium chloride (NaCl) using the relation : $C = \frac{n}{V}$

$n_{NaCl} = C \times V$

$n_{NaCl} = 0.012 \frac{mol}{L} \times 0.9L = 0.0108$ mole

ii) converting 0.0108 mole to mass gives 0.63 g of NaCl.

∴ Answer B

57. Which of the following solutes will produce the largest increase in boiling point upon addition to 1kg of water?

A. 1 mol of $Co(NO_3)_2$

B. 2mol of KCl

C. 3mol of ethylene glycol($C_2H_6O_2$)

D. 1 mol of $CaCl_2$

Explanation: The species given in Choices A,B and D are electrolytes.

✓ **Electrolytes** produce greater colligative effect than non-electrolytes like ethylene glycol

✓ **Electrolytes** produce more particles per mole and the largest number of particles is produced by two moles of KCl.4 moles of ions are obtained from 2 moles of KCl.

∴ Answer B

58. What is the percentage of HF molecules ionized in a 0.10M HF solution ($K_a = 6.8\times10^{-4}$)?

A. 2
B. 5
C. 8
D. 10

Explanation: Per cent dissociation $= \frac{\text{Amount ionized } (\frac{mol}{L})}{\text{Initial conc } (\frac{mol}{L})} \times 100$

	HF (aq)	$\rightleftharpoons$	H^+ (aq)	+	F^-(aq)
Initial	0.10 M		0		0
Change	- x		+x		+x
Equil	0.10 − x		x		x

$$K_a = \frac{[H^+][F^-]}{[HF]} = \frac{(X)(X)}{0.1-X}$$

$K_a = \frac{x^2}{0.1-x}$ ..... Assume $0.1 - x \approx 0.1$

$$6.8\times10^{-4} = \frac{x^2}{0.1} \Longrightarrow 0.1\times 6.8\times10^{-4} = x^2$$

$x^2 = 68\times 10^{-6}$

$x = 8.25\times 10^{-3} = [H^+][F^-]$

$$\%diss = \frac{\text{Amount dissociated } (\frac{mol}{L})}{\text{Initial conc'n}(\frac{mol}{L})} \times 100\%$$

$$\%diss = \frac{8.25\times10^{-3}\times10^{2}}{10^{-1}} = 8.25 \times 10^{-3}\times 10^{2}\times 10^{1}$$

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% diss = 8.25%

8% in choice **C** is closer to the calculated value 8.25%

∴ Answer C

59. How does the titration of a strong acid with a strong base differ from titration of a weak acid with a strong base with respect to the quantity of base required to reach the equivalence point?

A .The quantity of base required to reach the equivalence point is the same in both titrations.

B . The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is higher than for the titration of a weak acid with a strong base.

C .The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is lower than for the titration of a weak acid with a strong base.

D .The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is slightly higher than for the titration of a weak acid with a strong base.

***Explanation*:** The **equivalent point** is the point on the titration curve at which the number of moles of added OH^- equals the number of moles of H^+ ions originally present and the vice versa. That means the number of equivalents of the acid must be equal to the number of equivalents of the base for any combination.

∴ Answer A

60. Which of the following reactions is a redox reaction?

A .$3Se(s) + 4HNO_3(aq) \longrightarrow 3SeO_2(s) + 4NO(g) + H_2O(\ell)$

B $CaCl_2(aq) + 2AgNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$

C .$BaCO_3(s) \longrightarrow BaO(s) + CO_2(g)$

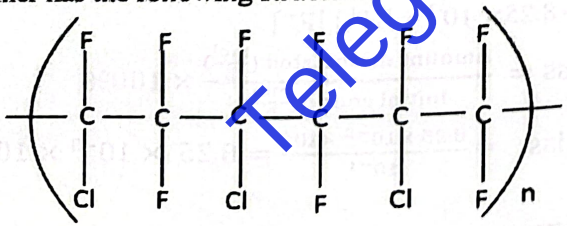
D .$3Mg(OH)_2(s) + 2NH_3(g) \longrightarrow Mg_3N_2(s) + 6H_2O(\ell)$

***Explanation*:** **A chemical reaction** in which a free element appears is always a redox reaction. The free element **selenium** is oxidized from zero to the +4 oxidation state in **SeO_2.**

Nitrogen is reduced from +5 in HNO_3 to +2 in NO.

∴ Answer A

61. A polymer has the following structure:



What is the monomer for this polymer?

A .$CFCl = CF_2$ C. CF_2Cl_2

B .CF_3Cl D. CF_4

***Explanation*:** The **monomer** in the structure of a polymer is the **repeating unit. In this polymer the repeating unit has the structure .**

∴ Answer A

62. How many significant figures are there in 0.0500?

A .One B. Three C. Two D. Four

***Explanation*:**

a) Zeros that precede the first non-zero digit are not significant.

b) Zeros after a non-zero digit preceeded by a decimal point are significant figures. Using this guide line for 0.0500 gives 3 significant figures.

∴ Answer B

63. If 16 grams of O_2 react with excess C_2H_6, how many grams of CO_2 will be formed ? The unbalanced equation is:

$C_2H_6(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$

A .22 grams B. 9 grams C. 13 grams D. 12.6 grams

***Explanation*:** To determine the mass of CO_2 produced balance the equation using the general formula

$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \longrightarrow xCO_2 + \frac{y}{2}H_2O$

$C_2H_6 + \left(2 + \frac{6}{4}\right)O_2 \longrightarrow 2\,CO_2 + \frac{6}{2}H_2O$

Multiply the whole equation by 2.

$2C_2H_6 + 7O_2 \longrightarrow 4CO_2 + 6H_2O$

✓ The given mass of oxygen which is 16 grams is 0.5 mole of oxygen.

✓ From the balanced equation 7 moles of O_2 produces 4 moles of CO_2.

✓ 0.5 moles of O_2 produces 0.286 mole of CO_2.

When this is multiplied by the molar of CO_2, that is , 0.286 mole × 44g mol = **12.57** which is roughly equal to 12.6 g of CO_2.

∴ Answer D

64. Hydrogen atoms are excited to the stationary state designated by the principal quantum number, n= 4. What will be the number of lines (frequencies) emitted by the hydrogen atoms?

A .1 B .4 C .6 D .16

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Explanation: For the transition of electrons

n= 4 to n=3	3 Spectral Lines	Total of 6 Spectral Lines
n= 4 to n=2		
n= 4 to n=1		
n= 3 to n=2	2 Spectral Lines	
n= 3 to n=1		
n= 2 to n=1	1 Spectral Lines	

∴ Answer C

65 .According to the molecular orbital theory, which one of the following is paramagnetic?

A .F_2 B .H_2 C .NO^+ D .NO

Explanation: Each of the species in A,B, and C have paired electrons in their molecular orbital's. There is unpaired electron in NO and it is paramagnetic.

∴ Answer D

66.For the reaction :$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$ the following rate data were obtained:

$[NO]_0$ (M):	0.10	0.10	0.40
$[O_2]_0$ (M):	0.10	0.20	0.20
Rate (M/s):	2.5×10^{-4}	5.0×10^{-4}	8.0×10^{-3}

What is the rate law for the reaction?

A .rate = $k[NO][O_2]$ C .rate = $k[NO]^2[O_2]$

B .rate = $k[NO][O_2]^2$ D. rate = $k[NO]^2[O_2]^2$

Explanation:

i) To determine the order of the reaction in [NO] use the last two sets of measurements for which $[O_2]$ is constant.

$$8.0 \times 10^{-3} = K(0.4)^x \;\Big\|\; \frac{8.0\times10^{-3}}{5.0\times10^{-4}} = \left(\frac{0.4}{0.1}\right)^x$$

$$5.0 \times 10^{-4} = K(0.1)^x$$

$$16 = 4^x$$

$$4^2 = 4^x \Longrightarrow x = 2$$

The reaction is second –order in [NO]

ii) To determine, the order of the reaction in $[O_2]$ use the first two sets of measurements for which $[O_2]$ is constant.

$$5.0 \times 10^{-4} = K(0.2)^x \;\Big\|\; \frac{5.0\times10^{-4}}{2.5\times10^{-4}} = \left(\frac{0.2}{0.1}\right)^x$$

$$2.5 \times 10^{-4} = K(0.1)^x$$

$$2 = 2^x \Longrightarrow x = 1$$

The reaction is first order in $[O_2]$.

The rate law for the reaction is **Rate = $K[NO]^2[O_2]$.**

∴ Answer C

67.The value of k_c for the reaction, $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ is 2.5×10^{-10} at 300K. Starting with pure $COCl_2$ (g) in a closed vessel, if the concentration of CO(g) is 1×10^{-5} M at equilibrium, what would be the equilibrium concentration of $COCl_2(g)$?

A .2.5M C .2.5×10^{-5}M

B .2.5×10^{5}M D . 0.4M

Explanation: The equilibrium constant expression for the reaction is

$$K_c = \frac{[CO][Cl_2]}{[COCl_2]}$$

✓ The equilibrium concentrations of CO and Cl_2 are the same equal to 1×10^{-5}.

✓ Substitute the known values in the above equation, we get :

$$2.5 \times 10^{-10} = \frac{1\times10^{-5}\times1.0\times10^{-5}}{[COCl_2]}$$

$$[COCl_2] = \frac{1\times10^{-10}}{2.5\times10^{-10}} = 0.4M$$

∴ Answer D

68. An aqueous solution freezes at − 0.186 °C (k_f = 1.86°C mol^{-1}; K_b = 0.512°C mol^{-1}). What is the elevation in boiling point (in °C)?

A .0.186 C. 0.512/1.86

B .0.512 D. 0.0512

Explanation: The elevation in boiling point (ΔT_b) is directly proportional to the molarity (m) of the solution.

That is: $\Delta T_b \propto m$ and $\Delta T_b = K_b m = \Delta T_b = K_b \left(\frac{\Delta T_f}{K_f}\right)$

$$\Delta T_b = \frac{0.512°C}{molal} \times \left(\frac{0.186}{1.86}\right) molal.$$

$$\therefore \Delta T_b = 0.0512°C$$

∴ Answer B

69. The pK_a of acetyl salicylic acid (aspirin) is 3.5.The PH of gastric juice in human stomach is about 2 to 3 and the PH in the small intestine is about 8.Which one of the following statement is correct? Aspirin is

A . unionized in the small intestine and in the stomach.

B .completely ionized in the small intestine and in the stomach.

C .ionized in the stomach and almost unionized in the small intestine.

D .ionized in the small intestine and almost unionized in the stomach.

Explanation: We can use the following general dissociation reaction equation for a hypothetical acid HA: $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

i) In the more acidic medium of the stomach the ionization of acetyl salicylic acid (Aspirin) is suppressed due to the common H^+ present in the gastric juice.

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ii) Aspirin is ionized in the slightly basic medium of the small intestine and almost unionized in the stomach.

∴ *Answer D*

70. A 10mL sample of tartaric acid (a diprotic acid) is titrated with 20mL of 1.0M NaOH . What is the molarity of the acid?

A .2.0 B .1.0 C. 4.0 D. 0.5

***Explanation*: A titration** reaches end point, when equivalent amounts of an acid and a base are mixed. In the process, one equivalent of an acid neutralizes one equivalent of a base and equivalent is equal to normality times volume.

$N_AV_A = N_BV_B$ .....For NaOH Normality = Molarity

$N_A \times 10mL = 1.0N \times 20ML \Longrightarrow N_A = \mathbf{2.0N}$

Normality = Molarity × number of ionizable hydrogen per molecule

2.0N= molarity × 2(a diprotic Acid) ⟹ **Molarity = 1.0M**

∴ *Answer B*

71. When the equation for the following reaction in basic solution is balanced, What is the sum of the coefficients?

$MnO_2 + HO_2^- \longrightarrow MnO_4^-$

A .11 B .31 C .14 D .9

***Explanation*:** To get the sum of the coefficients , the equation must be balanced to determine the coefficients. The half – reaction method is the convenient method for this reaction.

$MnO_2 + HO_2^- \longrightarrow MnO_4^-$

i) Split the skeletal equation into oxidation half – reaction and reduction half – reaction.

$MnO_2 \longrightarrow MnO_4^-$ (Oxidation) (+4 ⟶ +7).

$HO_2^- \longrightarrow MnO_4^-$ (Reduction) (-1 ⟶ -2)

ii) Balance atoms other than hydrogen and oxygen in each half – reaction.

$MnO_2 \longrightarrow MnO_4^-$

$MnO_4^- + HO_2^- \longrightarrow MnO_4^-$

iii) Balance oxygen atoms by adding H_2O for every oxygen need to the side deficient in oxygen.

$MnO_2 + 2H_2O \longrightarrow MnO_4^-$

$MnO_4^- + OH_2^- \longrightarrow MnO_4^- + 2H_2O$

iv) Balance hydrogen atoms by H_2O for each hydrogen needed to the side deficient in hydrogen and equal amount of OH^- ions to the opposite side since the reaction is in basic medium.

$MnO_2 + 2H_2O + 4OH^- \longrightarrow MnO_4^- + 4H_2O$

$MnO_4^- + HO_2H^- + 3H_2O \longrightarrow MnO_4^- + 2H_2O + 3OH^-$

V) Balance the charge by adding electrons to the more positive side in each case.

$2\times [MnO_2 + 2H_2O + 4OH^- \longrightarrow MnO_4^- + 4H_2O + 3e]$

$3\times [MnO_4^- + HO_2^- + 3H_2O + 2e^- \longrightarrow MnO_4^- + 2H_2O + 3OH^-]$

Vi) Multiply each half – reaction by appropriate coefficients to equalize the number of electrons gained or lost cancel out species that appear on both sides and you obtain the following balanced equation .

$2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + H_2O + OH^-$

The coefficient add up to 9.

∴ *Answer D*

72. The standard emf for the cell reaction, $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$, is 1.1 V. What would the emf for the cell reaction be when $0.1M Cu^{2+}$ and $0.1M Zn^{2+}$ solutions are used at 298 K?

A .1.0V B .0.10V C .– 1.10V D .– 0.110V

***Explanation*:** The dependence of cell potential on the concentrations of the ions is given by the **Nernst** equation: $E = E^0_{cell} - \frac{0.0592}{n}\log^Q$

At standard thermodynamic condition of 298 K.

For the zinc – carbon cell

$Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$

$$Q = \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$E = E^o_{cell} - \frac{0.0592}{2}\log^Q$$

$$E = 1.10V - \frac{0.0592}{2}\log\frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$E = 1.10V - \frac{0.0592}{2}\log\frac{(0.1M)}{(0.1M)} \text{.....} \log^{1.0} = 0$$

$\therefore \mathbf{E = 1.10\ V}$

∴ *Answer A*

73. Which one of the following is a polyamide molecule?

A .Terylene C. Nylon

B .Rayon D. Polystyrene.

***Explanation*:** It is only the polymer Nylon that contains amide linkage (bonding) and **Nylon** is a polyamide. The monomers are difunctional (diacid and diammine).

∴ *Answer C*

74. Which metal atoms can form ionic bonds by losing electrons from both the outermost and next to outermost principle energy levels?

A .Fe B. Mg C. Pb D. Ca

***Explanation*: Mg ,Ca and Pb** lose electrons only from their outer most shell in ionic bond formation. But **Fe** with $(n-1)d^6\ ns^2$ valence shell electron configuration can lose 2 electrons from the outer most

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shell and 1 electron from the (n-1) shell to form ionic bond. Therefore Fe can lose electrons from the n shell and n-1 shell to form ionic bond.

∴ ***Answer A***

75. The energy required to remove an electron from an iron atom amounts to 7.21×10^{-19} J. What is the maximum wavelength of light that can perform this?

A .250nm B .276 nm C. 289nm D. 300nm

Explanation: To calculate the maximum wavelength corresponding to the threshold frequency we use the fundamental equation of the quantum theory, that is : $\Delta E = h\nu_o$

From the relation : $V_o\lambda_{max} = C$

$$\Longrightarrow \Delta E = h\nu_{minimum}$$

$$\Longrightarrow \Delta E = h\frac{c}{\lambda_{max}}$$

$$\Delta E = \frac{hC}{\lambda_{max}}$$

$$\lambda_{max} = \frac{hC}{\Delta E} = \frac{6.626\times10^{-34}J.S\times3.0\times10^{8}ms^{-1}}{7.21\times10^{-19}J}$$

$$\lambda_{max} = 2.76\times10^{-7}m$$

$$\lambda_{max} = 2.76\times10^{-7}m\times\frac{10^{9}nm}{m}$$

$$\therefore \lambda_{max} = 2.76\times10^{2}nm = 276\ nm$$

∴ ***Answer B***

76. What is the molecular geometry of O_3?

A .Trigonal planar C. Trigonal pyramidal

B .Bent D. Tetrahedral

Explanation:

i) **Linear** if there is no lone pair on the central atom.

ii) **Angular or bent** if there is lone pair on the central atom. The lone pair on the central oxygen gives ozone (O_3) a bent or angular geometry

∴ ***Answer B***

77. At 295K, a 3.0 L flask contains 7 mol HI, 4mol H_2 , and 0.6 mol I_2 in equilibrium. What is the value of K at 295K?

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

A .10.2 B .20.4 C. 30.4 D. 61.2

Explanation: Calculate the equilibrium concentration of each species using the relation: $C = \frac{n}{V}$

$$HI: C = \frac{n}{V} = \frac{7\ mole\ HI}{3.0\ L} = 2.33M$$

$$H_2: C = \frac{n}{V} = \frac{4\ mole\ H_2}{3.0\ L} = 1.33\ M$$

$$I_2: C = \frac{n}{V} = \frac{0.6\ mole\ I_2}{3.0\ L} = 0.2\ M$$

$$K_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2.33)^2}{1.33\times0.2} = \frac{5.44}{0.266} = 20.4$$

∴ ***Answer B***

78. The equilibrium constant is 0.09 at 295 K for the reaction:

$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$

For which of the following sets of conditions is the system at equilibrium?

A .A 1.0L flask contains 1.0 mol of HOCl, 0.10 mol of Cl_2O and 0.1 mol of H_2O

B .A 2.0 L flask contains 0.084 mol of HOCl , 0.080 mol of Cl_2O, and 0.98 mol of H_2O.

C .A 3.0 L flask contains 0.25 mol of HOCl, 0.0010 mol of Cl_2O, and 0.50 mol of H_2O

D .A 4.0 L flask contains 0.25 mol of HOCl, 0.0010 mol of Cl_2O, and 0.56 mol of H_2O

Explanation: Calculate the value of the reaction quotient Q for each set of concentration. Use the relation: $C = \frac{n}{V}$ to calculate the concentrations for each species.

A .HOCl: $C = \frac{n}{V} = \frac{1.0\ mol}{1.0\ L} = 1.0\ M$

Cl_2O: $C = \frac{n}{V} = \frac{0.1\ mol}{1.0L} = 0.1M$

H_2O : $C = \frac{n}{V} = \frac{0.1\ mol}{1.0L} = 0.1\ M$

$$Q = \frac{[HOCl]^2}{[H_2O]\,[Cl_2O]} = \frac{(1)^2}{(0.1)(0.1)} = 100$$

B) HOCl: $C = \frac{n}{V} = \frac{0.084mol}{2.0L} = 0.042M$

Cl_2O: $C = \frac{n}{V} = \frac{0.08mol}{2.0L} = 0.04M$

H_2O: $C = \frac{n}{V} = \frac{0.98\ mol}{2.0\ L} = 0.49M$

$$Q = \frac{[HOCl]^2}{[H_2O][Cl_2O]} = \frac{0.001764}{(0.04)(0.49)} = \frac{0.001764}{0.0196} = 0.09$$

The equilibrium constant for the reaction at 295k = 0.09.

✓ The system is at equilibrium for the sets of conditions in B.

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✓ In a similar manner you can calculate the concentration and the value of the reaction quotient, Q

∴ ***Answer B***

79. 1000g of a 55% (mass percentage) nitric acid solution is to be diluted to 20% (mass percentage) nitric acid solution. How many liters of water should be added to the starting solution?

A .0.45 B .1.55 C .2.30 D .2.75

Explanation:

i) In 1000 g of 55% by mass nitric acid solution, there is 550g $HNO_3$and 450 g water.

ii) During dilution, the mass of the solute (HNO_3) remains to be 550g in 20% nitric solution.

iii) Let x represent the total mass of the diluted solution.

$$\text{Percentage of solution} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100\% = 20\,\%$$

$$\frac{550g}{X} \times 100 = 20$$

$$\frac{550g}{X} = \frac{20}{100}$$

$$\Rightarrow \mathbf{x = 2750g}$$

iv) Subtracting the mass of the starting solution (1000g) from the total mass of the diluted solution (2750g) gives 1750g.The mass of added water is 1750g .Assuming the density of water to be 1g/cm^3

$V_{water} = \frac{mass}{density} = 1750cm^3$

$\therefore$**1750cm^3 = 1.75L of water.**

∴ ***No Answer***

80. When pure methanol is mixed with water, the solution becomes warm. Is this an ideal solution?

A .Yes , for ideal solution $\Delta H_{solution} = 0$

B .No , for ideal solution $\Delta H_{solution} \neq 0$

C .Yes , for ideal solution $\Delta H_{solution} < 0$

D .No, for ideal solution $\Delta H_{solution} > 0$

Explanation: For an **ideal solution**, the solution process is neither **endothermic nor exothermic** since $\Delta H = 0$. But for methanol in water the solution becomes warm. The solution process is exothermic, $\Delta H < 0$.

∴ The solution of methanol in water is not ideal ,since $\Delta H = 0$ for an ideal solution.

∴ ***Answer B***

1. How is a buffer solution prepared?

A .By mixing a weak acid and a strong acid.

B .By mixing a weak acid and its conjugate base.

C .By mixing a strong acid and its conjugate base.

D .By mixing a strong base and its conjugate acid.

Explanation :A **buffer solution** is a solution which undergoes only a slight change in pH when an acid (H^+) or a base (OH^-) is added to it. It is a solution prepared by mixing a **weak acid** and its **conjugate base** or a **weak base** and **its conjugate acid** at approximately equal concentration.

∴ ***Answer*** B

2. Which of the following compounds would be the most basic?

A .0.1M acetic acid C. 0.1 M hydrochloric acid

B .0.1M sodium acetate. D. 0.1M ammonium chloride.

Explanation: 0.1M sodium acetate ($NaCH_3CO_2$) is the most basic. Because the acetate ion of the salt under goes hydrolysis giving rise to a basic solution.

$CH_3COO^-(aq) + H_2O(\ell) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$

The other three release H^+ ions (acidic solution)

∴ ***Answer B***

3. Consider the following reaction:

$$HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$$

What will happen if NaOCl is added to the above reaction at equilibrium?

A .The concentrations of both HOCl and H_3O^+ would increase.

B .The concentrations of both HOCl and H_3O^+ would decrease.

C .The concentration of HOCl would increase and the concentration of H_3O^+ would decrease.

D .The concentration of HOCl would decrease and the concentration of H_3O^+ would increase.

Explanation: If NaOCl is added to the equilibrium system

$HOCl(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OCl^-(aq)$,

the dissociation of the weak acid HOCl decreases due to the common ion, OCl^- and the equilibrium position shifts to the left. As a result H_3O^+ is consumed and HOCl is formed. This causes an increase in the concentration of HOCl and a decrease in the concentration of H_3O^+.

∴ ***Answer C***

4. A solution is labeled 0.500 M HCl. What is $[H^+]$?

A .0.5 B .1.5 C .1.6 D .1.69

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